Biochemistry MCQ Bank

1st Professional MBBS — Corrected Answers with Reasons

How to read: each True/False statement is marked **[T]** (true) or **[F]** (false). The italic line below gives the reason. For single-best-answer questions (Q11–20 in newer papers) the correct option is shown in green with the reason.

† after a question title means the original option wording or answer was faulty (typo, two correct options, or a wrong key) and has been corrected here.

Answers follow standard medical biochemistry textbook facts. Where a statement is debatable, the more accepted textbook position is used. Please verify against your own textbook/teacher if in doubt.

November 2025 — Paper I

Q1. HCO_3^- buffer system

a) The HCO_3^-/H_2CO_3 buffer is the major ECF buffer **[T]**
Reason: Bicarbonate is the most important ECF buffer (high concentration, open system via lungs and kidneys).

b) The ratio of HCO_3^- to H_2CO_3 in blood is 20:1 **[T]**
Reason: At pH 7.4 the ratio $HCO_3^-:H_2CO_3$ is 20:1 (24 : 1.2 mmol/L).

c) Carbonic anhydrase accelerates CO_2 hydration **[T]**
Reason: Carbonic anhydrase catalyses $CO_2 + H_2O \rightleftharpoons H_2CO_3$, so it is central to bicarbonate buffering.

d) Lungs regulate HCO_3^- concentration directly **[F]**
Reason: Lungs regulate CO_2 (H_2CO_3), not HCO_3^-; kidneys regulate HCO_3^- and H^+.

e) Kidney regulates H^+ ion excretion **[T]**
Reason: Kidneys excrete H^+ and reclaim/generate HCO_3^-.

Q2. In glycolysis the following statements are correct

a) The end product is lactate **[F]**
Reason: End product of glycolysis is pyruvate (aerobic); lactate forms only in anaerobic conditions.

b) The end product is pyruvate **[T]**
Reason: Aerobic glycolysis ends in pyruvate (to mitochondria); lactate is formed only anaerobically.

c) There are four irreversible reactions **[F]**
Reason: Glycolysis has 3 irreversible steps: hexokinase/glucokinase, PFK-1, pyruvate kinase.

d) Lactate conversion to pyruvate is oxidation **[T]**
Reason: Lactate → pyruvate removes 2H (LDH, NAD^+ → NADH): oxidation.

e) In the aerobic pathway 2 ATP are produced (net) **[F]**
Reason: Net ATP: anaerobic = 2; aerobic ≈ 7–8 (2 + NADH via shuttles).

Q3. Regarding hypothyroidism

a) Serum TSH is always increased in primary hypothyroidism **[T]**
Reason: Primary hypothyroidism: gland failure → ↓T_3/T_4 → ↑TSH (feedback).

b) Serum TSH is always decreased in secondary hypothyroidism **[T]**
Reason: Secondary hypothyroidism: pituitary failure → TSH low (or inappropriately normal) with low T_4.

c) Serum T_3, T_4 are always decreased in primary hypothyroidism **[F]**
Reason: In subclinical primary hypothyroidism T_3/T_4 stay normal with raised TSH.

d) Serum T_3, T_4 are always decreased in secondary hypothyroidism **[T]**
Reason: Secondary hypothyroidism has low T_3/T_4.

e) Serum T_3, T_4 can be normal in primary (subclinical) hypothyroidism **[T]**
Reason: Subclinical hypothyroidism: normal T_3/T_4, raised TSH.

Q4. Regarding ATP

a) It is a nucleotide derivative with 3 phosphate residues **[T]**
Reason: ATP = adenine + ribose + 3 phosphates.

b) It is a high-energy phosphate producing more than 7 kcal/mol of energy **[T]**
Reason: ATP hydrolysis ≈ –7.3 kcal/mol.

c) It is formed only in the respiratory chain **[F]**
Reason: ATP is also formed by substrate-level phosphorylation.

d) It is not formed at substrate level **[F]**
Reason: Substrate-level phosphorylation exists (PK, PGK, succinyl-CoA synthetase).

e) More than 50% of ATP is used to drive the osmotic (Na^+-K^+) pump **[F]**
Reason: Na^+-K^+ pump uses ~20–40% of ATP.

Q5. Insulin

a) Is a potent anabolic hormone **[T]**
Reason: Insulin promotes glycogen, fat and protein synthesis.

b) Is required for utilization of glucose in brain **[F]**
Reason: Brain uses GLUT-3, which is insulin-independent.

c) Increases glucose uptake in skeletal muscle **[T]**
Reason: Insulin recruits GLUT-4 in muscle and adipose tissue.

d) Increases plasma K^+ **[F]**
Reason: Insulin drives K^+ into cells (lowers plasma K^+); used to treat hyperkalaemia.

e) Mode of action is mediated by cyclic AMP **[F]**
Reason: Insulin acts via tyrosine kinase receptor.

Q6. Regarding cholesterol

a) HMG-CoA reductase is the rate-limiting enzyme in its synthesis **[T]**
Reason: HMG-CoA reductase is the rate-limiting step.

b) Is an important precursor of sex and adrenal steroids **[T]**
Reason: Cholesterol is the precursor of all steroid hormones.

c) Excess LDL cholesterol is a risk factor for atherogenesis **[T]**
Reason: LDL deposition in intima initiates atherosclerosis.

d) LDL cholesterol requires apo B-48 for receptor internalization **[F]**
Reason: LDL receptor binds apo B-100 (and E); B-48 lacks the receptor domain.

e) VLDL cholesterol is produced in the intestine **[F]**
Reason: VLDL is made in the liver; chylomicrons in intestine.

Q7. Carbohydrates †

a) All monosaccharides are reducing sugars **[T]**
Reason: All monosaccharides have a free aldehyde/ketone group → reducing.

b) Sucrose is a non-reducing sugar **[T]**
Reason: Sucrose has glucose C1 and fructose C2 linked, no free anomeric carbon.

c) Glucose shows mutarotation **[T]**
Reason: Free anomeric carbon interconverts α/β forms in solution.

d) Blood reducing sugars are detected by Benedict's test **[T]**
Reason: Benedict's test detects reducing sugars (glucose, fructose, lactose, maltose), not sucrose.

e) Aldose is a non-reducing sugar **[F]**
Reason: Aldoses have a free aldehyde → reducing.

Q8. Which of the following are true for ketone bodies

a) They are acetoacetate and acetone only **[F]**
Reason: Ketone bodies are acetoacetate, β-hydroxybutyrate and acetone.

b) They are acidic **[T]**
Reason: Acetoacetate and β-hydroxybutyrate are acids.

c) Acetoacetate spontaneously decarboxylates to acetone **[T]**
Reason: Acetoacetate → acetone + CO_2 spontaneously.

d) During starvation ketones are a major substrate for brain **[T]**
Reason: In starvation brain adapts to use ketones.

e) They are not produced in uncontrolled diabetes mellitus **[F]**
Reason: Lack of insulin → lipolysis → ketogenesis.

Q9. Choose the correct answer †

a) Achondroplasia is an example of an autosomal recessive disorder **[F]**
Reason: Achondroplasia is autosomal DOMINANT (FGFR3).

b) Autosomal recessive phenotypes appear in the heterozygous state **[F]**
Reason: AR traits appear only in homozygotes.

c) Mitochondrial DNA is inherited from the mother only **[T]**
Reason: mtDNA is maternally inherited (sperm mitochondria are lost).

d) The karyotype of Down syndrome is trisomy 21 **[T]**
Reason: Down syndrome = trisomy 21 (47,XX,+21).

e) Father-to-son transmission is possible in X-linked disease **[F]**
Reason: X-linked genes cannot pass father→son (son receives Y).

Q10. Regarding liver function tests

a) Serum gamma-glutamyl transpeptidase is elevated in viral hepatitis **[T]**
Reason: GGT rises in hepatobiliary disease, including viral hepatitis.

b) Serum albumin level is 3.5–5.4 g/dL **[T]**
Reason: Normal serum albumin 3.5–5.4 g/dL.

c) Prothrombin time is normal in acute liver disease **[F]**
Reason: Severe hepatocellular injury prolongs PT; PT is a synthetic function test.

d) A normal albumin–globulin ratio excludes chronic liver disease **[F]**
Reason: A:G is reversed in chronic liver disease; a normal ratio cannot exclude it.

e) Serum bilirubin indicates synthetic function of the liver **[F]**
Reason: Bilirubin = excretory function; synthetic = albumin, PT.

Q11. Which level of protein structure involves H-bonding?

a) Primary
b) Secondary ✓
c) Tertiary
d) Quaternary

Answer: b) Secondary

Reason: Secondary structure (α-helix, β-sheet) is stabilised mainly by H-bonds.

Q12. Which test reflects long-term glycemic control?

a) FBS
b) RBS
c) OGTT
d) HbA_{1c} ✓

Answer: d) HbA_{1c}

Reason: HbA_{1c} reflects average glycaemia of the past 2–3 months; FBS/RBS/OGTT are single-time tests.

Q13. The most specific biomarker for myocardial infarction is

a) CK
b) CK-MB
c) LDH
d) Troponin-I ✓

Answer: d) Troponin-I

Reason: Cardiac troponins (I and T) are highly cardiac-specific; CK-MB and LDH are less specific.

Q14. Which one is the most important role of bile salts?

a) Micelle formation
b) Fat absorption ✓
c) Fat digestion
d) Emulsification

Answer: b) Fat absorption

Reason: Bile salts form micelles that carry lipids to the mucosa for absorption.

Q15. Angiotensin II †

a) Causes vasoconstriction ✓
b) Is directly converted from angiotensinogen
c) Causes vasodilatation
d) Is secreted by the adrenal cortex

Answer: a) Causes vasoconstriction

Reason: Ang II is a potent vasoconstrictor formed by ACE from Ang I; it stimulates aldosterone.

Q16. Enzymes †

a) Are heat stable
b) Are biocatalysts ✓
c) Are non-protein in nature
d) Have only a linear protein structure

Answer: b) Are biocatalysts

Reason: Enzymes are protein biocatalysts that lower activation energy; they are heat labile.

Q17. Regarding diffusion, choose the correct answer

a) Movement of solute from higher to lower concentration ✓
b) Needs use of ATP for the process
c) Temperature has no effect on diffusion
d) A semipermeable membrane is needed for movement

Answer: a) Movement of solute from higher to lower concentration

Reason: Diffusion is passive movement down a concentration gradient.

Q18. Which of the following is true for glycolysis? †

a) In aerobic glycolysis pyruvate is produced as the end product ✓
b) Glycolysis occurs in the mitochondria of all cells
c) There are five unidirectional reactions
d) Two moles of ATP are produced in aerobic glycolysis

Answer: a) In aerobic glycolysis pyruvate is produced as the end product

Reason: Glycolysis is cytosolic with 3 irreversible steps; aerobically it ends in pyruvate.

Q19. Plasma

a) Contains 10 mmol/L urea normally
b) Has an osmolality in the range of 260–320 mosmol/L
c) Contains protein buffer ✓
d) Has an Na^+ concentration of 35–53 mEq/L

Answer: c) Contains protein buffer

Reason: Plasma contains bicarbonate/protein buffers; urea 2.5–6.5 mmol/L, osmolality 280–296, Na^+ 135–145 mEq/L.

Q20. LDL cholesterol †

a) Is synthesized from chylomicron
b) Contains apo B-100 ✓
c) Is metabolized in cardiac muscle
d) Requires HDL for metabolism

Answer: b) Contains apo B-100

Reason: LDL is formed from VLDL, carries apo B-100 and is cleared by hepatic LDL receptors.

November 2025 — Paper II

Q1. Nutrients having no caloric value

a) Vitamins **[T]**
Reason: Vitamins have no caloric value.

b) Carbohydrate **[F]**
Reason: Carbohydrate metabolism is mainly deranged in DM.

c) Minerals **[T]**
Reason: Minerals have no caloric value.

d) Dietary fibre **[T]**
Reason: Fibre is not digested; negligible energy.

e) Protein **[F]**
Reason: Protein supplies 4 kcal/g, so it has caloric value.

Q2. The following statements are true for ECF

a) Na^+ is the major cation of ECF **[T]**
Reason: Na^+ is the principal ECF cation (~140 mEq/L).

b) pH is 7.35–7.45 **[T]**
Reason: ECF pH is 7.35–7.45.

c) Is 60% of body weight in old persons **[F]**
Reason: ECF is ~20% of body weight; TBW is ~60%.

d) Can be measured by inulin **[T]**
Reason: Inulin is confined to ECF, so it measures ECF volume.

e) Interstitial fluid contains more protein than plasma **[F]**
Reason: Interstitial fluid has much less protein than plasma.

Q3. Hypovolemia

a) Produces scanty urine **[T]**
Reason: Hypovolemia raises ADH/aldosterone → oliguria.

b) Inhibits renin secretion **[F]**
Reason: Hypovolemia stimulates renin secretion.

c) Increases aldosterone secretion **[T]**
Reason: Low volume → renin–angiotensin → aldosterone.

d) Decreases ADH secretion **[F]**
Reason: Hypovolemia increases ADH.

e) Stimulates the thirst center **[T]**
Reason: Hypovolemia stimulates thirst.

Q4. Regarding vitamin A

a) Beta-carotene is the preformed vitamin obtained from liver **[F]**
Reason: β-carotene is the provitamin; preformed vitamin A (retinol) comes from liver/animal foods.

b) Retinoic acid is needed for visual pigment synthesis **[F]**
Reason: Visual pigment uses 11-cis-retinal; retinoic acid is for growth/differentiation.

c) Excess intake does not produce toxicity **[F]**
Reason: Fat-soluble vitamins (A, D) accumulate and cause hypervitaminosis.

d) Night blindness is the earliest sign of deficiency **[T]**
Reason: Rod dysfunction (nyctalopia) is the earliest sign of vitamin A deficiency.

e) Needed for immune function **[T]**
Reason: Vitamin A maintains epithelial integrity and immunity.

Q5. Homologous chromosomes

a) Contain the same number of genes **[T]**
Reason: Homologous chromosomes have the same genes at the same loci.

b) Contain different numbers of gene loci **[F]**
Reason: Homologous chromosomes have identical loci.

c) Have identical loci at identical positions **[T]**
Reason: Homologous pairs have the same loci positions.

d) Genes determine different characteristics **[F]**
Reason: Corresponding loci control the SAME trait.

e) Responsible for a trait **[T]**
Reason: Alleles on homologous chromosomes control a trait.

Q6. Codons

a) Are triplets of four types of nitrogenous bases **[T]**
Reason: A, U, G, C in triplets.

b) All codons code for specific amino acids **[F]**
Reason: 3 stop codons code for no amino acid.

c) There are 64 codons for 20 amino acids **[T]**
Reason: 4^3 = 64 codons, 61 sense + 3 stop.

d) Degeneracy is responsible for multiple codons for an amino acid **[T]**
Reason: Degeneracy = several codons per amino acid.

e) Specificity is responsible for mutations **[F]**
Reason: Specificity is unambiguity; mutation arises from changes in DNA.

Q7. During blood pH homeostasis

a) Immediate maintenance is done by the respiratory system **[F]**
Reason: Immediate defence is by chemical buffers; the respiratory system acts in minutes.

b) The $[HCO_3^-]/PCO_2$ ratio is targeted **[T]**
Reason: pH depends on the HCO_3^-/PCO_2 ratio (H-H equation).

c) The pK of the buffer has a negligible role **[F]**
Reason: Buffer capacity is maximum at pH = pK, so pK is crucial.

d) Carbonic anhydrase plays an important role **[T]**
Reason: Carbonic anhydrase catalyses $CO_2 + H_2O \rightleftharpoons H_2CO_3$, so it is central to bicarbonate buffering.

e) Alkali reserve is an important issue **[T]**
Reason: Alkali reserve (plasma HCO_3^-) is the key buffer reserve for pH homeostasis.

Q8. Nutritional status can be assessed by

a) BMI **[T]**
Reason: BMI is the first-line index of malnutrition/obesity.

b) BMR **[F]**
Reason: BMR is energy expenditure, not a nutritional-status measure.

c) DRI **[F]**
Reason: DRI is a dietary recommendation, not an assessment tool.

d) Skin fold thickness **[T]**
Reason: Skinfold thickness estimates body fat.

e) Mid-arm circumference **[T]**
Reason: MAC assesses muscle/fat reserve.

Q9. Balanced diet

a) Has a carbohydrate : protein : fat ratio of approximately 4:1:1 **[T]**
Reason: Balanced diet ratio ≈ 4:1:1 (carbohydrate : protein : fat).

b) Is required for maintenance of cellular growth **[T]**
Reason: Balanced diet supports growth and maintenance.

c) Does not contain fibre **[F]**
Reason: A balanced diet includes fibre.

d) Limits fat content **[T]**

e) May not contain essential amino acids **[F]**
Reason: Balanced diet supplies essential amino acids.

Q10. In diuresis

a) Urine volume is within 3 L/day **[F]**
Reason: Diuresis means urine volume above 3 L/day.

b) Uncontrolled diabetes causes osmotic diuresis **[T]**
Reason: Uncontrolled diabetes: glucose in tubule holds water → osmotic diuresis.

c) Decreased ADH concentration causes osmotic diuresis **[F]**
Reason: Low ADH causes WATER diuresis, not osmotic diuresis.

d) Medullary interstitial hyperosmolarity is decreased **[T]**
Reason: In diuresis the medullary gradient is washed out.

e) Excess hypotonic solution may cause diuresis **[T]**
Reason: Excess hypotonic fluid suppresses ADH → water diuresis.

Q11. Plasma osmolarity mainly depends upon

a) Sodium ✓
b) Potassium
c) Urea
d) Glucose

Answer: a) Sodium

Reason: Plasma contains bicarbonate/protein buffers; urea 2.5–6.5 mmol/L, osmolality 280–296, Na^+ 135–145 mEq/L.

Q12. A characteristic feature of metabolic acidosis is

a) Decreased pH only
b) Decreased PCO_2 only
c) Decreased plasma bicarbonate ✓
d) Decreased hydrogen ion concentration

Answer: c) Decreased plasma bicarbonate

Reason: Metabolic acidosis = primary fall in HCO_3^- (with low pH, compensatory low PCO_2).

Q13. The most common cause of anemia is

a) Folic acid deficiency
b) Vitamin C deficiency
c) Vitamin B_{12} deficiency
d) Iron deficiency ✓

Answer: d) Iron deficiency

Reason: Iron deficiency is the commonest cause worldwide.

Q14. A characteristic feature of kwashiorkor is

a) Edema present ✓
b) Body weight is 80–60% of standard
c) Mostly occurs after one year of age
d) Flaky skin

Answer: a) Edema present

Reason: Pitting edema with hypoalbuminaemia.

Q15. PCR differs from biological cloning by the following point

a) PCR is an in vitro type of cloning process ✓
b) PCR requires heat-labile DNA polymerase
c) PCR is highly insensitive
d) PCR causes arithmetic rise of DNA

Answer: a) PCR is an in vitro type of cloning process

Reason: PCR is cell-free in vitro amplification.

Q16. The most important enzyme for replication is

a) Helicase
b) Ligase
c) DNA polymerase ✓
d) Topoisomerase

Answer: c) DNA polymerase

Reason: DNA polymerase.

Q17. In renal failure, acidosis occurs because

a) Buffer bases cannot be filtered
b) Hydrogen ions cannot be secreted ✓
c) Bicarbonate cannot be reclaimed
d) Urinary pH rapidly reaches its limiting point

Answer: b) Hydrogen ions cannot be secreted

Reason: Failed H^+/NH_4^+ excretion.

Q18. Autoregulation of blood supply is an exceptional quality of kidney. The aim of this is to

a) Retain the nutrients
b) Maintain the GFR ✓
c) Excrete the waste products
d) Ensure the renal fraction

Answer: b) Maintain the GFR

Reason: Autoregulation keeps GFR constant despite BP changes.

Q19. Many enzymes are necessary for biotechnology; the one known as the 'blessing of genetics' is

a) Taq DNA polymerase
b) DNA ligase
c) Helicase
d) Restriction endonuclease ✓

Answer: d) Restriction endonuclease

Reason: Restriction endonucleases cut DNA specifically.

Q20. Synthesized from a monohydric alcohol (cholesterol), activated in the kidney, and maintains the balance of an important mineral

a) Vitamin D ✓
b) Vitamin A
c) Cholesterol ester
d) PTH

Answer: a) Vitamin D

Reason: Vitamin D (from cholesterol) → 25-OH in liver → 1,25-$(OH)_2$ in kidney.

May 2025 — Paper I

Q1. The following statements are true for pH †

a) Negative log of proton concentration **[T]**
Reason: pH = –log10 [H^+].

b) pH of pure water is 7.4 **[F]**
Reason: Pure water has [H^+] = 10^{-7} mol/L, so pH = 7.0 (not 7.4; 7.4 is blood pH).

c) If pH is 7, then proton concentration is 10-7 mol/L **[T]**
Reason: pH 7 means [H^+] = 10^{-7} mol/L.

d) Addition of alkali to a solution increases pH **[T]**
Reason: Alkali lowers [H^+], so pH rises.

e) pH + pOH = 14 **[T]**
Reason: At 25 °C, pH + pOH = 14 (Kw = 10^{-14}).

Q2. Protein †

a) In plasma is 6–8 g/dL **[T]**
Reason: Normal plasma protein is 6–8 g/dL.

b) Contains no peptide bond in primary structure **[F]**
Reason: Primary structure is the sequence of amino acids joined by peptide bonds.

c) Acts as carrier for different substances in plasma **[T]**
Reason: Plasma proteins (albumin, globulins) carry hormones, drugs, bilirubin, etc.

d) Cannot be separated by electrophoresis **[F]**
Reason: Plasma proteins are separated by electrophoresis (albumin, α1, α2, β, γ).

e) Does not need essential amino acids to maintain biological integrity **[F]**
Reason: Essential amino acids must be supplied in diet for protein synthesis.

Q3. The statements that can be applied for colloid

a) Has a diameter of 1–200 millimicron **[T]**
Reason: Colloid particle size is 1–200 millimicron (nm); crystalloids are < 1 nm.

b) Colloidal osmotic pressure is higher than that of crystalloids **[F]**
Reason: Colloids have few particles per volume, so their osmotic pressure is lower than that of crystalloids.

c) Has dielectric pH **[F]**
Reason: Colloids (proteins) have an ISOelectric pH; 'dielectric pH' does not exist.

d) Shows molecular movement in solution **[F]**
Reason: Colloids show Brownian movement (random particle movement); molecular/ionic movement is typical of crystalloids.

e) Sol–gel transformation occurs due to change of temperature **[T]**
Reason: Sol–gel transformation of colloids occurs with temperature change.

Q4. Co-enzyme

a) Are heat stable **[T]**
Reason: Coenzymes are low-molecular-weight and heat stable; enzymes (proteins) are heat labile.

b) Are inorganic **[F]**
Reason: Coenzymes are organic molecules (vitamin derivatives).

c) Are water-soluble vitamin derivatives **[T]**
Reason: Most coenzymes derive from water-soluble B vitamins.

d) Are protein in nature **[F]**
Reason: Coenzymes are non-protein organic molecules; the apoenzyme is the protein.

e) Act as co-substrate **[T]**
Reason: Coenzymes like NAD^+ are changed during reaction (co-substrates).

Q5. Isotopes †

a) Are atoms of different elements **[F]**
Reason: Isotopes are atoms of the SAME element with different mass numbers.

b) Are compounds having the same number of protons **[F]**
Reason: Isotopes are atoms, not compounds.

c) Radioactive isotopes can be used as a therapeutic tool **[T]**
Reason: Radioisotopes (I-131, Co-60) are used therapeutically.

d) Have different atomic weights **[T]**
Reason: Different neutron number → different atomic (mass) weight.

e) Do not have a definite stable state **[F]**
Reason: Some isotopes are stable, others radioactive.

Q6. The following statements can be applied for the H-H equation

a) pH = 6.1 + log $[HCO_3^-]/[H_2CO_3]$ **[T]**
Reason: Henderson–Hasselbalch for bicarbonate buffer: pH = 6.1 + log $[HCO_3^-]/[H_2CO_3]$.

b) pH = pK + log $[HCO_3^-]/[H_2CO_3]$ **[T]**
Reason: Henderson–Hasselbalch for bicarbonate buffer: pH = 6.1 + log $[HCO_3^-]/[H_2CO_3]$.

c) pH = pK at half neutralization **[T]**
Reason: When acid is half-neutralized [salt] = [acid], log 1 = 0, so pH = pK.

d) Is a derivative of the law of mass action **[T]**
Reason: H-H equation is derived from the law of mass action applied to Ka of a weak acid.

e) pH = pK + log salt/acid **[T]**
Reason: H-H equation: pH = pK + log [salt]/[acid].

Q7. Which of the following are inhibitors of ETC?

a) Carbon monoxide **[T]**
Reason: CO blocks cytochrome oxidase (complex IV).

b) Cyanide **[T]**
Reason: Cyanide blocks complex IV.

c) Fluoride **[F]**
Reason: Fluoride inhibits enolase (glycolysis), not ETC.

d) Free radicals **[F]**
Reason: Free radicals are by-products of ETC, not inhibitors.

e) Nitric oxide **[T]**
Reason: NO inhibits cytochrome oxidase.

Q8. Regarding carbohydrates †

a) Lactose has reducing properties **[T]**
Reason: Lactose has a free anomeric C1 of glucose.

b) Fructose and glucose combine to form sucrose **[T]**
Reason: Sucrose = glucose + fructose.

c) In animals, starch is the storage form of glucose **[F]**
Reason: Animals store glycogen (animal starch); starch is the plant store.

d) Shows mutarotation **[T]**
Reason: Sugars with free anomeric carbon show mutarotation.

e) Altrose is a non-reducing sugar **[F]**
Reason: Altrose is an aldohexose → reducing.

Q9. Ammonia is removed from the body by

a) NH4Cl formation **[F]**
Reason: Ammonia is detoxified mainly to urea.

b) Creatinine formation **[F]**
Reason: Creatinine is from creatine, not ammonia removal.

c) Glutamate formation **[T]**
Reason: NH_3 + α-KG → glutamate traps ammonia.

d) Urea formation **[T]**
Reason: Urea cycle is the major route.

e) Formation of NPN **[F]**
Reason: NPN includes urea but this is not ammonia removal.

Q10. Anabolism occurs during

a) Synthetic reactions **[T]**
Reason: Anabolism = biosynthesis.

b) Degradation processes **[F]**
Reason: Degradation is catabolism.

c) Gluconeogenesis **[T]**
Reason: Gluconeogenesis is an anabolic synthesis of glucose; occurs in fasting, liver/kidney, inhibited by insulin.

d) Oxidation of fatty acid **[F]**
Reason: Fatty acid oxidation is catabolic.

e) Utilization of acetone **[F]**
Reason: Acetone utilisation is catabolic.

Q11. Isotonic solution †

a) 0.9 g of NaCl in 100 mL is isotonic
b) Can be used in dehydration ✓
c) Causes RBC rupture
d) Has an osmolality of 260 mosmol/L

Answer: b) Can be used in dehydration
Reason: Isotonic saline (0.9% = 9 g/L) is used for dehydration; hypotonic fluids burst RBCs.

Q12. The confirmatory test for diabetes mellitus is

a) Fasting blood glucose
b) Random blood glucose
c) HbA_{1c}
d) OGTT ✓

Answer: d) OGTT
Reason: OGTT is the confirmatory diagnostic test.

Q13. The following is true for bile salts

a) Bile salt reduces the surface tension of lipid ✓
b) Bile salt increases the surface area of the lipid droplet only
c) It decreases lipid absorption by forming micelles
d) Micelle is an aggregate of phospholipid only

Answer: a) Bile salt reduces the surface tension of lipid
Reason: Bile salts lower surface tension and emulsify fat.

Q14. About properties of carbohydrate the following is true †

a) Glucose is a keto derivative
b) Fructose is dextrorotatory
c) Sucrose on hydrolysis yields invert sugar ✓
d) Glucose in solution does not show mutarotation

Answer: c) Sucrose on hydrolysis yields invert sugar
Reason: Sucrose hydrolysis gives invert sugar (glucose + fructose); glucose is an aldose, fructose is laevorotatory, glucose shows mutarotation.

Q15. The following statement is true for pH †

a) Negative logarithm of [OH-]
b) Negative log of H^+ ion concentration ✓
c) If pH of a solution is 7 then pOH is 9
d) If pH is 7.0, $[H^+]$ is 0.0001 mol/L

Answer: b) Negative log of H^+ ion concentration
Reason: pH = –log $[H^+]$; at pH 7 $[H^+] = 10^{-7}$ mol/L and pOH = 7.

Q16. Which of the following is mostly affected in diabetes mellitus?

a) Carbohydrate ✓
b) Lipid
c) Mineral
d) Protein

Answer: a) Carbohydrate
Reason: DM is primarily a disorder of carbohydrate metabolism (secondarily fat and protein).

Q17. Among the thyroid tests which one is most informative?

a) T_3
b) T_4
c) TRH
d) TSH ✓

Answer: d) TSH
Reason: TSH is the single most informative test; it rises in primary hypothyroidism and falls in hyperthyroidism.

Q18. Which proteolytic enzyme is more effective?

a) Chymotrypsin
b) Carboxypeptidase
c) Elastase
d) Trypsin ✓

Answer: d) Trypsin

Reason: Trypsin activates other zymogens and is the major pancreatic protease.

Q19. Enzyme test most specific for parenchymal liver disease

a) ALT ✓
b) ACP
c) LDH
d) AST

Answer: a) ALT

Reason: ALT is more liver-specific than AST (AST is also in heart/muscle).

Q20. Which of the following is an integral apoprotein of LDL? †

a) Apo-A
b) Apo-B-48
c) Apo-B-100 ✓
d) Apo-C

Answer: c) Apo-B-100

Reason: Apo B-100 is integral (non-exchangeable) in LDL; apo A and C are exchangeable.

May 2025 — Paper II

Q1. Southern blot technique requires

a) Restriction enzyme **[T]**
Reason: Southern blot: DNA is cut by restriction enzymes, run on gel, blotted, and probed.

b) Hybrid DNA **[T]**
Reason: Probe hybridises with target DNA on the membrane.

c) cDNA **[F]**
Reason: Southern blot uses a labelled DNA/RNA probe, cDNA is not required.

d) Electrophoresis **[T]**
Reason: DNA fragments are separated by gel electrophoresis.

e) Nitrocellulose paper **[T]**
Reason: DNA is transferred to nitrocellulose/nylon membrane.

Q2. Replication needs the following enzymes

a) Helicase **[T]**
Reason: Helicase unwinds DNA at the replication fork.

b) RNA primase **[T]**
Reason: Primase makes the RNA primer for DNA polymerase.

c) DNA polymerase **[T]**
Reason: DNA polymerase synthesises the new strand.

d) RNA polymerase **[F]**
Reason: RNA polymerase is for transcription (primase is a special RNA polymerase for primers).

e) DNA ligase **[T]**
Reason: Ligase joins Okazaki fragments.

Q3. PCR †

a) Needs Taq polymerase enzyme **[T]**
Reason: PCR uses heat-stable Taq polymerase.

b) Needs primers **[T]**
Reason: PCR needs two primers flanking the target.

c) Passes through proofreading activity **[F]**
Reason: Taq polymerase lacks 3'→5' proofreading, so PCR can introduce errors.

d) Needs high temperature **[T]**
Reason: Denaturation at ~94 °C.

e) Can give false-positive results **[T]**
Reason: Contamination can give false-positive PCR.

Q4. Hypovolemia is corrected by

a) ADH system **[T]**
Reason: ADH conserves water in hypovolemia.

b) RAAS **[T]**
Reason: RAAS retains Na^+ and water in hypovolemia.

c) ANP mechanism **[F]**
Reason: ANP is released in hypervolemia and promotes Na^+/water loss.

d) Producing less urine volume **[T]**
Reason: Reduced urine output conserves volume.

e) Decreasing thirst **[F]**
Reason: Thirst increases in hypovolemia.

Q5. Hyperkalemia

a) Mostly occurs in acidosis **[T]**
Reason: Acidosis moves K^+ out of cells (H^+–K^+ exchange), causing hyperkalaemia.

b) Is associated with renal failure **[T]**
Reason: Renal failure impairs K^+ excretion.

c) Is corrected by aldosterone hormone **[T]**

d) Is corrected by producing renin **[F]**
Reason: Renin acts through angiotensin–aldosterone, but the direct corrector is aldosterone.

e) Causes paralytic ileus **[F]**
Reason: Paralytic ileus is a feature of HYPOkalaemia.

Q6. Kwashiorkor

a) Occurs in babies due to abrupt weaning **[T]**
b) Shows severe muscle wasting **[F]**
Reason: Marasmus shows muscle wasting; kwashiorkor retains muscle bulk.
c) There are generalized edemas **[T]**
d) Baby's appetite is good **[F]**
Reason: Kwashiorkor: anorexia and apathy.
e) Is due to predominant protein deficiency **[T]**

Q7. The following statements are correct for obesity

a) A BMI of > 24.9 in Asian people is designated as obesity **[F]**
Reason: WHO Asia-Pacific: BMI 23–27.4 = overweight; ≥ 27.5 = obese.
b) A waist–hip ratio > 0.9 in males is designated as abdominal obesity **[T]**
c) Abdominal obesity, high triacylglycerol and insulin resistance together are risk factors for coronary heart disease **[T]**
d) Dietary fibre doesn't have any beneficial role in obesity **[F]**
Reason: Fibre gives satiety, delays absorption, lowers energy density.
e) Obesity is always associated with high energy intake **[F]**
Reason: Obesity can result from low expenditure/genetic factors.

Q8. Kidney produces the following hormones

a) Rennin **[F]**
Reason: Renin (JG cells) is the kidney enzyme/hormone; rennin (chymosin) is a gastric enzyme.
b) Angiotensin **[F]**
Reason: Angiotensinogen is liver-made; Ang II is formed in plasma.
c) 1,25-dihydroxycholecalciferol **[T]**
Reason: Kidney 1α-hydroxylase makes calcitriol.
d) Gastrin **[F]**
Reason: Gastrin comes from stomach G cells.
e) Erythropoietin **[T]**
Reason: Peritubular cells of kidney make EPO.

Q9. Vitamin K

a) Helps in post-translational maturation of clotting factors **[T]**
Reason: Vitamin K γ-carboxylates Glu residues of factors II, VII, IX, X.
b) Is abundant in breast milk for newborn baby **[F]**
Reason: Breast milk is poor in vitamin K → vitamin K prophylaxis at birth.
c) Menaquinone is the synthetic analogue **[F]**
Reason: Menaquinone (K2) is bacterial; menadione (K3) is synthetic.
d) Toxicity of the synthetic form causes hemolytic anemia **[T]**
Reason: High-dose synthetic menadione causes hemolysis in neonates.
e) Is used as antidote in warfarin poisoning **[T]**
Reason: Warfarin blocks vitamin K epoxide reductase; vitamin K reverses it.

Q10. Regarding metabolic acidosis

a) GIT cause predominant (diarrhea) **[T]**
b) Anion gap is always high **[F]**
Reason: Anion gap is high in some, normal in other metabolic acidosis.
c) Correction is done by hyperventilation (respiratory compensation) **[T]**
Reason: Hyperventilation (Kussmaul) is respiratory compensation that lowers PCO_2.
d) Primary defect is low HCO_3^- **[T]**
Reason: Primary defect in metabolic acidosis is low HCO_3^-.
e) Base excess becomes negative **[T]**
Reason: Base excess becomes negative (base deficit) in metabolic acidosis.

Q11. Metabolic acidosis is characterized by †

a) Low plasma bicarbonate level ✓
b) Increased bicarbonate to PCO_2 ratio
c) Increased acid production only
d) Increased bicarbonate utilization only

Answer: a) Low plasma bicarbonate level
Reason: Primary event is a fall in plasma bicarbonate.

Q12. The most common cause of hypokalemia is

a) Diarrhea, vomiting ✓
b) Excessive diuretic use
c) Low dietary intake
d) Metabolic alkalosis

Answer: a) Diarrhea, vomiting
Reason: Gastrointestinal loss (diarrhoea/vomiting) is the commonest cause.

Q13. Plasma osmolality mainly depends upon

a) Sodium ✓
b) Potassium
c) Glucose
d) Urea

Answer: a) Sodium
Reason: Na^+ (with its anions) contributes ~90% of plasma osmolality.

Q14. The primary regulator of low plasma sodium concentration is

a) RAAS
b) Aldosterone
c) ANP
d) ADH ✓

Answer: d) ADH
Reason: Hyponatraemia is mainly due to water excess; ADH regulates water.

Q15. Hyperventilation †

a) Decreases PCO_2 ✓
b) Occurs in metabolic alkalosis
c) Causes acidosis
d) Occurs in response to decreased H^+ concentration

Answer: a) Decreases PCO_2
Reason: Hyperventilation lowers PCO_2 (respiratory alkalosis).

Q16. The vitamin that is synthesized in significant amount in our body is

a) Vitamin D ✓
b) Vitamin K
c) Niacin
d) Biotin

Answer: a) Vitamin D
Reason: Skin synthesises vitamin D under UV-B.

Q17. The most common cause of microcytic hypochromic anemia is

a) Iron deficiency anemia ✓
b) Anemia of chronic disease
c) Thalassemia
d) Sideroblastic anemia

Answer: a) Iron deficiency anemia
Reason: Iron deficiency (↓Hb synthesis) is the commonest cause of microcytic hypochromic anaemia.

Q18. Which statement is WRONG for RFLP?

a) May be a DNA mutation effect
b) In intron helps in DNA fingerprinting
c) Can be detected by Southern blot technique
d) Detection doesn't require restriction enzymes ✓

Answer: d) Detection doesn't require restriction enzymes
Reason: RFLP needs restriction enzyme digestion followed by Southern blot.

Q19. Property of the genetic codon responsible for silent mutation

a) Commaless
b) Degeneracy ✓
c) Nonoverlapping
d) Specificity

Answer: b) Degeneracy

Reason: Degeneracy of the code lets different codons code for the same amino acid.

Q20. Acidification of urine is mostly done in

a) Distal convoluted tubule and collecting duct ✓
b) Glomerulus
c) Loop of Henle
d) Proximal convoluted tubule

Answer: a) Distal convoluted tubule and collecting duct

Reason: DCT/collecting-duct intercalated cells secrete H^+.

November 2024 — Paper II

Q1. Body water is maintained by

a) Hypothalamus **[T]**
Reason: Hypothalamus has osmoreceptors and makes ADH → water balance.

b) ADH **[T]**
Reason: ADH conserves water by increasing collecting-duct permeability.

c) Thyroxine **[F]**
Reason: Thyroxine regulates metabolism, not water balance.

d) Plasma osmolarity **[T]**
Reason: Plasma osmolarity is the main stimulus for ADH and thirst.

e) Thirst mechanism **[T]**
Reason: Thirst is stimulated by high osmolarity and low volume.

Q2. Regarding vitamin B_{12}

a) Is a cobalt-containing compound possessing the corrin ring **[T]**
Reason: B_{12} contains cobalt in a corrin ring.

b) Its deficiency causes microcytic hypochromic anemia **[F]**
Reason: B_{12} deficiency causes megaloblastic (macrocytic) anaemia; iron deficiency gives microcytic anaemia.

c) A strict vegetarian may develop B_{12} deficiency **[T]**
Reason: B_{12} occurs only in animal foods.

d) Its deficiency causes folic acid deficiency **[F]**
Reason: B_{12} deficiency traps folate as methyl-THF (functional folate deficiency).

e) Its main function is production of RBC from stem cells **[T]**
Reason: B_{12} is required for DNA synthesis in maturing erythroblasts.

Q3. Choose the correct answer

a) Vitamin C inhibits iron absorption from gut **[F]**
Reason: Vitamin C reduces Fe^{3+} to Fe^{2+} and ENHANCES absorption.

b) Selenium acts as an antioxidant **[T]**
Reason: Selenium is a component of glutathione peroxidase (antioxidant).

c) Calcitonin increases blood calcium level **[F]**
Reason: Calcitonin LOWERS blood calcium.

d) Iodine deficiency causes mental retardation **[T]**
Reason: Iodine is needed for thyroid hormone for brain development.

e) Retinol acts as a carrier / antioxidant vitamin **[T]**
Reason: Retinol/carotenoids have antioxidant action.

Q4. Folic acid has the following facts

a) Deficiency causes microcytic anemia **[F]**
Reason: Folate deficiency causes macrocytic (megaloblastic) anaemia.

b) Active form is tetrahydrofolate **[T]**
Reason: THF is the active coenzyme form of folate.

c) Acts as carrier of single carbon unit **[T]**
Reason: THF carries one-carbon groups for purine/thymidylate synthesis.

d) Vitamin B_{12} is not essential for folic acid metabolism **[F]**
Reason: B_{12} is needed to regenerate THF from methyl-THF.

e) Chronic alcoholism causes deficiency **[T]**
Reason: Alcohol impairs folate absorption and increases loss.

Q5. Criteria of autosomal dominant disorder

a) Skip generation is possible **[F]**
Reason: AD disease appears in every generation (vertical).

b) New mutation occurs **[T]**
Reason: AD diseases can arise from new mutations (e.g., achondroplasia).

c) Follows vertical transmission **[T]**
Reason: AD disease: vertical transmission.

d) Involves sex chromosome **[F]**
Reason: AD = autosomal.

e) Diabetes insipidus is an example **[F]**
Reason: Nephrogenic DI is X-linked recessive.

Q6. Dietary fibre

a) Decreases the incidence of constipation **[T]**
b) Increases the absorption of cholesterol **[F]**
Reason: Fibre binds bile acids, REDUCING cholesterol absorption.
c) Decreases the absorption of zinc **[T]**
d) Decreases the incidence of diverticulosis **[T]**
Reason: Fibre prevents diverticulosis.
e) Helps in better control of diabetes mellitus **[T]**

Q7. Statement related to transcription

a) Synthesis of protein following rRNA **[F]**
Reason: Transcription makes RNA; protein synthesis is translation.
b) Requires RNA polymerase **[T]**
Reason: PCR uses DNA polymerase.
c) Promoter sequence is needed **[T]**
Reason: RNA polymerase binds the promoter.
d) 5' cap is formed by polyadenylation **[F]**
Reason: 5' cap = 7-methylguanosine; polyadenylation is the 3' tail.
e) Has no proofreading property **[T]**
Reason: RNA polymerase lacks efficient proofreading.

Q8. Regarding BMI

a) 20 is within normal value **[T]**
Reason: Normal BMI 18.5–24.9.
b) 38 is obese **[T]**
Reason: BMI ≥ 30 = obese (≥ 27.5 in Asians).
c) Calculated by weight in kg / height in metre squared **[T]**
Reason: BMI = kg/m².
d) High value is a risk for DM **[T]**
Reason: Obesity ↑ insulin resistance.
e) Can determine malnutrition **[T]**
Reason: Low BMI = undernutrition.

Q9. The following are examples of post-translational modification †

a) 5' capping **[F]**
Reason: 5' capping is post-TRANSCRIPTIONAL.
b) Proteolytic cleavage **[T]**
Reason: Post-translational: e.g., zymogen activation.
c) Reversible phosphorylation and dephosphorylation **[T]**
Reason: Reversible phosphorylation is a post-translational modification.
d) Splicing **[F]**
Reason: Splicing is post-transcriptional.
e) Hydroxylation **[T]**
Reason: Hydroxylation (collagen) is post-translational.

Q10. Regarding trace elements †

a) Zn helps in insulin synthesis **[T]**
Reason: Zinc is needed for insulin storage/action.
b) Selenium is an antioxidant **[T]**
c) Ca normal is 2 mg/dL **[F]**
Reason: Normal calcium is 8.5–10.5 mg/dL (2.1–2.6 mmol/L).
d) Iron deficiency causes anemia **[T]**
Reason: Iron is needed for haemoglobin.
e) Na maintains the osmolality **[T]**
Reason: Na^+ is the main determinant of ECF osmolality.

Q11. Earliest feature of vitamin A deficiency †

a) Immune system disorder
b) Gastritis
c) Night blindness ✓
d) Toad skin
e) Keratomalacia

Answer: c) Night blindness

Reason: Night blindness is the earliest sign.

Q12. In replication †

a) Occurs only during cell division
b) It is DNA-directed DNA synthesis ✓
c) A primer is not needed
d) RNAP is needed
e) RNAP has proofreading

Answer: b) It is DNA-directed DNA synthesis

Reason: Replication is DNA-directed DNA synthesis by DNA polymerase.

Q13. Which of the following would be most appropriate for vitamin B_{12}? †

a) Absorption occurs in large gut
b) Is abundant in plant sources
c) Absorbed without intrinsic factor
d) Helps in RBC maturation ✓
e) Deficiency causes microcytic anemia

Answer: d) Helps in RBC maturation

Reason: B_{12} is needed for DNA synthesis → RBC maturation.

Q14. The worst effect of folic acid deficiency

a) Megaloblastic anemia
b) Peripheral neuropathy
c) Neural tube defect ✓
d) Lethargy
e) Infertility

Answer: c) Neural tube defect

Reason: Folate deficiency in pregnancy causes neural tube defects.

Q15. The most important feature of scurvy is

a) Rough skin
b) Bleeding from gum ✓
c) Poor wound healing
d) Loosening of teeth
e) Anemia

Answer: b) Bleeding from gum

Reason: Weak capillaries/collagen defect → gum bleeding.

Q16. Karyotyping is done in

a) Down syndrome ✓
b) Haemophilia
c) Cystic fibrosis
d) Familial hypercholesterolemia
e) Polycystic kidney disease

Answer: a) Down syndrome

Reason: Karyotype is used for chromosomal disorders like Down syndrome.

Q17. Best feature of mitochondrial DNA †

a) The sperm does not contain mitochondrial DNA
b) It accounts for about 1% of total human DNA
c) It encodes 37 genes ✓
d) Synthesizes respiratory chain enzymes
e) Shows variable expression

Answer: c) It encodes 37 genes

Reason: mtDNA encodes 37 genes (13 proteins, 22 tRNA, 2 rRNA).

Q18. Select the reliable diagnostic tool for PEM

a) Decreased protein level
b) Skin change
c) Hair change
d) BMI ✓
e) Edema

Answer: d) BMI

Reason: BMI/weight-for-height is the reliable anthropometric tool; protein levels and clinical signs vary.

Q19. Most important feature regarding mutation

a) Changes in DNA ✓
b) Germ line mutation
c) Mutation is somatic cell
d) Result in error in DNA replication during cell division
e) Changes in coding region

Answer: a) Changes in DNA

Reason: A mutation is a change in DNA sequence.

Q20. The most common karyotype of Turner syndrome is

a) Monosomy only
b) Mosaic
c) 45,XO ✓
d) Is a ring chromosome
e) Deletion

Answer: c) 45,XO

Reason: Turner syndrome is 45,X.

May 2024 — Paper I

Q1. The following are the normal levels in blood †

a) Na^+ level is 135–145 mEq/L **[T]**
b) K^+ level is 3.5–5.3 mEq/L **[T]**
c) HCO_3^- level is 22–26 mEq/L **[T]**
Reason: Normal plasma HCO_3^- is 22–26 mmol/L.
d) 97–105 mEq/L is for Cl^- **[T]**
e) Plasma osmolarity is 280–296 mosm/L **[T]**
Reason: Normal plasma osmolarity is ~280–296 mosmol/L.

Q2. Insulin

a) Is a potent anabolic hormone **[T]**
Reason: Insulin promotes glycogen, fat and protein synthesis.
b) Is required for utilization of glucose in brain **[F]**
Reason: Brain uses GLUT-3, which is insulin-independent.
c) Increases glucose uptake in skeletal muscle **[T]**
Reason: Insulin recruits GLUT-4 in muscle and adipose tissue.
d) Increases plasma K^+ **[F]**
Reason: Insulin drives K^+ into cells (lowers plasma K^+); used to treat hyperkalaemia.
e) Mode of action is mediated by cAMP **[F]**
Reason: Insulin acts via tyrosine kinase receptor.

Q3. Blood calcium level is regulated by

a) ADH **[F]**
Reason: ADH conserves water by increasing collecting-duct permeability.
b) PTH **[T]**
c) ACTH **[F]**
Reason: ACTH controls cortisol, not calcium.
d) Calcitonin **[T]**
e) Calcitriol **[T]**

Q4. Regarding colloids

a) Non-dialyzable **[T]**
Reason: Colloids are too large to pass through a dialysis membrane.
b) Particle size < 1 nm **[F]**
Reason: Colloid particle size is 1–200 nm; < 1 nm is crystalloid.
c) Exhibits Brownian movement **[T]**
d) Can pass through semipermeable membrane **[F]**
Reason: Crystalloids pass; colloids cannot pass through semipermeable membranes.
e) Has a definite isoelectric pH **[T]**
Reason: Colloids such as proteins have a definite isoelectric pH.

Q5. Regarding primary hypothyroidism †

a) Serum cholesterol increases **[T]**
Reason: Hypothyroidism reduces LDL receptors → ↑ cholesterol.
b) TSH always increases **[T]**
Reason: Primary hypothyroidism → raised TSH.
c) TSH is 5–20 µIU/mL in subclinical hypothyroidism **[T]**
Reason: Subclinical hypothyroidism: TSH 5–20 µIU/mL with normal T_3/T_4.
d) T_3 and T_4 may be normal **[T]**
Reason: In subclinical hypothyroidism T_3/T_4 are normal.
e) Strongly associated with infertility **[T]**
Reason: Hypothyroidism causes anovulation/menstrual disturbance → infertility.

Q6. Choose the correct answer

a) All amino acids undergo transamination **[F]**
Reason: Lysine, threonine, proline, hydroxyproline do not undergo transamination.

b) Ammonia is removed by intestinal bacterial degradation **[F]**
Reason: Intestinal bacteria PRODUCE ammonia (urease); it is removed as urea/glutamine.

c) Oxidative deamination is a catabolic process **[T]**
Reason: Oxidative deamination (glutamate dehydrogenase) releases NH_3 from amino acids.

d) Excretion of NH_3 with urine as NH_4^+ **[T]**
Reason: Urinary NH_3 is excreted as NH_4^+.

e) Proline can be synthesized from the urea cycle **[F]**
Reason: Proline is made from glutamate; it is not a urea-cycle product.

Q7. About digestion the following are true

a) Is a process of hydrolysis **[T]**

b) Lipid digestion begins in mouth **[T]**

c) Pepsin splits carbohydrate into glucose **[F]**
Reason: Pepsin digests protein.

d) Micelle is an aggregate of cholesterol only **[F]**
Reason: Micelles contain bile salts, phospholipid, cholesterol and FA — not cholesterol alone.

e) Bile salt reduces the surface tension of fat **[T]**
Reason: Bile salts are detergents that lower surface tension → emulsification.

Q8. Regarding oxidative phosphorylation

a) Requires electron transport chain **[T]**
Reason: Oxidative phosphorylation couples ETC to ATP synthase.

b) Occurs in the inter-membrane space **[F]**
Reason: Occurs at the inner mitochondrial membrane.

c) NADPH undergoes oxidation **[F]**
Reason: ETC oxidises NADH/$FADH_2$, not NADPH.

d) Produces ATP **[T]**
Reason: Glycolysis produces ATP.

e) Most accepted mechanism is the chemiosmotic hypothesis **[T]**
Reason: Mitchell's chemiosmotic hypothesis explains ATP synthesis.

Q9. The following are true in glycolysis

a) Glucose or glycogen is broken down **[T]**
Reason: Glycolysis starts from glucose or glycogen (via G6P).

b) The end product of glycolysis is pyruvate only **[F]**
Reason: Pyruvate is the end product aerobically; in anaerobic glycolysis it is reduced to lactate, so 'only' is wrong.

c) There are four irreversible reactions **[F]**
Reason: Glycolysis has 3 irreversible steps: hexokinase/glucokinase, PFK-1, pyruvate kinase.

d) Conversion of pyruvate to lactate is oxidation **[F]**
Reason: Pyruvate → lactate is REDUCTION (NADH → NAD^+).

e) Net 2 ATP are produced in the aerobic pathway **[F]**
Reason: Aerobic net ≈ 7–8 ATP; 2 ATP is anaerobic.

Q10. Ketoacidosis could be a finding of

a) Uncontrolled diabetes mellitus **[T]**
Reason: Absent insulin → ketogenesis.

b) Prolonged starvation **[T]**
Reason: Low insulin/glucose in starvation → ketosis.

c) Severe diarrhea **[F]**
Reason: Diarrhoea gives non-ketotic metabolic acidosis.

d) Persistent vomiting **[F]**
Reason: Vomiting causes metabolic alkalosis (loss of HCl), not ketoacidosis.

e) Renal failure **[F]**
Reason: Renal failure causes uraemic (high-AG) acidosis without ketones.

Q11. For glucose 6-phosphate †

a) Can freely cross the cell membrane
b) Is produced by glucose-6-phosphatase
c) Feeds glycolysis, glycogenesis and the HMP shunt pathway ✓
d) Is produced without ATP
e) Activates hexokinase

Answer: c) Feeds glycolysis, glycogenesis and the HMP shunt pathway

Reason: G6P is the branch-point for glycolysis, glycogenesis and HMP shunt.

Q12. Most important enzyme of the feed state is

a) Glycogen synthase ✓
b) Phosphoprotein phosphatase and phosphodiesterase
c) Lipoprotein lipase
d) Acetyl-CoA carboxylase
e) Pyruvate kinase

Answer: a) Glycogen synthase

Reason: Glycogen synthase is activated by insulin in the fed state.

Q13. For the TCA cycle

a) Central pathway for oxidation of carbohydrate, protein and fat ✓
b) Supports gluconeogenesis
c) Has limited capacity in fasting condition
d) Depends upon oxaloacetate
e) Produces oxaloacetate

Answer: a) Central pathway for oxidation of carbohydrate, protein and fat

Reason: TCA is the final common oxidative pathway for fuels.

Q14. Regarding acute hepatitis †

a) Low serum AST
b) Low serum bilirubin
c) High serum ALT ✓
d) Low serum albumin
e) Prolonged prothrombin time

Answer: c) High serum ALT

Reason: ALT is the most specific marker of hepatocellular injury.

Q15. The most important laboratory finding in primary hypothyroidism

a) Increased TSH ✓
b) Decreased T_3
c) Decreased T_4
d) Increased anti-TPO
e) Increased anti-TG

Answer: a) Increased TSH

Reason: Raised TSH is the hallmark.

Q16. A young woman complains of cold intolerance, tiredness and weight gain. Laboratory tests show low T_3, T_4, high TSH and normal urinary iodine excretion. The most likely cause is

a) Hashimoto's thyroiditis ✓
b) Thyroidectomy
c) Pituitary disease
d) Hypothalamic disease
e) Thyroid hormone resistance

Answer: a) Hashimoto's thyroiditis

Reason: High TSH + low T_3/T_4 + normal urinary iodine = autoimmune (Hashimoto's) thyroiditis.

Q17. The commonest cause of glycosuria is

a) Uncontrolled diabetes mellitus ✓
b) Renal tubular defect
c) Alimentary glycosuria
d) Pregnancy
e) Chronic pancreatitis

Answer: a) Uncontrolled diabetes mellitus

Reason: Hyperglycaemia exceeds the renal threshold in uncontrolled DM.

Q18. The value of IGT (2-h OGTT) is †

a) ≥ 11.1 mmol/L
b) 7.8 to < 11.1 mmol/L ✓
c) < 7.8 mmol/L
d) < 4.4 mmol/L
e) 4.4 to 6.1 mmol/L

Answer: b) 7.8 to < 11.1 mmol/L

Reason: IGT = 2-h OGTT glucose 7.8 to < 11.1 mmol/L.

Q19. A middle-aged obese man on routine checkup is found to have very high serum cholesterol. Which of the following enzymes should be inhibited to initiate drug therapy?

a) HMG-CoA synthase
b) HMG-CoA reductase ✓
c) Thiolase
d) Squalene synthase
e) Lanosterol synthase

Answer: b) HMG-CoA reductase

Reason: Statins inhibit HMG-CoA reductase, the rate-limiting enzyme.

Q20. Preferential fuel for cardiac muscle is

a) Fatty acid ✓
b) Glucose
c) Acetoacetic acid
d) β-hydroxybutyric acid
e) Lactate

Answer: a) Fatty acid

Reason: Cardiac muscle gets most energy from fatty-acid β-oxidation.

May 2024 — Paper II

Q1. Choose the correct answer

a) If plasma pH is 7.1, it is acidosis **[T]**

b) Mutation of genes may result in loss of function **[T]**

c) If pH is 7.1 and anion gap is 12 mmol/L, it is metabolic acidosis **[F]**

Reason: A pH of 7.1 with normal AG doesn't prove metabolic acidosis (need HCO_3^-/PCO_2).

d) Aldosterone is a lifesaving hormone **[T]**

Reason: Aldosterone retains Na^+/water; deficiency is fatal.

e) Hyposecretion of aldosterone may cause Conn syndrome **[F]**

Reason: Conn's syndrome is aldosterone EXCESS.

Q2. Body water is maintained by

a) Hypothalamus **[T]**

Reason: Hypothalamus has osmoreceptors and makes ADH → water balance.

b) ADH **[T]**

Reason: ADH conserves water by increasing collecting-duct permeability.

c) Thyroxine **[F]**

Reason: Thyroxine regulates metabolism, not water balance.

d) Plasma osmolarity **[T]**

Reason: Plasma osmolarity is the main stimulus for ADH and thirst.

e) Thirst mechanism **[T]**

Reason: Thirst is stimulated by high osmolarity and low volume.

Q3. ADH deficiency causes

a) Hypervolemia **[F]**

Reason: ADH deficiency causes loss of water (hypovolemia), not hypervolemia.

b) Urinary hyperosmolarity **[F]**

Reason: Lack of ADH gives dilute (hypo-osmolar) urine.

c) Diuresis **[T]**

Reason: ADH deficiency → polyuria (water diuresis).

d) Diabetes mellitus **[F]**

Reason: ADH deficiency causes diabetes INsipidus.

e) Increased thirst **[T]**

Reason: Polyuria in ADH deficiency causes thirst/polydipsia.

Q4. Regarding codon

a) They are 3 consecutive bases of mRNA **[T]**

Reason: Codon = 3 consecutive bases of mRNA.

b) Present in tRNA **[F]**

Reason: tRNA carries the anticodon; codon is in mRNA.

c) Responsible for synthesis of specific amino acids **[T]**

Reason: Each codon specifies an amino acid.

d) UAG, UGA, UAA are stop codons **[T]**

Reason: UAA, UAG, UGA are stop codons.

e) AGU codes for methionine **[F]**

Reason: AUG = methionine (start); AGU = serine.

Q5. During DNA replication

a) DNA helicase separates double-stranded DNA **[T]**

Reason: Helicase unwinds the double helix.

b) DNA is synthesized in one direction (5'→3') **[T]**

Reason: DNA is synthesised 5'→3' only.

c) Okazaki fragments are synthesized **[T]**

Reason: Lagging strand is made as Okazaki fragments.

d) DNA is synthesized in a conservative manner **[F]**

Reason: Replication is SEMIconservative.

e) Uracil is paired to adenine **[F]**

Reason: DNA has thymine; uracil pairs with adenine in RNA.

Q6. High-fibre diet

a) Reduces incidence of constipation **[T]**
b) Increases blood cholesterol **[F]**
Reason: Fibre lowers cholesterol.
c) Reduces risk of bowel malignancy **[T]**
d) Increases incidence of diverticulosis **[F]**
Reason: Fibre prevents diverticulosis.
e) Decreases absorption of certain minerals **[T]**

Q7. Obesity

a) Diagnosed by BMI **[T]**
Reason: BMI is the diagnostic index.
b) Genetic mechanisms play a major role **[T]**
Reason: Genetics contributes substantially.
c) Hormone leptin is involved **[T]**
Reason: Leptin from adipocytes regulates satiety.
d) There is hypoglycemia **[F]**
Reason: Obesity is associated with hyperglycaemia/insulin resistance, not hypoglycaemia.
e) Correlated with type-1 diabetes mellitus **[F]**
Reason: Obesity is linked to type 2 DM (insulin resistance).

Q8. Bicarbonate buffer is most effective because †

a) It is most abundant in ECF **[T]**
Reason: HCO_3^- is the most abundant ECF buffer.
b) It can act through both respiratory and renal systems **[T]**
Reason: Open buffer system.
c) Base concentration is less than acid concentration **[F]**
Reason: At pH 7.4 HCO_3^- (24) : H_2CO_3 (1.2) = 20:1, so base exceeds acid.
d) It has a high pK value **[F]**
Reason: pK of the bicarbonate buffer is 6.1 (low).
e) It works along with the hemoglobin buffer **[T]**
Reason: HCO_3^- and Hb buffer work together.

Q9. Deficiency of vitamins is associated with

a) Vitamin A – toad skin **[T]**
Reason: Phrynoderma (toad skin) = vitamin A deficiency.
b) Vitamin K – hemorrhagic disorder **[T]**
Reason: Vitamin K deficiency → hemorrhagic disease.
c) Vitamin B_{12} – neural tube defect **[F]**
Reason: Neural tube defect is folate deficiency.
d) Vitamin C – RBC maturation defect **[F]**
Reason: Vitamin C is for collagen; RBC maturation needs B_{12}/folate.
e) Vitamin D – osteoporosis **[F]**
Reason: Vitamin D deficiency causes rickets/osteomalacia, not osteoporosis.

Q10. Statements related to transcription

a) Synthesis of protein following rRNA **[F]**
Reason: Transcription makes RNA; protein synthesis is translation.
b) Requires RNA polymerase **[T]**
Reason: PCR uses DNA polymerase.
c) Promoter sequence is needed **[T]**
Reason: RNA polymerase binds the promoter.
d) 5' cap is formed by polyadenylation **[F]**
Reason: 5' cap = 7-methylguanosine; polyadenylation is the 3' tail.
e) Has no proofreading property **[T]**
Reason: RNA polymerase lacks efficient proofreading.

Q11. Nucleosome contains †

a) sRNA
b) mRNA
c) tRNA
d) Histone ✓

Answer: d) Histone

Reason: Nucleosome = ~146 bp DNA wrapped around a histone octamer.

Q12. Vitamin B complex †

a) Is chemically related to one another
b) Comes from endogenous synthesis only
c) Acts as coenzyme ✓
d) Is stored in large amounts in the body

Answer: c) Acts as coenzyme

Reason: B-vitamins function as coenzymes.

Q13. The backbone of nucleic acid structure is constructed by

a) Hydrogen bond
b) Disulfide bond
c) Phosphodiester bond ✓
d) Ionic interaction

Answer: c) Phosphodiester bond

Reason: Nucleotides are joined by 3'–5' phosphodiester bonds.

Q14. Recombinant DNA technology

a) Requires restriction enzyme from virus
b) Needs Southern blot technique
c) Produces chimeric DNA
d) Transfer of gene from one species to another ✓

Answer: d) Transfer of gene from one species to another

Reason: rDNA = joining DNA from different species into a vector (chimeric DNA).

Q15. Mutation

a) Polymorphism is another name
b) Imbalance between DNA damage and repair ✓
c) Occurs in gene only
d) Transmitted to offspring

Answer: b) Imbalance between DNA damage and repair

Reason: Mutation = failure of repair relative to damage.

Q16. The longest phase of the cell cycle is

a) G0
b) G1 ✓
c) G2
d) S

Answer: b) G1

Reason: G1 is the longest and most variable phase.

Q17. Which one of the following does NOT contribute to the purine ring synthesis? †

a) CO_2
b) Aspartate
c) N10-formyl tetrahydrofolate
d) Alanine ✓

Answer: d) Alanine

Reason: Purine ring atoms come from glycine, aspartate, glutamine, CO_2 and N10-formyl-THF; alanine is not a donor.

Q18. Best feature of mitochondrial DNA

a) The sperm does not contribute mitochondrial DNA to the zygote
b) It accounts for 1% of total human DNA
c) Shows variable expression
d) Encodes respiratory chain enzymes ✓

Answer: d) Encodes respiratory chain enzymes

Reason: mtDNA encodes 37 genes (13 proteins, 22 tRNA, 2 rRNA).

Q19. Select the reliable diagnostic tool for PEM

a) Decreased protein level
b) Skin change
c) Hair change
d) BMI ✓

Answer: d) BMI

Reason: BMI/weight-for-height is the reliable anthropometric tool; protein levels and clinical signs vary.

Q20. The worst effect of folic acid deficiency

a) Megaloblastic anemia
b) Peripheral neuropathy
c) Neural tube defect ✓
d) Lethargy

Answer: c) Neural tube defect

Reason: Folate deficiency in pregnancy causes neural tube defects.

November 2023 — Paper II

Q1. Criteria of autosomal dominant disorder

a) Skip generation is possible **[F]**
Reason: AD disease appears in every generation (vertical).

b) New mutation occurs **[T]**
Reason: AD diseases can arise from new mutations (e.g., achondroplasia).

c) Involves sex chromosome **[F]**
Reason: AD = autosomal.

d) Follows vertical transmission **[T]**
Reason: AD disease: vertical transmission.

e) Diabetes insipidus is an example **[F]**
Reason: Nephrogenic DI is X-linked recessive.

Q2. Folic acid has the following facts

a) Deficiency causes microcytic anemia **[F]**
Reason: Folate deficiency causes macrocytic (megaloblastic) anaemia.

b) Active form is tetrahydrofolate **[T]**
Reason: THF is the active coenzyme form of folate.

c) Acts as carrier of single carbon unit **[T]**
Reason: THF carries one-carbon groups for purine/thymidylate synthesis.

d) Vitamin B_{12} is not essential for folic acid metabolism **[F]**
Reason: B_{12} is needed to regenerate THF from methyl-THF.

e) Chronic alcoholism causes deficiency **[T]**
Reason: Alcohol impairs folate absorption and increases loss.

Q3. Choose the correct answer

a) Vitamin C inhibits iron absorption from gut **[F]**
Reason: Vitamin C reduces Fe^{3+} to Fe^{2+} and ENHANCES absorption.

b) Selenium acts as an antioxidant **[T]**
Reason: Selenium is a component of glutathione peroxidase (antioxidant).

c) Calcitonin increases blood calcium level **[F]**
Reason: Calcitonin LOWERS blood calcium.

d) Iodine deficiency causes mental retardation **[T]**
Reason: Iodine is needed for thyroid hormone for brain development.

e) Retinol acts as an antioxidant vitamin **[T]**
Reason: Retinol/carotenoids have antioxidant action.

Q4. Choose the correct answer

a) Fat-soluble vitamins may produce toxicity if taken in excess **[T]**
Reason: Fat-soluble vitamins accumulate in fat stores.

b) Anion gap is raised in diarrhea **[F]**
Reason: Diarrhoea causes normal-AG acidosis (Cl^- rises as HCO_3^- is lost).

c) Respiratory alkalosis produces tetany **[T]**
Reason: Alkalosis lowers ionized Ca^{2+} (more protein-bound) → tetany.

d) Warfarin toxicity is treated by vitamin K **[T]**
Reason: Vitamin K reverses warfarin toxicity.

e) Riboflavin deficiency causes pellagra **[F]**
Reason: Pellagra is niacin deficiency; riboflavin gives cheilosis/glossitis.

Q5. Regarding serum electrolytes

a) Hyperkalemia causes cardiac arrest **[T]**
Reason: High K^+ depolarises heart → diastolic arrest.

b) Sodium is the major electrolyte of ECF **[T]**
Reason: Na^+ is the principal ECF cation.

c) Hyperkalemia is treated by glucagon **[F]**
Reason: Hyperkalaemia is treated with calcium gluconate, insulin + glucose, salbutamol.

d) Aldosterone secretion increases in hypokalemia **[F]**
Reason: Aldosterone rises in HYPERkalaemia and falls in hypokalaemia.

e) Sea water drowning causes hypernatremia **[T]**
Reason: Sea water is hypertonic → hypernatraemia.

Q6. Polymerase chain reaction

a) Needs two primers **[T]**
Reason: Forward and reverse primers.

b) Requires cleavage of the DNA by restriction endonuclease **[F]**
Reason: PCR does not use restriction digestion.

c) Is an in vitro process **[T]**
Reason: PCR amplifies DNA in a tube.

d) Requires Taq DNA polymerase **[T]**
Reason: Taq polymerase is heat-stable.

e) Can produce 10^6 products in 20 cycles **[T]**
Reason: $2^{20} \approx 10^6$ copies in 20 cycles.

Q7. Hypokalemia may be due to

a) Uncontrolled DM **[T]**
Reason: Osmotic diuresis in uncontrolled DM causes K^+ loss.

b) Diarrhea **[T]**
Reason: Diarrhoea fluid is rich in K^+ and HCO_3^-.

c) Decreased aldosterone secretion **[F]**
Reason: Aldosterone excretes K^+; deficiency gives hyperkalaemia.

d) Acidosis **[F]**
Reason: Acidosis shifts K^+ out of cells → hyperkalaemia, not hypokalaemia.

e) Alkalosis **[T]**
Reason: Alkalosis shifts K^+ into cells → hypokalaemia.

Q8. Regarding vitamin C

a) Has no role in Fe absorption **[F]**
Reason: Vitamin C reduces $Fe^{3+} \rightarrow Fe^{2+}$ and enhances absorption.

b) Deficiency causes defective collagen synthesis **[T]**
Reason: Vitamin C is needed for prolyl/lysyl hydroxylase → collagen.

c) Milk is the main dietary source **[F]**
Reason: Citrus fruits/vegetables are the main source; milk is poor.

d) Synthesis occurs in the human body **[F]**
Reason: Humans lack L-gulonolactone oxidase; vitamin C is dietary.

e) Does not act as a co-factor (co-enzyme) **[F]**
Reason: Vitamin C is a cofactor (reducing agent) of hydroxylases.

Q9. Transcellular fluid is

a) Fluid in aqueous humor **[T]**
Reason: Transcellular fluids (CSF, aqueous humor, synovial, pleural, peritoneal) are formed by epithelial transport.

b) Blood **[F]**
Reason: Plasma is intravascular fluid, not transcellular.

c) Synovial fluid **[T]**
Reason: Synovial fluid is a transcellular fluid.

d) Interstitial fluid **[F]**
Reason: Interstitial fluid is extracellular but not transcellular.

e) Peritoneal fluid **[T]**
Reason: Peritoneal fluid is a transcellular fluid.

Q10. Basal metabolic rate will increase in

a) Overactivity of the Na^+-K^+ pump **[T]**
Reason: Na^+-K^+ pump uses ATP; more activity → ↑BMR.

b) Hyperthyroidism **[T]**
Reason: Thyroid hormones raise BMR.

c) Hot climate **[F]**
Reason: Cold climate raises BMR; hot climate lowers it.

d) Starvation **[F]**
Reason: BMR falls in starvation.

e) Febrile condition **[T]**
Reason: Each °C of fever raises BMR ~13%.

Q11. Regarding proteinuria

a) Detected by heat coagulation test only
b) Presence of protein in urine more than 150 mg/day ✓
c) Microalbuminuria indicates early diabetic nephropathy only
d) Its consequences develop ankle edema only

Answer: b) Presence of protein in urine more than 150 mg/day
Reason: Proteinuria > 150 mg/day is abnormal.

Q12. Regarding metabolic acidosis †

a) Anion gap is always increased
b) HCO_3^- is < 22 mmol/L ✓
c) Due to primary increase in plasma bicarbonate
d) DKA is the most common cause

Answer: b) HCO_3^- is < 22 mmol/L
Reason: Metabolic acidosis = HCO_3^- < 22 mmol/L.

Q13. Free water clearance gives information regarding

a) Hypertonicity of urine
b) Excretion of isotonic urine
c) Renal concentrating ability ✓
d) Diluted urine only

Answer: c) Renal concentrating ability
Reason: Free water clearance reflects renal diluting/concentrating ability.

Q14. Mutation

a) Polymorphism is another name
b) Imbalance between DNA damage and repair ✓
c) Occurs in gene only
d) Transmitted to offspring

Answer: b) Imbalance between DNA damage and repair
Reason: Mutation = failure of repair relative to damage.

Q15. Restriction endonuclease †

a) Enzyme cleaves RNA at its middle
b) Enzyme cleaves DNA at random sites
c) Cleaves DNA at a specific nucleotide sequence which is palindromic ✓
d) Enzymes are named according to the scientists who first discovered them

Answer: c) Cleaves DNA at a specific nucleotide sequence which is palindromic
Reason: Type II restriction enzymes cut DNA at palindromic sequences.

Q16. Vitamin C †

a) Is a fat-soluble vitamin
b) Citrus fruits are the only source
c) The most prominent action is hydroxylation of prolyl and lysyl residues of collagen protein ✓
d) Deficiency causes scurvy only

Answer: c) The most prominent action is hydroxylation of prolyl and lysyl residues of collagen protein
Reason: Vitamin C hydroxylates proline and lysine in collagen.

Q17. Regarding plasma osmolarity, choose the correct answer †

a) Normal plasma osmolarity is 270 to 310 mosmol/L only
b) Serum sodium is the most contributing factor ✓
c) Glucose and urea are the major contributors
d) Hyposmolarity stimulates ADH secretion

Answer: b) Serum sodium is the most contributing factor
Reason: Na^+ is the main osmotically active solute.

Q18. The characteristics of trace elements include †

a) Calcium is an example
b) Sodium is an example
c) Are microminerals with daily requirement less than 100 mg ✓
d) Zinc acts as an anti-oxidant

Answer: c) Are microminerals with daily requirement less than 100 mg

Reason: Trace elements are needed in < 100 mg/day.

Q19. Regarding translation

a) Synthesis of protein according to the base sequence of tRNA
b) Synthesis of protein according to the base sequence of mRNA ✓
c) Ribosome may or may not be required
d) tRNA only acts as linker molecules

Answer: b) Synthesis of protein according to the base sequence of mRNA

Reason: mRNA codons determine the protein sequence.

Q20. Regarding obesity †

a) Is measured by BMI only
b) Is associated with primary hypothyroidism only
c) Is a disorder of body weight regulatory system characterized by excess fat accumulation ✓
d) Only hypertrophic change occurs in adipocytes

Answer: c) Is a disorder of body weight regulatory system characterized by excess fat accumulation

Reason: Obesity is excess body fat from disturbed energy balance.

May 2023 — Paper I (New Curriculum)

Q1. Buffer

a) Shows higher capacity when its total concentration is more **[T]**
Reason: Buffer capacity increases with total buffer concentration.

b) Components ratio 1:1 means pH = pK **[T]**
Reason: Equal acid and salt gives pH = pK (maximum buffering).

c) Base converts strong acid to its pair weak acid **[T]**
Reason: A buffer base converts a strong acid into its weak conjugate acid, limiting pH change.

d) Can completely return pH back to normal **[F]**
Reason: A buffer only resists change; it cannot restore pH completely (lungs/kidneys do that).

e) Capacity depends upon the compartment it is present in **[T]**
Reason: Buffer capacity depends on the compartment, e.g., HCO_3^- in ECF, phosphate/protein in ICF.

Q2. Enzymes

a) Are reaction- and substrate-specific in action **[T]**
Reason: Enzymes are reaction and substrate specific.

b) Always need coenzyme for their activity **[F]**
Reason: Many enzymes need no coenzyme (e.g., pepsin, trypsin).

c) Are non-dialyzable **[T]**
Reason: Enzymes are large proteins (colloids).

d) Show maximum velocity at their Km **[F]**
Reason: Km is the substrate concentration at HALF Vmax.

e) Are competitively inhibited by allosteric modifiers **[F]**
Reason: Allosteric modifiers bind away from the active site (non-competitive).

Q3. Regarding disaccharides †

a) Insoluble in fat **[T]**
Reason: Disaccharides are water-soluble and fat-insoluble.

b) General formula is $Cn(H_2O)n$ **[F]**
Reason: Disaccharide formula is C12H22O11 = $Cn(H_2O)n–1$.

c) Cannot form ester **[F]**
Reason: Sugar –OH groups can form esters (e.g., phosphates).

d) Some act as reducing agents in alkaline media **[T]**
Reason: Maltose, lactose are reducing; sucrose is not.

e) Sorbitol is a disaccharide **[F]**
Reason: Sorbitol is a sugar alcohol, not a disaccharide.

Q4. Regarding colloid

a) Osmotically less active **[T]**
Reason: Colloids are large, few in number, hence osmotically less active.

b) Not seen by ultramicroscope **[F]**
Reason: Colloid particles are visible by ultramicroscope (Tyndall effect).

c) Cellulose is a dietary colloid **[T]**
Reason: Cellulose is dietary fibre, a colloid that is not digested.

d) Unconjugated bilirubin is a hydrophilic colloid **[F]**
Reason: Unconjugated bilirubin is water-insoluble (hydrophobic) and travels bound to albumin.

e) Can produce a true solution **[F]**
Reason: Colloids form colloidal, not true, solutions.

Q5. The following are rate-limiting reactions of certain metabolic pathways

a) Lipogenesis – acetyl-CoA carboxylase **[T]**
Reason: ACC is rate-limiting for fatty acid synthesis.

b) Glycolysis – PFK-1 **[T]**
Reason: PFK-1 is rate-limiting for glycolysis.

c) Cholesterol biosynthesis – hormone-sensitive lipase **[F]**
Reason: HMG-CoA reductase is rate-limiting for cholesterol synthesis.

d) Glycogenesis – glycogen phosphorylase **[F]**
Reason: Glycogen synthase is the rate-limiting enzyme of glycogenesis.

e) Lipolysis – HMG-CoA reductase **[F]**
Reason: Hormone-sensitive lipase is rate-limiting for lipolysis.

Q6. Choose the correct answers

a) All amino acids undergo transamination **[F]**
Reason: Lysine, threonine, proline, hydroxyproline do not undergo transamination.

b) Ammonia is removed by intestinal bacterial degradation **[F]**
Reason: Intestinal bacteria PRODUCE ammonia (urease); it is removed as urea/glutamine.

c) Oxidative deamination is a catabolic process **[T]**
Reason: Oxidative deamination (glutamate dehydrogenase) releases NH_3 from amino acids.

d) NH_3 is excreted with urine as NH_4^+ **[T]**
Reason: Urinary NH_3 is excreted as NH_4^+.

e) Proline can be synthesized from the urea cycle **[F]**
Reason: Proline is made from glutamate; it is not a urea-cycle product.

Q7. About digestion the following are true

a) Is a process of hydrolysis **[T]**

b) Lipid digestion begins in mouth **[T]**

c) Pepsin splits carbohydrate into glucose **[F]**
Reason: Pepsin digests protein.

d) Micelle is an aggregate of cholesterol only **[F]**
Reason: Micelles contain bile salts, phospholipid, cholesterol and FA — not cholesterol alone.

e) Bile salt reduces the surface tension of fat **[T]**
Reason: Bile salts are detergents that lower surface tension → emulsification.

Q8. Insulin regulates metabolism by

a) Enhancing GLUT2-induced glucose uptake **[F]**
Reason: Insulin acts via GLUT-4; GLUT-2 is insulin-independent (liver, β-cell).

b) Reversible dephosphorylation of enzymes **[T]**
Reason: Insulin activates phosphatases, dephosphorylating key enzymes.

c) Enzyme induction **[T]**
Reason: Insulin induces glucokinase, PFK, pyruvate kinase, lipogenic enzymes.

d) Stimulating the tyrosine kinase receptor pathway **[T]**
Reason: Insulin receptor is a tyrosine kinase.

e) Stimulating pancreatic glucagon secretion **[F]**
Reason: Insulin inhibits glucagon secretion.

Q9. In glycolysis the following statements are correct

a) Glucose or glycogen is broken down **[T]**
Reason: Glycolysis starts from glucose or glycogen (via G6P).

b) The end product of glycolysis is pyruvate only **[F]**
Reason: Pyruvate is the end product aerobically; in anaerobic glycolysis it is reduced to lactate, so 'only' is wrong.

c) There are four irreversible reactions **[F]**
Reason: Glycolysis has 3 irreversible steps: hexokinase/glucokinase, PFK-1, pyruvate kinase.

d) Conversion of lactate to pyruvate is oxidation **[T]**

e) In the aerobic pathway 2 ATP is produced (net) **[F]**
Reason: Net ATP: anaerobic = 2; aerobic ≈ 7–8 (2 + NADH via shuttles).

Q10. Ketoacidosis could be a finding of

a) Uncontrolled diabetes mellitus **[T]**
Reason: Absent insulin → ketogenesis.

b) Prolonged starvation **[T]**
Reason: Low insulin/glucose in starvation → ketosis.

c) Severe diarrhea **[F]**
Reason: Diarrhoea gives non-ketotic metabolic acidosis.

d) Persistent vomiting **[F]**
Reason: Vomiting causes metabolic alkalosis (loss of HCl), not ketoacidosis.

e) Renal failure **[F]**
Reason: Renal failure causes uraemic (high-AG) acidosis without ketones.

Q11. The characteristic blood chemistry of viral hepatitis is

a) Increased serum total bilirubin only
b) Increased serum ALT activity (markedly) ✓
c) Increased serum ALP activity (markedly)
d) Increased serum LDH only
e) Increased prothrombin time only

Answer: b) Increased serum ALT activity (markedly)
Reason: ALT (and AST) rises markedly in viral hepatitis.

Q12. The most common cause of hyperglycemia is

a) Cushing syndrome
b) Diabetes mellitus ✓
c) Gigantism
d) Acromegaly
e) Pheochromocytoma

Answer: b) Diabetes mellitus
Reason: Diabetes mellitus.

Q13. The EMP (glycolysis) pathway †

a) Produces ATP ✓
b) Evolves CO_2
c) Produces NADPH only
d) Needs no coenzyme
e) Occurs in mitochondria

Answer: a) Produces ATP
Reason: Glycolysis generates ATP by substrate-level phosphorylation and is cytosolic.

Q14. Which of the following is true for β-oxidation?

a) End product is acetyl-CoA only
b) FAD and NAD^+ are oxidized
c) Glucagon stimulates β-oxidation ✓
d) Thiokinase is the rate-limiting enzyme
e) Activation of the pathway occurs after a fatty meal

Answer: c) Glucagon stimulates β-oxidation
Reason: Glucagon induces lipolysis and mitochondrial β-oxidation (via CPT-I).

Q15. Regarding GI hormones †

a) Oxyntic cells secrete gastrin
b) Secretin stimulates enzyme-rich pancreatic secretion
c) CCK stimulates HCO_3^--rich pancreatic secretion
d) Gastrin stimulates HCl secretion only
e) Somatostatin inhibits gastrin secretion ✓

Answer: e) Somatostatin inhibits gastrin secretion
Reason: Somatostatin inhibits gastrin, secretin, CCK.

Q16. A young woman complains of weight gain, cold intolerance and tiredness. Laboratory tests show low T_3, low T_4 and high TSH, high anti-TPO antibody. What is the more likely cause?

a) Hypothalamic disease
b) Pituitary disease
c) Subclinical hypothyroidism
d) Hashimoto's thyroiditis ✓
e) Thyroid hormone resistance

Answer: d) Hashimoto's thyroiditis
Reason: Low T_3/T_4, high TSH and anti-TPO = Hashimoto's thyroiditis.

Q17. All of the following are characteristics of diabetic ketoacidosis EXCEPT †

a) Low blood pH
b) Low blood HCO_3^- concentration
c) Increased plasma bicarbonate ✓
d) High anion gap
e) Acetone breath

Answer: c) Increased plasma bicarbonate

Reason: Low HCO_3^- (not raised) is the feature of DKA.

Q18. The value of IGT is †

a) ≥ 11.1 mmol/L
b) 7.8 to < 11.1 mmol/L ✓
c) < 7.8 mmol/L
d) 4.1 mmol/L
e) 4.4 to 6.1 mmol/L

Answer: b) 7.8 to < 11.1 mmol/L

Reason: IGT = 2-h OGTT glucose 7.8 to < 11.1 mmol/L.

Q19. Phospholipids are mostly carried by

a) VLDL
b) HDL ✓
c) IDL
d) LDL
e) CM

Answer: b) HDL

Reason: HDL is phospholipid-rich (~25–30%).

Q20. To diagnose obstructive jaundice, which enzyme is mostly preferred?

a) AST
b) ALT
c) ALP ✓
d) γ-GT
e) LDH

Answer: c) ALP

Reason: ALP is the preferred enzyme.

May 2023 — Paper II (New Curriculum)

Q1. Choose the correct answer

a) Fat-soluble vitamins may produce toxicity if taken in excess **[T]**
Reason: Fat-soluble vitamins accumulate in fat stores.

b) Plasma volume is measured by mannitol **[F]**
Reason: Plasma volume is measured by Evans blue/I-131 albumin; mannitol measures ECF.

c) Sodium is the major electrolyte of ICF **[F]**
Reason: Na^+ is the major ECF cation; K^+ is the major ICF cation.

d) Hypoalbuminemia is the main biochemical feature of kwashiorkor **[T]**

e) Vitamin C inhibits Fe absorption from gut **[F]**
Reason: Vitamin C ENHANCES iron absorption.

Q2. Body water is maintained by

a) Hypothalamus **[T]**
Reason: Hypothalamus has osmoreceptors and makes ADH → water balance.

b) ADH **[T]**
Reason: ADH conserves water by increasing collecting-duct permeability.

c) Thyroxine **[F]**
Reason: Thyroxine regulates metabolism, not water balance.

d) Plasma osmolarity **[T]**
Reason: Plasma osmolarity is the main stimulus for ADH and thirst.

e) Thirst mechanism **[T]**
Reason: Thirst is stimulated by high osmolarity and low volume.

Q3. Regarding trace elements

a) RDA is less than 100 mg **[T]**
Reason: Trace elements are needed in < 100 mg/day.

b) Selenium acts as an antioxidant **[T]**
Reason: Selenium is a component of glutathione peroxidase (antioxidant).

c) RDA of Fe is 1 mg/day for male **[F]**
Reason: RDA of iron in men is ~10 mg/day (1 mg is the absorbed amount).

d) Iodine deficiency causes mental retardation **[T]**
Reason: Iodine is needed for thyroid hormone for brain development.

e) Hypocalcaemia causes tetany **[T]**
Reason: Low ionized Ca^{2+} increases neuromuscular excitability.

Q4. In metabolic acidosis

a) Diarrhoea is a major cause **[T]**
Reason: Bicarbonate-rich intestinal fluid is lost in diarrhoea → metabolic acidosis.

b) PCO_2 increased **[F]**
Reason: In metabolic acidosis PCO_2 falls (respiratory compensation); raised PCO_2 is respiratory acidosis.

c) HCO_3^- decreased **[T]**
Reason: Primary defect in metabolic acidosis is low HCO_3^-.

d) Normal anion gap only **[F]**
Reason: Metabolic acidosis may have high or normal anion gap.

e) Compensated by the renal system **[F]**
Reason: Metabolic acidosis is compensated by the lungs (hyperventilation); kidneys correct it slowly.

Q5. Dietary fibre

a) Decreases the incidence of constipation **[T]**

b) Increases the absorption of cholesterol **[F]**
Reason: Fibre binds bile acids, REDUCING cholesterol absorption.

c) Decreases the absorption of zinc **[T]**

d) Decreases the incidence of diverticulosis **[T]**
Reason: Fibre prevents diverticulosis.

e) Helps in better control of diabetes mellitus **[T]**

Q6. ECF differs from ICF in that

a) Volume is greater **[F]**
Reason: ICF volume is larger than ECF.

b) pH is lower **[F]**
Reason: ECF pH (7.4) is higher than ICF (7.0).

c) Tonicity is lower **[F]**
Reason: ECF and ICF are iso-osmotic at steady state.

d) Anions are mainly inorganic **[T]**
Reason: ECF anions are mainly Cl^- and HCO_3^- (inorganic); ICF anions are organic (proteins, phosphates).

e) Sodium : potassium ratio is higher **[T]**
Reason: ECF has high Na^+ and low K^+; ICF is the reverse.

Q7. Central dogma of genetic information

a) Was originally described by Mendel **[F]**
Reason: Central dogma was proposed by Crick.

b) Does not include DNA replication **[F]**
Reason: Central dogma includes replication, transcription, translation.

c) May include synthesis of DNA from RNA **[T]**

d) Includes translation **[T]**

e) Transfer of information from DNA to RNA to protein **[T]**

Q8. Normal plasma values are

a) Na^+ = 140 mEq/L **[T]**

b) K^+ = 3.5 mEq/L **[T]**

c) Calcium 2.1–2.6 mmol/L **[T]**

d) Creatinine 3 mg/dL **[F]**
Reason: Normal serum creatinine is 0.6–1.2 mg/dL.

e) HCO_3^- = 18 mmol/L **[F]**
Reason: Normal plasma HCO_3^- is 22–26 mmol/L.

Q9. Transcellular fluid is

a) Fluid in aqueous humor **[T]**
Reason: Transcellular fluids (CSF, aqueous humor, synovial, pleural, peritoneal) are formed by epithelial transport.

b) Blood **[F]**
Reason: Plasma is intravascular fluid, not transcellular.

c) Synovial fluid **[T]**
Reason: Synovial fluid is a transcellular fluid.

d) Interstitial fluid **[F]**
Reason: Interstitial fluid is extracellular but not transcellular.

e) Peritoneal fluid **[T]**
Reason: Peritoneal fluid is a transcellular fluid.

Q10. Thiamine deficiency causes

a) Beriberi **[T]**
Reason: Thiamine deficiency → beriberi.

b) Stomatitis **[F]**
Reason: Stomatitis is riboflavin/niacin deficiency, not thiamine.

c) Wernicke's encephalopathy **[T]**
Reason: Thiamine deficiency in alcoholics → Wernicke's encephalopathy.

d) Neuropathy **[T]**
Reason: Dry beriberi causes peripheral neuropathy.

e) Korsakoff psychosis **[T]**
Reason: Korsakoff psychosis is thiamine deficiency.

Q11. Regarding renal function tests

a) Benedict's test is done for presence of glucose **[T]**
Reason: Benedict's test detects reducing sugars (glucose, fructose, lactose, maltose), not sucrose.

b) Blood urea and serum creatinine are the most sensitive tests **[F]**
Reason: GFR/clearance tests are more sensitive; urea/creatinine rise late.

c) Serum electrolytes are sensitive tests **[F]**
Reason: Electrolytes change late in renal disease.

d) Estimation of specific gravity is done in diabetes insipidus **[T]**

Q12. Vitamin D

a) Steroid derivative **[T]**
Reason: Vitamin D is a secosteroid derived from 7-dehydrocholesterol.

b) Synthesized in skin with the help of UV light **[T]**
Reason: UV-B converts 7-dehydrocholesterol → cholecalciferol in skin.

c) The most prominent actions are to regulate the plasma level of calcium and phosphorus **[T]**
Reason: Calcitriol raises plasma Ca^{2+} and phosphate.

d) Deficiency causes rickets **[T]**
Reason: Vitamin D deficiency → rickets (children), osteomalacia (adults).

Q13. Regarding replication

a) Is a semiconservative process **[T]**
Reason: Each daughter DNA has one old and one new strand.

b) Is DNA-directed DNA synthesis with the help of DNA polymerase **[T]**
Reason: DNA polymerase copies a DNA template.

c) Primer may or may not be required **[F]**
Reason: DNA polymerase always needs a primer.

d) Lagging strand is constructed by Okazaki fragments **[T]**
Reason: Lagging strand = Okazaki fragments.

Q14. X-linked recessive disorder includes

a) Is the disease of X chromosome expressed only in homozygous state **[F]**
Reason: XLR disease is expressed in hemizygous males too.

b) Haemophilia is an example **[T]**

c) Affects males mainly **[T]**
Reason: XLR disease mainly affects males.

d) Oblique or diagonal transmission **[T]**
Reason: XLR: carrier mother → affected sons (oblique/criss-cross).

Q15. Mutation

a) Polymorphism is another name **[F]**
Reason: Polymorphism is a common variant (> 1%); mutation is rare.

b) Imbalance between DNA damage and repair **[T]**
Reason: Mutation = failure of repair relative to damage.

c) Occurs in gene only **[F]**
Reason: Mutations occur in coding and non-coding DNA.

d) Transmitted to offspring **[T]**
Reason: Only germline mutations are inherited.

Q16. Recombinant DNA technology

a) Requires restriction enzymes from virus **[F]**
Reason: Restriction enzymes come from BACTERIA.

b) Needs Southern blot technique **[F]**
Reason: rDNA technology does not require Southern blot.

c) Produces chimeric DNA **[T]**
Reason: Recombinant DNA is chimeric.

d) Transfer of genes from one species to another **[T]**
Reason: Gene from one species into another.

Q17. Regarding plasma osmolarity, choose the correct answer †

a) Normal plasma osmolarity is 280–296 mosmol/L **[T]**
Reason: Normal plasma osmolarity is ~280–296 mosmol/L.

b) Serum sodium is the most contributing factor **[T]**
Reason: Na^{+} and accompanying anions determine most of plasma osmolarity.

c) Glucose and urea are also responsible **[T]**
Reason: Glucose and urea contribute little (~5% each) normally.

d) Hyperosmolarity stimulates ADH secretion **[T]**
Reason: High osmolarity stimulates ADH.

Q18. The characteristics of kwashiorkor include

a) Protein deficiency with adequate calories **[T]**

b) Skin changes are the most characteristic feature **[F]**

Reason: Edema/hypoalbuminaemia are the most characteristic features.

c) Hypoproteinaemia and edema are the most characteristic features **[T]**

d) Mental changes **[T]**

Q19. Regarding potassium

a) Hyperkalaemia stimulates aldosterone secretion in short-term regulation **[T]**

Reason: High K^+ directly stimulates aldosterone release from zona glomerulosa.

b) ECF hyperosmolarity causes influx of K^+ **[F]**

Reason: Hyperosmolarity draws water and K^+ OUT of cells (efflux).

c) Is concerned with the action potential **[T]**

Reason: K^+ efflux repolarises the membrane.

d) Hypokalaemia inhibits the release of insulin **[T]**

Reason: Hypokalaemia reduces insulin secretion.

Q20. Regarding iron

a) Absorption occurs in large gut **[F]**

Reason: Iron is absorbed in duodenum and proximal jejunum.

b) Is absorbed in ferric form **[F]**

Reason: Fe^{2+} (ferrous) is the absorbed form.

c) Vitamin C is rarely needed for absorption **[F]**

Reason: Vitamin C promotes iron absorption.

d) Binds with ferritin **[T]**

Reason: Iron is stored as ferritin.

May 2023 — Paper I (Old Curriculum)

Q1. Regarding buffer

a) Consists of a weak acid and its conjugate base **[T]**
Reason: A buffer = weak acid + its conjugate base (salt).

b) Is more effective when pH = pK **[T]**
Reason: Buffering is maximal at pH = pK (±1).

c) Converts weak acid into strong acid **[F]**
Reason: A buffer does the opposite: it converts a strong acid into a weak acid.

d) Only present intracellularly **[F]**
Reason: Buffers are present in both ECF and ICF.

e) The pK of the HCO_3^- buffer is 6.1 **[T]**

Q2. Colloid

a) Diameter is 1–200 mμ **[T]**
Reason: Colloid particle size is 1–200 millimicron (mμ = nm); crystalloids are < 1 nm.

b) Non-dialyzable **[T]**
Reason: Colloids are too large to pass through a dialysis membrane.

c) Has an isoelectric pH **[T]**
Reason: Colloids such as proteins have a definite isoelectric pH.

d) Shows molecular movement in solution **[F]**
Reason: Colloids show Brownian movement (random particle movement); molecular/ionic movement is typical of crystalloids.

e) DNA is an example **[T]**
Reason: DNA is a macromolecule that forms colloidal solution.

Q3. The statements can be applied for the H-H equation

a) pH = pK + log salt/acid **[T]**
Reason: H-H equation: pH = pK + log [salt]/[acid].

b) pH = 6.1 + log HCO_3^-/H_2CO_3 **[T]**
Reason: Henderson–Hasselbalch for bicarbonate buffer: pH = 6.1 + log $[HCO_3^-]/[H_2CO_3]$.

c) pH = 6.1 + log $[HCO_3^-]/[H_2CO_3]$ **[T]**
Reason: Henderson–Hasselbalch for bicarbonate buffer: pH = 6.1 + log $[HCO_3^-]/[H_2CO_3]$.

d) pH = pK at half neutralization **[T]**
Reason: When acid is half-neutralized [salt] = [acid], log 1 = 0, so pH = pK.

e) Is a derivative of the law of mass action **[T]**
Reason: H-H equation is derived from the law of mass action applied to Ka of a weak acid.

Q4. Regarding carbohydrates

a) Are derivatives of polyhydric aldehydes and ketones **[T]**
Reason: Carbohydrates = polyhydroxy aldehydes/ketones and derivatives.

b) Show optical isomerism **[T]**
Reason: Asymmetric carbons give optical isomerism.

c) Fructose is an invert sugar **[F]**
Reason: Invert sugar is a glucose + fructose mixture from sucrose hydrolysis.

d) Sucrose exhibits mutarotation **[F]**
Reason: Sucrose has no free anomeric carbon, so no mutarotation.

e) Prevent ketosis **[T]**
Reason: Carbohydrate is anti-ketogenic (supplies oxaloacetate).

Q5. Denaturation of protein involves

a) Change of tertiary structure **[T]**
Reason: Denaturation alters 2°, 3° and 4° structures.

b) Loss of biological activity **[T]**
Reason: Denaturation destroys the native conformation, so activity is lost.

c) Loss of primary structure **[F]**
Reason: Denaturation disrupts 2°–4° structure only; peptide bonds (primary) stay intact.

d) Release of some amino acids **[F]**
Reason: Denaturation does not hydrolyse peptide bonds.

e) Adding acid or alkali into it **[T]**
Reason: Acid, alkali, heat, and urea denature proteins.

Q6. Regarding body fluid volume measurement

a) Plasma volume measured by mannitol **[F]**
Reason: Plasma volume is measured by Evans blue/I-131 albumin; mannitol measures ECF.

b) RBC volume measured by ^{51}Cr **[T]**
Reason: Cr-51-tagged RBCs measure RBC volume.

c) ECF volume measured by inulin **[T]**
Reason: Inulin stays extracellular → ECF volume.

d) Plasma volume measured by aminopyrine **[F]**
Reason: Aminopyrine/antipyrine measure TBW; plasma volume uses Evans blue.

e) TBW measured by heavy water **[T]**
Reason: D2O/antipyrine distribute in TBW.

Q7. Enzymes with their clinical significance

a) ALT is highly specific for liver **[T]**
Reason: ALT is more liver-specific than AST.

b) CPK-MM is highly specific for muscle **[T]**
Reason: CPK-MM is the skeletal muscle isoenzyme; CK-MB is cardiac.

c) SGOT is a heart disease marker **[T]**
Reason: AST (SGOT) rises in MI.

d) ALP is increased in prostate cancer **[F]**
Reason: ACP (not ALP) rises in prostate cancer.

e) Aldolase is increased in brain tumor **[F]**
Reason: Aldolase rises in muscle disease (muscular dystrophy).

Q8. Regarding fatty acids

a) Are aliphatic monocarboxylic acids **[T]**
Reason: Fatty acids are aliphatic monocarboxylic acids.

b) 95% exist as free fatty acid **[F]**
Reason: Raised free FFA inhibits GH.

c) Are simple lipids **[F]**
Reason: Fatty acids are components/derived lipids.

d) Require micelle for absorption **[T]**
Reason: Long-chain fatty acids need bile-salt micelles.

e) Act as precursor of eicosanoids **[T]**
Reason: Arachidonic acid → eicosanoids.

Q9. Membrane transport process for colloid is

a) Simple diffusion **[F]**
Reason: Colloids are too large for diffusion; they are taken up by endocytosis.

b) Facilitated diffusion **[F]**
Reason: Colloids are too large for carrier-mediated transport.

c) Secondary active transport **[F]**
Reason: Colloids are taken in by endocytosis, not by carriers.

d) Endocytosis **[T]**
Reason: Large colloids enter cells by endocytosis.

e) Exocytosis **[T]**
Reason: Large colloids leave cells by exocytosis.

Q10. Vitamin D

a) Is an amino acid derivative **[F]**
Reason: Vitamin D is a steroid derivative.

b) Is not soluble in alcohol **[F]**
Reason: Vitamin D is fat-soluble, soluble in alcohol/organic solvents.

c) Is produced from 7-dehydrocholesterol in skin **[T]**
Reason: Skin 7-dehydrocholesterol → cholecalciferol.

d) Produces calcitriol which acts as a hormone **[T]**
Reason: 1,25-$(OH)_2$-D3 (calcitriol) is the active hormone.

e) Deficiency causes softening of bones **[T]**
Reason: Vitamin D deficiency = osteomalacia (soft bones).

Q11. Regarding coenzymes

a) Only protein in nature **[F]**
Reason: Coenzymes are NON-protein organic molecules.

b) Catalyze chemical reactions **[F]**
Reason: Enzymes catalyse; coenzymes only assist as carriers.

c) Most of them are derivatives of fat-soluble vitamins **[F]**
Reason: Most coenzymes derive from water-soluble (B-complex) vitamins.

d) Thiamine pyrophosphate is an example **[T]**
Reason: TPP (from vitamin B1) is a coenzyme for oxidative decarboxylation and transketolase.

e) NADPH is a coenzyme for reductive synthesis **[T]**
Reason: NADPH donates reducing equivalents for biosynthesis (fatty acid, cholesterol).

Q12. Choose the correct answers

a) Gastric juice contains pepsinogen **[T]**

b) Bile contains lipolytic enzymes **[F]**
Reason: Bile has salts, not enzymes; lipase comes from pancreas.

c) Pancreatic juice contains colipase **[T]**

d) HCl secretion produces alkaline tide **[T]**

e) Chylomicron is synthesized in liver **[F]**
Reason: Chylomicrons are made in intestinal mucosa.

Q13. Dietary fibres

a) Are polysaccharides **[T]**

b) Are not absorbed from the gut **[T]**

c) Increase the incidence of colon cancer **[F]**
Reason: Fibre REDUCES colon cancer risk.

d) Lower the blood cholesterol level **[T]**

e) Do not prevent constipation **[F]**
Reason: Fibre adds bulk and prevents constipation.

Q14. In metabolic acidosis

a) $[H^+]$ decreased **[F]**
Reason: In acidosis $[H^+]$ rises and pH falls.

b) PCO_2 increased **[F]**
Reason: In metabolic acidosis PCO_2 falls (respiratory compensation); raised PCO_2 is respiratory acidosis.

c) HCO_3^- decreased **[T]**
Reason: Primary defect in metabolic acidosis is low HCO_3^-.

d) Normal anion gap only **[F]**
Reason: Metabolic acidosis may have high or normal anion gap.

e) pH is < 7.35 **[T]**
Reason: Acidaemia is pH < 7.35; alkalosis has pH > 7.45.

Q15. Regarding anion gap

a) Is equal to unmeasured anion **[T]**
Reason: AG represents the unmeasured anions (albumin, phosphate, sulphate, organic acids).

b) Measured anions are sulphate and phosphate **[F]**
Reason: Sulphate/phosphate are UNmeasured anions; measured anions are Cl^- and HCO_3^-.

c) Normal value is 8–16 mEq/L **[T]**
Reason: Normal anion gap is 8–16 mEq/L.

d) Anion gap = measured cation – measured anion **[T]**
Reason: AG = Na^+ – (Cl^- + HCO_3^-), i.e., measured cations – measured anions (= unmeasured anions – unmeasured cations).

e) Is increased in metabolic acidosis (high-AG type) **[T]**

Q16. The characteristics of marasmus include

a) Inadequate protein and calorie intake **[T]**

b) Most commonly seen in the first year of life **[T]**

c) Edema is present **[F]**
Reason: Edema is a feature of kwashiorkor, not marasmus.

d) Severe growth retardation **[T]**

e) Body weight/age is 60–80% **[F]**
Reason: Marasmus: weight/age < 60% of standard (kwashiorkor 60–80% with oedema).

Q17. Antioxidant vitamins are

a) Vitamin K **[F]**
Reason: Vitamin K is not an antioxidant; it is for clotting.

b) β-carotene **[T]**
Reason: β-carotene is an antioxidant.

c) Vitamin E **[T]**
Reason: Vitamin E is the main lipid-phase antioxidant.

d) Vitamin B_{12} **[F]**
Reason: B_{12} is not an antioxidant.

e) Vitamin C **[T]**
Reason: Vitamin C is the cofactor for collagen hydroxylation.

Q18. Isotopes

a) Are atoms of different elements **[F]**
Reason: Isotopes are atoms of the SAME element with different mass numbers.

b) Are compounds having the same number of protons **[F]**
Reason: Isotopes are atoms, not compounds.

c) Radioactive ones can be used as a therapeutic tool **[T]**
Reason: Radioisotopes (I-131, Co-60) are used therapeutically.

d) Have different atomic numbers **[F]**
Reason: Isotopes have the SAME atomic number, different mass number.

e) ^{131}I is used to diagnose thyroid disorder **[T]**
Reason: I-131 is taken up by the thyroid, used in diagnosis/therapy.

Q19. About digestive juice

a) Trypsin is an amylolytic enzyme **[F]**
Reason: Trypsin is proteolytic.

b) Succus entericus pH is 7.5 **[T]**

c) Lecithin is a phospholipid and amphipathic **[T]**

d) Salivary amylase acts on all types of starch **[F]**
Reason: Salivary amylase acts on cooked starch; raw starch poorly.

e) Enterokinase is present in gastric juice **[F]**
Reason: Enterokinase is in duodenal mucosa.

Q20. Regarding nutrition the following are true

a) BMI is weight in kg / height in square metre **[T]**
Reason: BMI = kg/m².

b) Normal range of BMI in male is 19–24 **[T]**
Reason: Normal BMI is 18.5–24.9 kg/m².

c) Abnormal excess accumulation of fat is an obesity disorder **[T]**
Reason: Obesity is excess body fat.

d) Marasmus is a protein deficiency disorder **[F]**
Reason: Marasmus = protein–calorie deficiency (energy).

e) Vitamin D deficiency causes osteoporosis **[F]**
Reason: Vitamin D deficiency causes osteomalacia, not osteoporosis.

May 2023 — Paper II (Old Curriculum)

Q1. Regarding TCA cycle

a) It is an amphibolic pathway **[T]**
Reason: TCA cycle is both catabolic and anabolic.

b) It oxidizes acetyl-CoA derived from fatty acid oxidation **[T]**
Reason: β-oxidation → acetyl-CoA → TCA.

c) Produces 24 ATP from each molecule of acetyl-CoA **[F]**
Reason: One acetyl-CoA gives ~10 ATP (12 in older texts).

d) Oxaloacetate is the catalyst **[T]**
Reason: Oxaloacetate is regenerated, acting catalytically.

e) Total ATP production is at substrate level **[F]**
Reason: Most ATP comes via oxidative phosphorylation.

Q2. Pyruvate

a) Produces alanine by transamination **[T]**
Reason: Pyruvate + glutamate ⇄ alanine + α-KG.

b) Produces acetyl-CoA by oxidative deamination **[F]**
Reason: Pyruvate → acetyl-CoA is oxidative DEcarboxylation (PDH), not deamination.

c) Produces lactate in anaerobic condition **[T]**
Reason: Anaerobic: pyruvate → lactate (LDH).

d) Comes from glucose by anaerobic condition **[T]**
Reason: Glycolysis works without O2.

e) Comes from lactate by oxidation **[T]**
Reason: Lactate → pyruvate by LDH.

Q3. Regarding gluconeogenesis

a) De novo synthesis of glucose **[T]**
Reason: Gluconeogenesis = new glucose from non-carbohydrate precursors.

b) Occurs in well-fed condition **[F]**
Reason: Gluconeogenesis occurs in fasting; insulin inhibits it.

c) Takes place in adipose tissues **[F]**
Reason: Gluconeogenesis occurs only in liver and kidney.

d) Insulin inhibits gluconeogenesis **[T]**
Reason: Insulin suppresses PEPCK, G6Pase, FBPase.

e) Glycerol is a substrate **[T]**
Reason: Glycerol → DHAP → glucose.

Q4. The following statements are true

a) Nucleic acid is a polymer of nucleotide **[T]**
Reason: DNA/RNA are polymers of nucleotides.

b) Uracil is a purine base **[F]**
Reason: Uracil is a pyrimidine.

c) Nucleotide connects by 3'–5' phosphodiester bond **[T]**
Reason: DNA polymerase forms 3'–5' phosphodiester bonds.

d) Guanine is a nucleotide base in RNA **[T]**
Reason: RNA has A, G, C, U.

e) Amino acid is not a part of DNA **[T]**
Reason: DNA = nucleotides only.

Q5. In β-oxidation of palmitic acid

a) End product is acetyl-CoA only **[T]**
Reason: Palmitate β-oxidation: acetyl-CoA + NADH + $FADH_2$.

b) In each cycle 1 NADH and 1 $FADH_2$ are generated **[T]**
Reason: Each β-oxidation cycle yields 1 $FADH_2$, 1 NADH, 1 acetyl-CoA.

c) Produces 7 acetyl-CoA **[F]**
Reason: Palmitic acid (C16) gives 8 acetyl-CoA after 7 cycles.

d) Complete oxidation releases net 129 ATP **[T]**
Reason: Palmitate yields net 129 ATP (106 by revised values).

e) 80% of energy is utilized by skeletal muscle **[F]**
Reason: Not 80% by skeletal muscle.

Q6. Features of myxedema

a) Puffy face **[T]**
Reason: Myxoedema → mucopolysaccharide deposition → puffiness.

b) Husky voice **[T]**
Reason: Myxoedema affects vocal cords.

c) Retarded growth **[F]**
Reason: Myxoedema is adult hypothyroidism; growth retardation is cretinism.

d) Diarrhoea **[F]**
Reason: Hypothyroidism causes constipation.

e) Bagginess under the eye **[T]**
Reason: Periorbital edema in myxoedema.

Q7. Features of Addison's disease are

a) Hypoglycemia **[T]**
Reason: Cortisol deficiency (Addison's) → hypoglycaemia.

b) Hyponatremia **[T]**
Reason: Aldosterone deficiency → Na^+ loss.

c) Hypokalemia **[F]**
Reason: Addison's → HYPERkalaemia (aldosterone lack); Cushing's → hypokalaemia (mineralocorticoid effect of cortisol).

d) Hypopigmentation **[F]**
Reason: ↑ACTH/MSH → hyperpigmentation in Addison's.

e) Hypotension **[T]**
Reason: Aldosterone/cortisol lack → salt loss, hypotension; Cushing's causes hypertension.

Q8. Carbohydrate is metabolized by

a) Krebs urea cycle **[F]**
Reason: Krebs cycle (TCA) and the urea cycle are separate.

b) Glycolysis **[T]**
Reason: Glycolysis: PFK-1 is rate limiting.

c) HMP shunt pathway **[T]**

d) Oxidative deamination **[F]**
Reason: Pyruvate → acetyl-CoA is oxidative DEcarboxylation (PDH), not deamination.

e) Gluconeogenesis **[T]**
Reason: Gluconeogenesis is an anabolic synthesis of glucose; occurs in fasting, liver/kidney, inhibited by insulin.

Q9. Growth hormone secretion is stimulated in

a) Hyperglycemia **[F]**
Reason: Hyperglycaemia suppresses GH; hypoglycaemia stimulates it. (In DKA hyperglycaemia is a diagnostic feature.)

b) Stress **[T]**
Reason: Stress, exercise, sleep and hypoglycaemia raise GH.

c) Increased plasma free fatty acid **[F]**
Reason: Raised free FFA inhibits GH.

d) The first few hours of deep sleep **[T]**
Reason: GH peaks during slow-wave sleep.

e) Increased plasma insulin **[F]**
Reason: Hyperinsulinaemia (via hypoglycaemia) is stimulatory but insulin itself is not the trigger.

Q10. The diagnostic features of diabetic ketoacidosis

a) High pH **[F]**
Reason: DKA has acidosis (low pH).

b) Low HCO_3^- **[T]**

c) Ketonaemia **[T]**

d) Hyperglycemia **[T]**
Reason: Hyperglycaemia suppresses GH; hypoglycaemia stimulates it. (In DKA hyperglycaemia is a diagnostic feature.)

e) Low anion gap **[F]**
Reason: DKA has HIGH anion gap.

Q11. Thyroid function tests

a) Normal TSH level is 0.4–4.5 µIU/mL **[T]**
Reason: Normal TSH is ~0.4–4.5 µIU/mL.

b) Normal TSH excludes hypothyroidism and hyperthyroidism (primary) **[T]**
Reason: A normal TSH excludes primary hypo/hyperthyroidism.

c) TSH, T_3 and T_4 increase in secondary hypothyroidism **[F]**
Reason: In secondary hypothyroidism TSH, T_3, T_4 are all low.

d) Normal free T_3 level is 3 to 9 pmol/L **[T]**
Reason: Normal free T_3 ≈ 3–7 pmol/L.

e) TSH level more than 20 µIU/mL indicates hyperthyroidism **[F]**
Reason: TSH > 20 µIU/mL indicates hypothyroidism.

Q12. Regarding lipoproteins

a) Chylomicrons are synthesized in the intestinal mucosa **[T]**
Reason: Chylomicrons are made in intestinal mucosa.

b) VLDL is synthesized in the liver **[T]**
Reason: VLDL is made in liver.

c) Chylomicrons are absorbed through portal circulation **[F]**
Reason: Chylomicrons enter lymph (lacteals), not the portal vein.

d) HDL acts as cholesterol scavenger **[T]**
Reason: HDL carries cholesterol from tissues to liver.

e) LDL contains more triacylglycerol but less cholesterol **[F]**
Reason: LDL is cholesterol-rich, TG-poor.

Q13. Regarding codon

a) They are 3 consecutive bases of mRNA **[T]**
Reason: Codon = 3 consecutive bases of mRNA.

b) Present in tRNA **[F]**
Reason: tRNA carries the anticodon; codon is in mRNA.

c) Responsible for synthesis of specific amino acids **[T]**
Reason: Each codon specifies an amino acid.

d) UAG, UGA, UAA are stop codons **[T]**
Reason: UAA, UAG, UGA are stop codons.

e) AGU codes for methionine **[F]**
Reason: AUG = methionine (start); AGU = serine.

Q14. During DNA replication

a) DNA helicase separates the double-stranded DNA **[T]**
Reason: Helicase unwinds the double helix.

b) DNA is synthesized in one direction **[T]**
Reason: DNA is synthesised 5'→3' only.

c) Okazaki fragments are synthesized **[T]**
Reason: Lagging strand is made as Okazaki fragments.

d) DNA is synthesized in a conservative manner **[F]**
Reason: Replication is SEMIconservative.

e) Uracil is paired to adenine **[F]**
Reason: DNA has thymine; uracil pairs with adenine in RNA.

Q15. Regarding blotting techniques

a) Northern blot technique is for DNA analysis **[F]**
Reason: Northern = RNA; Southern = DNA; Western = protein.

b) Southern blot technique is for RNA analysis **[F]**
Reason: Southern = DNA.

c) Western blot technique is for protein **[T]**
Reason: Western = protein.

d) Nucleic acid probe is used in Southern blot **[T]**
Reason: Southern blot uses a nucleic acid probe.

e) Antibody probe is used in Northern blot technique **[F]**
Reason: Antibody probe is for Western blot.

Q16. Diagnostic criteria of IGT

a) Fasting plasma glucose is > 7 mmol/L **[F]**
Reason: FPG ≥ 7 mmol/L is diabetes; IGT has FPG < 7.

b) 2-h post-glucose load level is 7.8–11 mmol/L **[T]**
Reason: IGT = 2-h glucose 7.8–11.0 mmol/L.

c) 2-h post-glucose load level is > 11 mmol/L **[F]**
Reason: ≥ 11.1 mmol/L is diabetes, not IGT.

d) HbA_{1c} value is 5.7–6.4% **[T]**
Reason: HbA_{1c} 5.7–6.4% = pre-diabetes.

e) HbA_{1c} value is ≥ 6.5% **[F]**
Reason: HbA_{1c} ≥ 6.5% = diabetes.

Q17. Disorders of amino acid metabolism

a) Phenylketonuria **[T]**
Reason: PKU is autosomal recessive.

b) Gaucher's disease **[F]**
Reason: Gaucher's is a lysosomal storage disease.

c) Albinism **[T]**

d) Wilson's disease **[F]**
Reason: Wilson's disease is a copper disorder.

e) Alkaptonuria **[T]**

Q18. Mechanism of edema

a) Increased colloid osmotic pressure **[F]**
Reason: Reduced colloid osmotic pressure causes edema.

b) Increased capillary hydrostatic pressure **[T]**

c) Salt retention in renal failure **[T]**

d) Lymphatic obstruction **[T]**

e) Decreased capillary permeability **[F]**
Reason: Increased permeability causes edema.

Q19. Mutation

a) Errors in the DNA occur at a slow rate **[T]**
Reason: Spontaneous mutation rate is low.

b) Only occurs in the X chromosome **[F]**
Reason: Mutations occur anywhere in the genome.

c) Occurs due to error in replication **[T]**
Reason: Replication errors cause mutation.

d) β-thalassemia is an example of nonsense mutation **[F]**
Reason: β-thalassaemia is mostly point/splice mutations; nonsense mutations also occur.

e) In any region of DNA is expressed phenotypically **[F]**
Reason: Mutations in non-coding DNA may be silent.

Q20. The following are autosomal dominant disorders

a) Achondroplasia **[T]**
Reason: Achondroplasia is autosomal dominant.

b) Familial hypercholesterolemia **[T]**
Reason: FH is autosomal dominant (LDL receptor).

c) Diabetes insipidus **[F]**
Reason: Nephrogenic DI is X-linked recessive.

d) Phenylketonuria **[F]**
Reason: PKU is autosomal recessive.

e) Sickle cell anaemia **[F]**
Reason: Sickle cell anaemia is autosomal recessive.

November 2022 — Paper I

Q1. About pH

a) Is the log of $[H^+]$ **[F]**
Reason: pH is the NEGATIVE logarithm of $[H^+]$.

b) pH = pK + log $[HCO_3^-]/[H_2CO_3]$ **[T]**
Reason: Henderson–Hasselbalch for bicarbonate buffer: pH = 6.1 + log $[HCO_3^-]/[H_2CO_3]$.

c) pH = pK at half neutralization **[T]**
Reason: When acid is half-neutralized [salt] = [acid], log 1 = 0, so pH = pK.

d) pH of ECF is 7.1 **[F]**
Reason: ECF/blood pH is 7.35–7.45 (7.4).

e) Hyperventilation increases pH **[T]**
Reason: Hyperventilation blows off CO_2, lowers H_2CO_3 and raises pH (respiratory alkalosis).

Q2. Regarding crystalloid

a) The diameter is < 1 nm **[T]**
Reason: Crystalloid particles are < 1 nm.

b) Movement is Brownian **[F]**
Reason: Crystalloids show molecular movement; Brownian movement is typical of colloids.

c) Can pass through semipermeable membrane **[T]**
Reason: Crystalloids pass; colloids cannot pass through semipermeable membranes.

d) Osmotic pressure is 32 mmHg **[F]**
Reason: Crystalloid osmotic pressure is large; colloid osmotic pressure is ~25–32 mmHg.

e) Is dialyzable **[T]**
Reason: Crystalloids are dialyzable; colloids/enzymes are not.

Q3. Enzymes

a) Are crystalloids **[F]**
Reason: Enzymes are proteins (colloids), not crystalloids.

b) Are dialyzable **[F]**
Reason: Enzymes are proteins (colloids) and are non-dialyzable.

c) Protein in nature **[T]**
Reason: Coenzymes are non-protein organic molecules; the apoenzyme is the protein.

d) Reaction specific **[T]**
Reason: Enzymes are reaction and substrate specific.

e) Inactivated by heat **[T]**
Reason: Heat denatures enzyme proteins.

Q4. Choose the correct answer for body fluid

a) Na^+ is the main cation of ECF **[T]**
Reason: Na^+ is the principal ECF cation (~140 mEq/L).

b) Cl^- is the main anion of ICF **[F]**
Reason: Cl^- is the main ECF anion; ICF anions are phosphate and proteins.

c) ECF is the major compartment **[F]**
Reason: ICF (2/3 of TBW) is the major compartment; ECF is 1/3.

d) pH of ICF is lower than that of ECF **[T]**
Reason: ICF pH (~7.0) is lower than ECF (7.4).

e) Ca^{2+} is the major divalent ion of ICF **[F]**
Reason: Mg^{2+} is the major intracellular divalent cation.

Q5. Regarding protein

a) Peptide bond is –CONH2– **[F]**
Reason: Peptide bond is –CO–NH– (not CONH2).

b) Peptide bond is an expression of gene **[F]**
Reason: Peptide bond is a covalent amide bond, not a gene expression.

c) Denaturation causes loss of primary structure **[F]**
Reason: Denaturation disrupts 2°–4° structure only; peptide bonds (primary) stay intact.

d) Amino acids are zwitterions **[T]**
Reason: At isoelectric pH amino acids carry both + and – charges.

e) Concentration in plasma is 6–8 g/dL **[T]**
Reason: Normal plasma protein is 6–8 g/dL.

Q6. Lipids have the following facts

a) Esters of polyhydric alcohols only **[F]**
Reason: Lipids include esters of mono- and polyhydric alcohols plus others (cholesterol, etc.).

b) Have higher specific gravity **[F]**
Reason: Lipids are lighter than water.

c) Iodine number indicates molecular weight **[F]**
Reason: Iodine number measures degree of unsaturation.

d) Open air oxidation causes rancidity **[T]**
Reason: Oxidation of unsaturated fats gives rancidity.

e) Polyunsaturated fats are available in fishes **[T]**
Reason: Fish oils are rich in PUFA (ω-3).

Q7. Regarding diffusion

a) Is an active transport process **[F]**
Reason: Diffusion is passive.

b) Diffusion does not need carrier protein **[T]**
Reason: Simple diffusion needs no carrier or energy.

c) Facilitated diffusion is bi-directional **[T]**
Reason: Facilitated diffusion goes down the gradient in either direction, as per gradient.

d) Temperature can facilitate diffusion **[T]**
Reason: Higher temperature increases diffusion.

e) Simple diffusion can be blocked competitively **[F]**
Reason: Simple diffusion has no carrier, so it cannot be competitively blocked.

Q8. Fatty acid

a) Is an –OH group-containing aromatic acid **[F]**
Reason: Fatty acids are aliphatic.

b) Mustard oil is a source of essential fatty acid **[T]**
Reason: Mustard oil contains linoleic acid, an essential fatty acid.

c) Melting point is low in PUFA **[T]**
Reason: Double bonds lower melting point.

d) Forms soap with alcohol **[F]**
Reason: Soap forms by saponification with alkali.

e) PUFA are available in fishes **[T]**
Reason: Fish oils are rich in PUFA (ω-3).

Q9. Renal buffers

a) Protein buffer and ammonia buffer **[F]**
Reason: Renal buffers are phosphate and ammonia.

b) Are needed for H^+ secretion in renal tubules **[T]**
Reason: Urinary buffers let tubules secrete H^+ without lowering urine pH below ~4.5.

c) Are not found in glomerular filtrate in renal failure **[F]**
Reason: Phosphate/ammonia buffers are present in filtrate; in renal failure they are inadequate, not absent.

d) Help in HCO_3^- generation **[T]**
Reason: Each H^+ buffered and excreted generates a new HCO_3^-.

e) Decrease plasma HCO_3^- concentration **[F]**
Reason: Renal buffers help generate HCO_3^-, they do not lower it.

Q10. Regarding carbohydrate

a) Hexose is a monosaccharide **[T]**
Reason: Hexoses are 6-carbon monosaccharides.

b) Lactose is a reducing sugar **[T]**
Reason: Lactose has a free anomeric carbon.

c) Sucrose is a reducing sugar **[F]**
Reason: Sucrose has no free anomeric carbon → non-reducing.

d) Amylopectin is a highly branched starch **[T]**
Reason: Amylopectin has α-1,6 branches every 24–30 residues.

e) Cellulose gives energy **[F]**
Reason: Humans lack cellulase; cellulose gives no energy.

Q11. Vitamins associated with some clinical conditions

a) Cobalamin – dementia **[T]**
Reason: B_{12} deficiency causes neuropsychiatric features.

b) Thiamine – cardiac failure **[T]**
Reason: Wet beriberi → high-output cardiac failure.

c) Phylloquinone – defect in blood clotting **[T]**
Reason: Phylloquinone (vit K1) is needed for clotting.

d) Ascorbic acid – gum bleeding **[T]**
Reason: Scurvy causes gum bleeding.

e) Pyridoxal phosphate – spina bifida **[F]**
Reason: Spina bifida/neural tube defects are due to FOLATE deficiency.

Q12. Apo-lipoproteins

a) Are the structural components of lipoprotein **[T]**
Reason: Apolipoproteins are structural and functional components of lipoproteins.

b) Provide solubility to blood lipids **[T]**
Reason: Apolipoproteins help solubilise lipids in plasma.

c) Act as ligand **[T]**
Reason: Apo B-100/E are ligands for receptors.

d) Act as coenzyme **[T]**
Reason: Apo C-II activates LPL; apo A-I activates LCAT.

e) Can be transferred from one lipoprotein to another **[T]**
Reason: Apo C and E are exchangeable between lipoproteins.

Q13. Dietary fibers

a) Are monosaccharides **[F]**
Reason: Dietary fibres are polysaccharides (cellulose, pectin) – non-starch.

b) Are not absorbed from the gut **[T]**

c) Increase the incidence of colon cancer **[F]**
Reason: Fibre REDUCES colon cancer risk.

d) Lower the blood cholesterol level **[T]**

e) Prevent constipation **[T]**

Q14. Cholesterol

a) Is synthesized in the liver only **[F]**
Reason: Cholesterol is also made in intestine, adrenals, skin.

b) Should be < 200 mg/dL in plasma **[T]**
Reason: Desirable total cholesterol < 200 mg/dL.

c) Is a simple lipid **[F]**
Reason: Cholesterol is a derived lipid.

d) Contains cyclopentano-perhydrophenanthrene ring **[T]**
Reason: Cholesterol has the cyclopentanoperhydrophenanthrene ring.

e) Synthesis is inhibited by prostaglandin **[F]**
Reason: Cholesterol synthesis is inhibited by cholesterol (feedback), statins, insulin/glucagon control.

Q15. Regarding PEM

a) In marasmus edema generalizes **[F]**
Reason: Marasmus has no edema.

b) In kwashiorkor appetite is usually good **[F]**
Reason: Kwashiorkor has anorexia.

c) In kwashiorkor skin is pigmented but hair is normal **[F]**
Reason: Kwashiorkor: skin changes and hair changes (flag sign).

d) In kwashiorkor liver may be palpable **[T]**

e) Severe growth retardation occurs in marasmus **[T]**

Q16. Hypokalemia

a) Means plasma K^+ level < 3.5 mmol/L **[T]**
Reason: Hypokalaemia = plasma K^+ < 3.5 mmol/L.

b) Causes transmembrane K^+ efflux **[F]**
Reason: Hypokalaemia causes K^+ shift INTO cells (influx).

c) Increases blood pH by excreting H^+ through urine **[T]**
Reason: Hypokalaemia → H^+ moves into cells, H^+ excreted in urine → alkalosis.

d) May cause cardiac arrest at systole **[T]**
Reason: Hypokalaemia can cause cardiac arrest in SYSTOLE; hyperkalaemia causes arrest in diastole.

e) Is a common finding of persistent vomiting **[T]**
Reason: Vomiting causes loss of H^+, K^+, Cl^- → hypokalaemic alkalosis.

Q17. Folic acid

a) The active form is dihydrofolate **[F]**
Reason: THF is the active form.

b) RDA is 600–800 mg/day **[F]**
Reason: RDA of folate is ~400 µg/day (600 µg in pregnancy), not mg.

c) Is required for the synthesis of nucleotides **[T]**
Reason: THF supplies carbon for purines and dTMP.

d) Acts as a carrier of single carbon unit **[T]**
Reason: THF carries one-carbon groups for purine/thymidylate synthesis.

e) Deficiency causes megaloblastic anemia **[T]**
Reason: Folate deficiency → megaloblastic anaemia.

Q18. Balanced diet

a) The ratio of carbohydrate, protein and fat is approximately 4:1:1 **[T]**
Reason: Balanced diet ratio ≈ 4:1:1 (carbohydrate : protein : fat).

b) Required for maintenance of cellular growth **[T]**
Reason: Balanced diet supports growth and maintenance.

c) Does not contain fibres **[F]**
Reason: A balanced diet includes fibre.

d) Fat is limited **[T]**
Reason: Balanced diet limits fat (~20–25% energy).

e) May not contain essential amino acids **[F]**
Reason: Balanced diet supplies essential amino acids.

Q19. Regarding zymogens

a) Zymogens remain active in digestive juice **[F]**
Reason: Zymogens are inactive precursors.

b) Chymotrypsinogen is a gastric enzyme **[F]**
Reason: Chymotrypsinogen is a pancreatic zymogen.

c) Enterokinase activates pepsinogen **[F]**
Reason: Enterokinase activates trypsinogen.

d) Trypsin is activated by auto-catalysis **[T]**
Reason: Trypsinogen → trypsin by enterokinase and autocatalysis.

e) Pepsin is an endopeptidase **[T]**
Reason: Pepsin cleaves internal peptide bonds.

Q20. In uncompensated metabolic acidosis

a) pH is decreased **[T]**
Reason: Acidosis lowers pH.

b) pH is increased **[F]**
Reason: Acidosis lowers pH; alkalosis raises it.

c) PCO_2 is increased **[F]**
Reason: Raised PCO_2 occurs in respiratory acidosis or as compensation of metabolic alkalosis, not as a feature of uncompensated metabolic acidosis.

d) HCO_3^- is decreased **[T]**
Reason: Primary defect in metabolic acidosis is low HCO_3^-.

e) HCO_3^- remains unchanged **[F]**
Reason: In respiratory acidosis the kidneys retain HCO_3^-.

November 2022 — Paper II

Q1. Regarding glycolysis the following statements are correct

a) The end product is lactate **[F]**
Reason: End product of glycolysis is pyruvate (aerobic); lactate forms only in anaerobic conditions.

b) The end product is pyruvate **[T]**
Reason: Aerobic glycolysis ends in pyruvate (to mitochondria); lactate is formed only anaerobically.

c) There are four irreversible reactions **[F]**
Reason: Glycolysis has 3 irreversible steps: hexokinase/glucokinase, PFK-1, pyruvate kinase.

d) Lactate conversion to pyruvate is oxidation **[T]**
Reason: Lactate → pyruvate removes 2H (LDH, NAD^+ → NADH): oxidation.

e) In the aerobic pathway 2 ATP are produced (net) **[F]**
Reason: Net ATP: anaerobic = 2; aerobic ≈ 7–8 (2 + NADH via shuttles).

Q2. The following are observed in diabetic ketoacidosis

a) pH > 7.5 **[F]**
Reason: DKA has low pH (acidosis).

b) PCO_2 is normal **[F]**
Reason: In DKA PCO_2 is low due to Kussmaul breathing.

c) HCO_3^- < 22 mEq/L **[T]**
Reason: DKA lowers HCO_3^- below 22 mEq/L.

d) There is > 16 mEq/L of anion gap **[T]**

e) Increased K^+ level **[T]**
Reason: Insulin lack and acidosis shift K^+ out; plasma K^+ is initially high.

Q3. Regarding liver function tests

a) Serum gamma-glutamyl transpeptidase is elevated in viral hepatitis **[T]**
Reason: GGT rises in hepatobiliary disease, including viral hepatitis.

b) Serum albumin level is 3.5–5 g/dL **[T]**
Reason: Normal serum albumin 3.5–5 g/dL.

c) Prothrombin time is normal in acute liver disease **[F]**
Reason: Severe hepatocellular injury prolongs PT; PT is a synthetic function test.

d) Serum albumin : globulin ratio is altered in chronic liver disease **[T]**
Reason: A:G is reversed in chronic liver disease.

e) Serum bilirubin indicates synthetic function of the liver **[F]**
Reason: Bilirubin = excretory function; synthetic = albumin, PT.

Q4. Insulin

a) Is a potent anabolic hormone **[T]**
Reason: Insulin promotes glycogen, fat and protein synthesis.

b) Is required for utilization of glucose in brain **[F]**
Reason: Brain uses GLUT-3, which is insulin-independent.

c) Increases glucose uptake in skeletal muscle **[T]**
Reason: Insulin recruits GLUT-4 in muscle and adipose tissue.

d) Increases plasma K^+ **[F]**
Reason: Insulin drives K^+ into cells (lowers plasma K^+); used to treat hyperkalaemia.

e) Mode of action is mediated by cyclic AMP **[F]**
Reason: Insulin acts via tyrosine kinase receptor.

Q5. Biochemical findings

a) Decreased blood osmolarity in diabetes mellitus **[F]**
Reason: Hyperglycaemia raises osmolarity.

b) Respiratory acidosis in diabetes mellitus **[F]**
Reason: DM causes metabolic acidosis.

c) Increased serum creatinine in renal failure **[T]**
Reason: Reduced GFR raises creatinine.

d) Increased serum bilirubin in viral hepatitis **[T]**
Reason: Hepatocellular damage raises bilirubin.

e) Normal anion gap in uncontrolled diabetes mellitus **[F]**
Reason: Diarrhoea/renal tubular acidosis give normal-AG (hyperchloremic) acidosis; DKA, lactic acidosis give high AG.

Q6. TSH values

a) 0.1 mU/L indicates thyrotoxicosis values **[T]**
Reason: TSH < 0.1 mU/L suggests thyrotoxicosis.

b) > 10 mU/L indicates clinical hypothyroidism **[T]**
Reason: TSH > 10 mU/L with low T_4 = clinical hypothyroidism.

c) Can be used in diagnosis of thyroid disease **[T]**
Reason: TSH is the first-line thyroid screening test.

d) 5–20 mU/L indicates subclinical hypothyroidism **[T]**
Reason: Subclinical hypothyroidism: TSH 5–20 μIU/mL with normal T_3/T_4.

e) Should always be done in diagnosis of thyroid disease **[T]**
Reason: TSH is the first-line test for thyroid disease.

Q7. HMP shunt pathway †

a) Occurs in mitochondria **[F]**
Reason: HMP shunt is cytosolic.

b) No ATP production **[T]**
Reason: HMP shunt produces NADPH and pentoses, no ATP.

c) H2 receptor is NAD^+ **[F]**
Reason: HMP shunt reduces $NADP^+$ to NADPH.

d) First part of reaction is non-oxidative and reversible **[F]**
Reason: First phase of HMP is oxidative and irreversible; second is non-oxidative reversible.

e) NADPH deficiency presents as hemolytic anemia **[T]**
Reason: High-dose synthetic menadione causes hemolysis in neonates.

Q8. Regarding autosomal dominant disorder, the following are true

a) Achondroplasia **[T]**
Reason: Achondroplasia is autosomal dominant.

b) Haemophilia **[F]**
Reason: Haemophilia is X-linked recessive.

c) Cystic fibrosis **[F]**
Reason: Cystic fibrosis is autosomal recessive.

d) Vitamin D resistant rickets **[F]**
Reason: Vitamin D-resistant rickets is X-linked DOMINANT.

e) Expressed in heterozygous state **[T]**
Reason: AD disorders manifest in heterozygotes.

Q9. In the anaerobic glycolysis pathway

a) Glucose-6-phosphate dehydrogenase is an enzyme **[F]**
Reason: G6PD is the HMP-shunt enzyme, not glycolytic.

b) ATP is produced **[T]**

c) Glycerol is produced **[F]**
Reason: Glycolysis does not produce glycerol.

d) Pyruvate is produced **[T]**
Reason: Glycolysis produces pyruvate.

e) Lactate is produced **[T]**
Reason: Anaerobic glycolysis produces lactate.

Q10. Choose the correct answer

a) HMG-CoA reductase is a rate-limiting enzyme **[T]**
Reason: HMG-CoA reductase is the rate-limiting step.

b) Bile acid is synthesized from cholesterol **[T]**
Reason: Bile acids are made from cholesterol by 7α-hydroxylase.

c) Serum LDL cholesterol level is increased by thyroxine **[F]**
Reason: Thyroxine up-regulates LDL receptors and LOWERS LDL.

d) Deoxycholic acid is a primary bile acid **[F]**
Reason: Deoxycholic and lithocholic acids are SECONDARY bile acids (bacterial 7-dehydroxylation).

e) Taurochenodeoxycholic acid is a secondary bile acid **[F]**
Reason: Taurochenodeoxycholic acid is a conjugated PRIMARY bile acid.

Q11. Choose the correct answer

a) Terminal codons are AUG, UAG and UGA **[F]**
Reason: AUG is the START codon; stop codons are UAA, UAG, UGA.

b) Mitochondrial DNA is maternal in origin **[T]**

c) Familial hypercholesterolemia is autosomal recessive **[F]**
Reason: FH is autosomal dominant (LDL receptor).

d) Hemophilia is an X-linked dominant disorder **[F]**
Reason: Haemophilia is X-linked RECESSIVE.

e) Mutation of genes may result in sudden termination of translation **[T]**

Q12. Choose the correct answer for ketone bodies

a) Acetone is a lipid-derived compound **[T]**
Reason: Acetone is derived from acetoacetate (lipid metabolism).

b) Acetoacetate is the parent ketone body **[T]**
Reason: Acetoacetate is the primary ketone body.

c) Is synthesized from pyruvate **[F]**
Reason: Ketone bodies come from acetyl-CoA of fatty acids.

d) Ketosis causes low pH **[T]**
Reason: Ketoacids lower pH.

e) Rothera's test identifies it in urine **[T]**
Reason: Rothera's nitroprusside test detects acetoacetate/acetone.

Q13. Regarding lipoprotein, the following statements are correct

a) Chylomicron transports endogenous triglyceride **[F]**
Reason: Chylomicrons carry EXOGENOUS (dietary) TG; VLDL carries endogenous TG.

b) VLDL transports exogenous triglyceride **[F]**
Reason: VLDL carries ENDOGENOUS TG.

c) VLDL is produced from liver **[T]**
Reason: VLDL is synthesised in liver.

d) Cholesteryl ester transfer protein transfers cholesteryl ester from HDL to LDL **[T]**
Reason: CETP moves cholesteryl ester from HDL to VLDL/LDL.

e) Apolipoproteins are enzyme cofactors **[T]**
Reason: Apolipoproteins are enzyme cofactors/activators.

Q14. Ammonia

a) Is produced during oxidation of uric acid **[F]**
Reason: Ammonia is not produced during uric acid oxidation.

b) Is added to the amino acid pool **[F]**
Reason: Ammonia leaves the amino acid pool.

c) Is produced in transamination **[F]**
Reason: Transamination moves NH2 without releasing NH_3.

d) Can combine with CO_2 to form carbamoyl phosphate **[T]**
Reason: NH_3 + CO_2 + 2ATP → carbamoyl phosphate.

e) Is increased in portal hypertension **[T]**
Reason: Portal shunting gives hyperammonaemia.

Q15. Conjugated bilirubin

a) Is conjugated with albumin **[F]**
Reason: Bilirubin is conjugated with glucuronic acid; unconjugated travels on albumin.

b) Contains glucuronic acid **[T]**

c) Is increased in hemolytic jaundice **[F]**
Reason: Hemolytic jaundice: unconjugated hyperbilirubinaemia.

d) Is excreted in urine **[T]**

e) Is conjugated in bone marrow **[F]**
Reason: Bilirubin is conjugated in liver.

Q16. Characteristic features of mRNA include

a) Carries message from DNA **[T]**
Reason: mRNA carries the genetic message.

b) Contains anticodon **[F]**
Reason: Anticodon is in tRNA.

c) Requires modification after synthesis **[T]**
Reason: Eukaryotic mRNA is capped, spliced, polyadenylated.

d) Acts as adaptor molecule **[F]**
Reason: tRNA is the adaptor.

e) Is part of a gene **[T]**
Reason: mRNA is a transcript of a gene.

Q17. Enzymes of gluconeogenesis

a) Glucokinase **[F]**
Reason: Glucokinase is glycolytic/liver glucose sensor, not gluconeogenic.

b) Phosphoenolpyruvate carboxykinase **[T]**
Reason: PEPCK is a key gluconeogenic enzyme.

c) Glucose-6-phosphatase **[T]**
Reason: Glucose-6-phosphatase releases free glucose in gluconeogenesis.

d) Alanine aminotransferase **[F]**
Reason: ALT only supplies pyruvate from alanine.

e) Carbamoyl phosphate synthetase **[F]**
Reason: CPS-I is a urea-cycle enzyme.

Q18. Complications of GDM

a) Macrosomia **[T]**
Reason: Maternal hyperglycaemia → fetal hyperinsulinism → macrosomia.

b) Acromegaly **[F]**
Reason: Acromegaly is GH excess, not a GDM complication.

c) Still birth **[T]**
Reason: GDM raises risk of stillbirth.

d) Development of DM in mother **[T]**
Reason: GDM predisposes to later type 2 DM.

e) Congenital anomaly of newborn **[T]**
Reason: Hyperglycaemia in early gestation is teratogenic.

Q19. Transcellular fluid is

a) Fluid in aqueous humor **[T]**
Reason: Transcellular fluids (CSF, aqueous humor, synovial, pleural, peritoneal) are formed by epithelial transport.

b) Blood **[F]**
Reason: Plasma is intravascular fluid, not transcellular.

c) Synovial fluid **[T]**
Reason: Synovial fluid is a transcellular fluid.

d) Interstitial fluid **[F]**
Reason: Interstitial fluid is extracellular but not transcellular.

e) Peritoneal fluid **[T]**
Reason: Peritoneal fluid is a transcellular fluid.

Q20. About anion gap

a) Is equal to unmeasured anions **[T]**
Reason: AG represents the unmeasured anions (albumin, phosphate, sulphate, organic acids).

b) Measured anions are sulphate and phosphate **[F]**
Reason: Sulphate/phosphate are UNmeasured anions; measured anions are Cl^- and HCO_3^-.

c) Normal value of anion gap is 8–16 mmol/L **[T]**
Reason: Normal anion gap is 8–16 mEq/L.

d) Anion gap is increased in metabolic acidosis (high-AG type) **[T]**
Reason: High-AG metabolic acidosis (DKA, lactic acidosis, renal failure) raises the gap.

e) Anion gap = measured cation – measured anion **[T]**
Reason: AG = Na^+ – (Cl^- + HCO_3^-), i.e., measured cations – measured anions (= unmeasured anions – unmeasured cations).

May 2022 — Paper I

Q1. Regarding buffer

a) Mixture of weak acid and its conjugate base **[T]**
Reason: A buffer = weak acid + its conjugate base (salt).

b) Potent plasma buffer is phosphate buffer **[F]**
Reason: Bicarbonate/plasma protein are plasma buffers; phosphate is mainly intracellular/renal.

c) Potent renal buffer is bicarbonate buffer **[F]**
Reason: Renal buffers are phosphate and ammonia.

d) pK of bicarbonate buffer is 6.1 **[T]**
Reason: pK of the bicarbonate buffer is 6.1.

e) Buffer action is maximum when pH = pK **[T]**
Reason: Buffer action is maximal when pH = pK.

Q2. Regarding colloid †

a) Diameter is 1–200 mμ **[T]**
Reason: Colloid particle size is 1–200 millimicron (mμ = nm); crystalloids are < 1 nm.

b) Movement is molecular **[F]**
Reason: Colloids show Brownian (particle) movement, not molecular movement.

c) Can't pass through semipermeable membrane **[T]**
Reason: Colloids cannot pass through a semipermeable membrane.

d) Non-dialyzable **[T]**
Reason: Colloids are too large to pass through a dialysis membrane.

e) Osmotic pressure is higher than that of crystalloid **[F]**
Reason: Colloids have few particles per volume, so their osmotic pressure is lower than that of crystalloids.

Q3. Denaturation of protein is characterized by

a) Uncoiling of peptide chain **[T]**
Reason: Denaturation uncoils the chain, losing biological activity.

b) Cannot be precipitated **[F]**
Reason: Denatured protein is less soluble and easily precipitated.

c) Loss of biological activity **[T]**
Reason: Denaturation destroys the native conformation, so activity is lost.

d) Diffusibility is increased **[F]**
Reason: Denatured protein is less diffusible.

e) Solubility is increased **[F]**
Reason: Denatured protein becomes less soluble.

Q4. Regarding isotopes †

a) Co-60 is used in radiation therapy **[T]**
Reason: Co-60 emits γ-rays for radiotherapy.

b) Helium is an isotope of hydrogen **[F]**
Reason: Helium is a separate element; hydrogen isotopes are deuterium and tritium.

c) I-131 is used to measure CSF volume **[F]**
Reason: I-131 albumin measures plasma volume; not CSF volume.

d) C-12 is used to measure atomic weight **[T]**
Reason: C-12 is the reference standard for atomic mass.

e) Cr-51 is used to measure RBC volume **[T]**
Reason: Cr-51-tagged RBCs measure RBC volume.

Q5. Body water is maintained by

a) Hypothalamus **[T]**
Reason: Hypothalamus has osmoreceptors and makes ADH → water balance.

b) ADH **[T]**
Reason: ADH conserves water by increasing collecting-duct permeability.

c) Thyroxine **[F]**
Reason: Thyroxine regulates metabolism, not water balance.

d) Plasma osmolarity **[T]**
Reason: Plasma osmolarity is the main stimulus for ADH and thirst.

e) Thirst mechanism **[T]**
Reason: Thirst is stimulated by high osmolarity and low volume.

Q6. Neutral fat

a) Is an ester of monohydric alcohol **[F]**
Reason: Neutral fat is an ester of glycerol (trihydric).

b) Has charge at body pH **[F]**
Reason: Neutral fat is uncharged.

c) Glycerol is the backbone alcohol **[T]**
Reason: Triacylglycerol = glycerol + 3 fatty acids.

d) Has polyunsaturated fatty acids only **[F]**
Reason: Neutral fat contains mixed saturated/unsaturated fatty acids.

e) Is a derived lipid **[F]**
Reason: Neutral fat is a simple lipid.

Q7. Regarding osmosis

a) Needs semipermeable membrane **[T]**
Reason: Osmosis = solvent movement across a semipermeable membrane.

b) Only the solutes cross the membrane **[F]**
Reason: In osmosis the solvent moves; solutes cannot cross.

c) Both solutes and solvent cross the membrane **[F]**
Reason: In osmosis only the solvent crosses.

d) Depends on the concentration of the solutes **[T]**
Reason: Osmotic pressure depends on the number of solute particles.

e) Depends on the movement of the solvent **[T]**
Reason: Osmosis is by definition solvent movement.

Q8. Regarding carbohydrate

a) D-variety of sugars are common in nature **[T]**
Reason: Natural sugars are mostly D-forms.

b) Maltose is a non-reducing sugar **[F]**
Reason: Maltose has a free anomeric carbon → reducing.

c) Hydrolysis of maltose gives rise to glucose only **[T]**
Reason: Maltose = glucose + glucose.

d) Inulin is composed of L-fructose units **[F]**
Reason: Inulin is a polymer of D-fructose.

e) Dextran is used as plasma volume expander **[T]**
Reason: Dextran is a glucose polymer used as plasma expander.

Q9. Concerning coenzyme

a) Coenzyme A transfers acetyl group **[T]**
Reason: CoA (from pantothenic acid) carries acyl/acetyl groups.

b) Thiamine pyrophosphate transfers CO_2 **[F]**
Reason: TPP transfers 2-carbon aldehyde groups (decarboxylation), not CO_2.

c) Can bind with the active site of enzyme **[T]**
Reason: Coenzymes bind the enzyme at/near active site.

d) Is heat labile **[F]**
Reason: Enzymes are proteins and heat labile.

Q10. Choose the correct answer

a) Bile contains lipolytic enzymes **[F]**
Reason: Bile has salts, not enzymes; lipase comes from pancreas.

b) Gastric juice contains gastrin **[F]**
Reason: Gastrin is a hormone secreted into blood, not an enzyme of juice.

c) Pancreatic juice contains α-1 antitrypsin **[F]**
Reason: Pancreatic juice contains trypsin inhibitor; α1-antitrypsin is plasma.

d) HCl secretion produces alkaline tide **[T]**

e) Chylomicron is synthesized in liver **[F]**
Reason: Chylomicrons are made in intestinal mucosa.

Q11. About vitamin A deficiency

a) Deficiency impairs immunity **[T]**
Reason: Vitamin A deficiency impairs immunity and epithelia.

b) Nyctalopia is a failure of dark adaptation **[T]**
Reason: Rod dysfunction → failure of dark adaptation.

c) Results in dry, thick skin **[T]**
Reason: Vitamin A deficiency causes follicular hyperkeratosis (toad skin).

d) Causes osteomalacia **[F]**
Reason: Osteomalacia is vitamin D deficiency; vitamin A deficiency causes xerophthalmia.

e) Causes repeated respiratory tract infection **[T]**
Reason: Keratinisation of respiratory epithelium predisposes to infection.

Q12. The characteristics of marasmus

a) Inadequate protein and calorie intake **[T]**

b) Most commonly seen in first year of life **[T]**

c) Oedema is present **[F]**
Reason: Edema is a feature of kwashiorkor, not marasmus.

d) Severe growth retardation occurs **[T]**

e) Body weight/age is 80–60% **[F]**
Reason: Marasmus: weight/age < 60% (Wellcome classification).

Q13. Functions of respective fat-soluble vitamins

a) Vitamin A – vision **[T]**
Reason: Retinal is the visual pigment chromophore.

b) Vitamin D – calcium and phosphate metabolism **[T]**
Reason: Calcitriol raises Ca^{2+} and phosphate absorption.

c) Vitamin E – blood clotting **[F]**
Reason: Blood clotting is vitamin K; vitamin E is an antioxidant.

d) Vitamin K – antioxidant **[F]**
Reason: Vitamin K carboxylates clotting factors; vitamin E is the antioxidant.

e) Vitamin A – reproduction **[T]**
Reason: Retinol is needed for spermatogenesis and reproduction.

Q14. An increase in ECF osmolarity

a) Stimulates ADH release **[T]**
Reason: High osmolarity stimulates osmoreceptors and ADH release.

b) Inhibits the sensation of thirst **[F]**
Reason: High ECF osmolarity STIMULATES thirst.

c) Stimulates the osmoreceptors **[T]**
Reason: Osmoreceptors in hypothalamus are stimulated by high osmolarity.

d) Is due to gain of ECF volume **[F]**
Reason: Gain of water would dilute (lower) osmolarity.

e) Is due to loss of body fluid **[T]**
Reason: Water loss concentrates ECF.

Q15. Regarding eicosanoids

a) PG1, PGI_2 and LT are eicosanoids **[T]**

b) The dietary precursor of PG is linoleic acid **[T]**

c) Eicosanoids contain 18 carbons **[F]**
Reason: Eicosanoids are 20-carbon derivatives.

d) PG acts as a local hormone **[T]**

e) Kidney is the major site of PG degradation **[F]**
Reason: Lung is the major site of PG degradation.

Q16. About GIT hormones

a) CCK causes gall bladder contraction **[T]**

b) Secretin stimulates biliary bicarbonate secretion **[T]**

c) Gastric inhibitory peptide inhibits insulin release **[F]**
Reason: GIP stimulates insulin release (incretin).

d) Secretin stimulates pancreatic bicarbonate secretion **[T]**

e) CCK stimulates gastric emptying **[F]**
Reason: CCK DELAYS gastric emptying.

Q17. One osmolar solution of NaCl

a) Contains 58.5 g NaCl in 1 L **[F]**
Reason: NaCl dissociates into 2 particles; 1 osmolar needs 29.25 g/L.

b) Causes shrinkage of RBC **[T]**
Reason: 1 osmolar NaCl is hypertonic → RBC shrink.

c) Gives osmotic pressure < 280 mosm/L **[F]**
Reason: 1 osmolar = 1000 mosm/L, far above plasma.

d) Is similar to a 0.5 M solution of NaCl **[T]**
Reason: NaCl gives two osmoles per mole, so 1 osmolar = 0.5 M.

e) Is similar to 0.9% NaCl solution **[F]**
Reason: 0.9% NaCl is ~0.3 osmolar.

Q18. Regarding iron

a) Absorption occurs in the large gut **[F]**
Reason: Iron is absorbed in duodenum/upper jejunum.

b) Is absorbed in ferric form **[F]**
Reason: Fe^{2+} (ferrous) is the absorbed form.

c) Vitamin D favors absorption **[F]**
Reason: Vitamin C favours iron absorption.

d) Binds with transferrin **[T]**
Reason: Fe^{3+} circulates bound to transferrin.

e) Is stored as ferritin **[T]**
Reason: Intracellular iron is stored as ferritin.

Q19. Characteristics of dehydration in severe diarrhoea

a) Decreased osmolarity **[F]**
Reason: Diarrhoea causes loss of hypotonic fluid → hyperosmolar dehydration.

b) PCO_2 is increased **[F]**
Reason: Raised PCO_2 occurs in respiratory acidosis or as compensation of metabolic alkalosis, not as a feature of uncompensated metabolic acidosis.

c) $[HCO_3^-]$ is decreased **[T]**

d) Normal anion gap **[T]**
Reason: Diarrhoea/renal tubular acidosis give normal-AG (hyperchloremic) acidosis; DKA, lactic acidosis give high AG.

e) pH is < 7.35 **[T]**
Reason: Acidaemia is pH < 7.35; alkalosis has pH > 7.45.

Q20. In respiratory acidosis

a) pH is decreased **[T]**
Reason: Acidosis lowers pH.

b) pH is increased **[F]**
Reason: Acidosis lowers pH; alkalosis raises it.

c) PCO_2 is increased **[T]**
Reason: Raised PCO_2 occurs in respiratory acidosis or as compensation of metabolic alkalosis, not as a feature of uncompensated metabolic acidosis.

d) $[HCO_3^-]$ is increased (compensation) **[T]**
Reason: Kidneys retain HCO_3^- to compensate respiratory acidosis.

e) $[HCO_3^-]$ remains unchanged **[F]**
Reason: In respiratory acidosis HCO_3^- rises with compensation.

May 2022 — Paper II

Q1. Regarding lipoprotein, the following statements are correct

a) Chylomicron transports endogenous triglyceride **[F]**
Reason: Chylomicrons carry EXOGENOUS (dietary) TG; VLDL carries endogenous TG.

b) VLDL transports exogenous triglyceride **[F]**
Reason: VLDL carries ENDOGENOUS TG.

c) VLDL is produced from liver **[T]**
Reason: VLDL is synthesised in liver.

d) Cholesteryl ester transfer protein transfers cholesteryl ester from HDL to LDL **[T]**
Reason: CETP moves cholesteryl ester from HDL to VLDL/LDL.

e) Apolipoproteins are enzyme cofactors **[T]**
Reason: Apolipoproteins are enzyme cofactors/activators.

Q2. Regarding protein metabolism

a) Site of deamination is liver and kidney **[T]**
Reason: Deamination occurs mainly in liver and kidney.

b) Site of transamination is cytoplasm only **[F]**
Reason: Transamination occurs in cytosol and mitochondria.

c) Site of transamination is liver only **[F]**
Reason: Transamination occurs in many tissues.

d) CO_2 liberated by TCA cycle is used in urea cycle **[F]**
Reason: Urea-cycle CO_2 is from HCO_3^-/bicarbonate, not directly from TCA.

e) Dehydrogenation occurs in non-oxidative deamination **[F]**
Reason: Non-oxidative deamination uses dehydratase/lyase.

Q3. In glycolysis the following statements are correct

a) The end product is lactate **[F]**
Reason: End product of glycolysis is pyruvate (aerobic); lactate forms only in anaerobic conditions.

b) The end product is pyruvate **[T]**
Reason: Aerobic glycolysis ends in pyruvate (to mitochondria); lactate is formed only anaerobically.

c) There are four irreversible reactions **[F]**
Reason: Glycolysis has 3 irreversible steps: hexokinase/glucokinase, PFK-1, pyruvate kinase.

d) Lactate conversion to pyruvate is oxidation **[T]**
Reason: Lactate → pyruvate removes 2H (LDH, NAD^+ → NADH): oxidation.

e) In the aerobic pathway 2 ATP are produced (net) **[F]**
Reason: Net ATP: anaerobic = 2; aerobic ≈ 7–8 (2 + NADH via shuttles).

Q4. Regarding hypothyroidism

a) Serum TSH is always increased in primary hypothyroidism **[T]**
Reason: Primary hypothyroidism: gland failure → ↓T_3/T_4 → ↑TSH (feedback).

b) Serum TSH is always decreased in secondary hypothyroidism **[T]**
Reason: Secondary hypothyroidism: pituitary failure → TSH low (or inappropriately normal) with low T_4.

c) Serum T_3, T_4 are always decreased in primary hypothyroidism **[F]**
Reason: In subclinical primary hypothyroidism T_3/T_4 stay normal with raised TSH.

d) Serum T_3, T_4 are always decreased in secondary hypothyroidism **[T]**
Reason: Secondary hypothyroidism has low T_3/T_4.

e) Serum T_3, T_4 can be normal in primary (subclinical) hypothyroidism **[T]**
Reason: Subclinical hypothyroidism: normal T_3/T_4, raised TSH.

Q5. Regarding ATP

a) Is a nucleotide derivative with 3 phosphate residues **[T]**
Reason: ATP = adenine + ribose + 3 phosphates.

b) Is a high-energy phosphate producing more than 7 kcal/mol of energy **[T]**
Reason: ATP hydrolysis ≈ –7.3 kcal/mol.

c) Is formed only in the respiratory chain **[F]**
Reason: ATP is also formed by substrate-level phosphorylation.

d) Is not formed at substrate level **[F]**
Reason: Substrate-level phosphorylation exists (PK, PGK, succinyl-CoA synthetase).

e) More than 50% of ATP is used to drive the osmotic pump **[F]**
Reason: Na^+-K^+ pump uses ~20–40% of ATP.

Q6. Fat-soluble vitamins

a) Require bile salts for absorption from gut **[T]**
Reason: Fat-soluble vitamins are absorbed in micelles.

b) May produce toxicity if taken in excess **[T]**
Reason: Fat-soluble vitamins (A, D) accumulate in the body.

c) Have no antioxidant properties **[F]**
Reason: Vitamins A, E (and carotenes) are antioxidants.

d) Vitamin K is required for blood coagulation **[T]**
Reason: Vitamin K γ-carboxylates factors II, VII, IX, X.

e) Are adequately synthesized in body **[F]**
Reason: None is adequately synthesised except vitamin D (skin) partly.

Q7. Dietary fibre

a) Are oligosaccharides **[F]**
Reason: Dietary fibres are non-starch polysaccharides.

b) Are not absorbed from the gut **[T]**

c) Increase the incidence of gut cancer **[F]**
Reason: Fibre REDUCES colon cancer risk.

d) Lower the blood cholesterol level **[T]**

e) Prevent constipation **[T]**

Q8. Beta-oxidation of fatty acid

a) Occurs in cytosol **[F]**
Reason: β-oxidation occurs in mitochondria (activation in cytosol).

b) Generates ATP **[T]**
Reason: β-oxidation generates abundant ATP.

c) Occurs only in saturated fatty acids **[F]**
Reason: Unsaturated and odd-chain FA are also oxidised.

d) Can't occur in odd-chain-carbon-containing fatty acids **[F]**
Reason: Cardiac muscle prefers fatty acids as fuel.

e) Requires nicotinamide (niacin-derived coenzyme) **[T]**
Reason: β-oxidation needs NAD^+ (niacin).

Q9. Fates of acetyl-CoA

a) Produces CO_2, H_2O and ATP in TCA cycle **[T]**
Reason: Acetyl-CoA is oxidised in the TCA cycle.

b) Formation of ketone bodies in prolonged starvation **[T]**
Reason: Excess acetyl-CoA → ketone bodies.

c) Directly converts to acetone **[F]**
Reason: Acetone arises from acetoacetate, not directly.

d) Synthesizes cholesterol **[T]**
Reason: Acetyl-CoA → HMG-CoA → cholesterol.

e) Gives rise to prostaglandin **[F]**
Reason: Prostaglandins derive from arachidonic acid.

Q10. Down syndrome

a) Karyotype is trisomy 22 **[F]**
Reason: Down syndrome is trisomy 21.

b) Is associated with primary hypothyroidism **[T]**
Reason: Down syndrome has an increased risk of hypothyroidism.

c) Is due to translocation (in all cases) **[F]**
Reason: ~95% of Down syndrome is trisomy; ~4% translocation.

d) In 75% of cases non-disjunction occurs during spermatogenesis **[F]**
Reason: ~95% of non-disjunction is maternal (oogenesis).

e) Occurs due to meiotic non-disjunction **[T]**
Reason: Meiotic non-disjunction (mostly maternal) causes trisomy 21.

Q11. The following are facts in the TCA cycle

a) An amphibolic cycle **[T]**
Reason: TCA cycle is both catabolic and anabolic.

b) Produces NADPH **[F]**
Reason: TCA produces NADH and $FADH_2$.

c) Produces ATP at substrate level **[T]**
Reason: Succinyl-CoA synthetase makes GTP/ATP at substrate level.

d) Involves the participation of lipoic acid **[T]**
Reason: α-KG dehydrogenase uses lipoic acid.

e) Citrate combines with oxaloacetate **[F]**
Reason: Oxaloacetate + acetyl-CoA → citrate.

Q12. PCR

a) Requires RNA polymerase **[F]**
Reason: PCR uses DNA polymerase.

b) Is a hybridization followed by amplification **[T]**
Reason: PCR cycles: denaturation, primer annealing (hybridization), extension.

c) Requires a primer **[T]**
Reason: DNA polymerase needs a primer.

d) Polymerization occurs by formation of phosphodiester bond **[T]**
Reason: DNA polymerase forms 3'–5' phosphodiester bonds.

e) DNA ligase is necessary **[F]**
Reason: Ligase is not used in PCR.

Q13. Thyroid hormone

a) Is an amino acid hormone **[T]**
Reason: T_3/T_4 are tyrosine derivatives.

b) Increases BMR **[T]**
Reason: Thyroid hormones raise BMR.

c) Increases Na^+-K^+ ATPase activity **[T]**
Reason: Thyroid hormone increases Na^+-K^+ ATPase → heat/O2 consumption.

d) Secretion is regulated by TSH **[T]**
Reason: TSH drives thyroid hormone secretion.

e) Mode of action is mediated by cyclic AMP **[F]**
Reason: Insulin acts via tyrosine kinase receptor.

Q14. Aldosterone

a) Is a powerful vasoconstrictor **[F]**
Reason: Angiotensin II is the vasoconstrictor; aldosterone retains Na^+.

b) Is a steroid hormone **[T]**
Reason: Aldosterone is a mineralocorticoid steroid.

c) Renin–angiotensin system induces its secretion strongly **[T]**
Reason: Angiotensin II stimulates aldosterone release.

d) Is a life-saving hormone **[T]**
Reason: Cortisol/aldosterone deficiency is fatal.

e) Hyposecretion may cause Conn's syndrome **[F]**
Reason: Conn's syndrome is aldosterone EXCESS.

Q15. Choose the correct answer

a) Hemophilia is an autosomal dominant disorder **[F]**
Reason: Haemophilia is X-linked recessive.

b) Permanent structural and functional changes in gene lead to mutation **[T]**
Reason: Mutation = permanent change in DNA.

c) Male is the sufferer, female is the carrier in hemophilia **[T]**
Reason: In XLR disease, males are affected, females are carriers.

d) Down syndrome is due to trisomy 21 **[T]**
Reason: Down syndrome = trisomy 21.

e) Transcription and translation constitute the central dogma **[T]**

Q16. Causes of dwarfism are

a) Growth hormone deficiency **[T]**
Reason: GH deficiency → pituitary dwarfism.

b) End-organ unresponsiveness to GH **[T]**
Reason: Laron dwarfism: GH-receptor defect.

c) Destruction of anterior pituitary **[T]**
Reason: Loss of GH → dwarfism.

d) Growth hormone deficiency (isolated) **[T]**
Reason: GH deficiency → pituitary dwarfism.

e) Overproduction of IGF-1 **[F]**
Reason: Excess IGF-1 → gigantism/acromegaly, not dwarfism.

Q17. ADH

a) Is synthesized in the anterior pituitary **[F]**
Reason: ADH is made in hypothalamic neurons and stored in the posterior pituitary.

b) Is passed to posterior pituitary by portal blood vessel **[F]**
Reason: ADH travels by axons (hypothalamo-hypophyseal tract).

c) Angiotensin is its most potent stimulus **[F]**
Reason: Osmolarity is the main stimulus; Ang II is a minor stimulus.

d) Increased K^+ concentration is its inhibitory stimulus **[F]**
Reason: ADH release is stimulated by high osmolarity/low volume.

e) Is a potent vasoconstrictor **[T]**
Reason: At high doses ADH constricts vessels (vasopressin).

Q18. Cell cycle

a) Controlled by alternate activation and deactivation of key enzymes **[T]**
Reason: Cyclin–CDK control the cell cycle.

b) In S phase replication of chromosomal DNA and histones occurs **[T]**
Reason: DNA and histones are synthesised in S phase.

c) G2 phase is the preparatory phase of mitosis **[T]**
Reason: G2 prepares for mitosis.

d) In M phase chromosomes are inactive here **[F]**
Reason: Chromosomes are condensed and transcriptionally silent in M phase.

e) G1 phase is the preparatory phase for replication **[T]**
Reason: G1 prepares for replication.

Q19. X-linked recessive disorders include

a) Phenylketonuria **[F]**
Reason: PKU is autosomal recessive.

b) β-thalassemia **[F]**
Reason: β-thalassaemia is autosomal recessive.

c) Hemophilia A **[T]**
Reason: Haemophilia A is X-linked recessive.

d) Agammaglobulinaemia (Bruton) **[T]**
Reason: Bruton's agammaglobulinaemia is X-linked recessive.

e) Nephrogenic diabetes insipidus **[T]**
Reason: Nephrogenic DI is X-linked recessive.

Q20. Insulin

a) Synthesized in β-cells of pancreatic islet **[T]**
Reason: Insulin is made in β-cells of islets.

b) Composed of two polypeptide chains **[T]**
Reason: Insulin has A (21 aa) and B (30 aa) chains linked by disulphides.

c) Is secreted in response to a fall in plasma glucose concentration **[F]**
Reason: Insulin is secreted when glucose RISES.

d) Activates glycogen phosphorylase in liver **[F]**
Reason: Insulin inhibits phosphorylase and activates glycogen synthase.

e) Absent in maturity-onset diabetes **[F]**
Reason: In MODY/type 2, insulin is present but insufficient/resistant.

November 2021 — Paper I

Q1. pH of a buffer solution

a) Is the negative logarithm of $[H^+]$ **[T]**
b) Is the negative logarithm of the dissociation constant **[F]**
Reason: pK is –log Ka; pH is –log $[H^+]$.
c) Depends only on the concentration of salt **[F]**
Reason: pH depends on the RATIO of salt to acid, not on concentration alone.
d) Depends only on the concentration of salt and acid (not on their ratio) **[F]**
Reason: pH depends on the RATIO of salt to acid, not on concentration alone.
e) Is equivalent to pK **[F]**
Reason: pH = pK only at equal salt and acid concentrations.

Q2. Regarding crystalloids

a) Non-dialyzable **[F]**
Reason: Colloids are too large to pass through a dialysis membrane.
b) Diameter is < 1 nm **[T]**
Reason: Crystalloid particles are < 1 nm.
c) Can pass through semipermeable membrane **[T]**
Reason: Crystalloids pass; colloids cannot pass through semipermeable membranes.
d) Having Brownian movement **[F]**
Reason: Crystalloids do not show Brownian movement; colloids do.
e) Having osmotic pressure higher than colloids **[T]**
Reason: Crystalloids are numerous small particles and exert greater osmotic pressure than colloids.

Q3. Enzymes are

a) Protein in nature **[T]**
Reason: Coenzymes are non-protein organic molecules; the apoenzyme is the protein.
b) Dialyzable **[F]**
Reason: Colloids are non-dialyzable.
c) Heat labile **[T]**
Reason: Enzymes are heat labile proteins; coenzymes are heat stable.
d) Reaction specific **[T]**
Reason: Enzymes are reaction and substrate specific.
e) Crystalloids **[F]**
Reason: Enzymes are colloids (proteins), not crystalloids.

Q4. Isotopes

a) Are atoms of the same element **[T]**
Reason: Isotopes: same atomic number, different neutrons.
b) Are atoms having the same number of protons **[T]**
Reason: Isotopes have the same number of protons.
c) Atomic weights are different **[T]**
Reason: Different neutron number → different atomic weight.
d) All have definite stable states **[F]**
Reason: Some isotopes are stable, others radioactive.
e) Radioactivity can be used as a therapeutic tool **[T]**
Reason: Radioisotopes (I-131, Co-60) are used therapeutically.

Q5. Regarding weak acid

a) Dissociates partially **[T]**
b) Is a proton acceptor **[F]**
Reason: A weak acid is a proton donor.
c) Has a weak conjugate base **[F]**
Reason: A weak acid has a relatively strong conjugate base.
d) Forms a buffer solution with its conjugate base **[T]**
e) H_2CO_3 is a weak acid **[T]**

Q6. Regarding lipid

a) Is soluble in alcohol **[T]**
Reason: Lipids dissolve in organic solvents, including alcohol.

b) Phospholipid reduces surface tension in lungs **[T]**
Reason: Dipalmitoyl lecithin (surfactant) lowers alveolar surface tension.

c) Saturated fatty acid is good for health **[F]**
Reason: Saturated fats raise LDL and cardiovascular risk.

d) Saponification number gives an idea about chain length of fatty acid **[T]**
Reason: Saponification number is inversely related to chain length.

e) Lipid peroxidation causes atherosclerosis **[T]**
Reason: Lipid peroxidation of LDL contributes to atherosclerosis.

Q7. Simple diffusion of a substance

a) Can occur against a concentration gradient **[F]**
Reason: Simple diffusion is passive, only down a concentration gradient.

b) Can occur against an electrochemical gradient **[F]**
Reason: Simple diffusion cannot go against an electrochemical gradient (needs active transport).

c) Varies directly with temperature change **[T]**
Reason: Diffusion rate rises with temperature (more kinetic energy).

d) Semipermeable membrane is essential **[F]**
Reason: Simple diffusion can occur without a semipermeable membrane.

e) Is inversely proportional to the solubility of solute **[F]**
Reason: Diffusion is directly proportional to lipid solubility.

Q8. Regarding carbohydrates

a) Glucose and mannose are epimers **[T]**
Reason: Glucose and mannose differ at C-2 (epimers).

b) Fructose is a ketose sugar **[T]**
Reason: Fructose has a keto group at C-2.

c) Do not show isomerism **[F]**
Reason: Carbohydrates show many isomerisms.

d) Sour in taste **[F]**
Reason: Sugars taste sweet.

e) Prevent ketosis **[T]**
Reason: Carbohydrate is anti-ketogenic (supplies oxaloacetate).

Q9. Characteristics of ICF

a) Major fluid volume compartment **[T]**
Reason: ICF holds ~2/3 of total body water.

b) K^+ is the major cation **[T]**
Reason: K^+ is the principal ICF cation.

c) Cl^- is the major anion **[F]**
Reason: Cl^- is the main ECF anion; ICF anions are phosphate and proteins.

d) Total ionic concentration is less than ECF **[F]**
Reason: ECF and ICF are iso-osmotic, with equal total ionic concentration.

e) pH is lower than ECF **[T]**
Reason: ICF pH (~7.0) is lower than ECF (7.4).

Q10. About digestion the following are true

a) Is a process of hydrolysis **[T]**

b) Lipid digestion begins in mouth **[T]**

c) Pepsin splits carbohydrate into glucose **[F]**
Reason: Pepsin digests protein.

d) Micelle is an aggregate of cholesterol only **[F]**
Reason: Micelles contain bile salts, phospholipid, cholesterol and FA — not cholesterol alone.

e) Bile salt reduces the surface tension of fat **[T]**
Reason: Bile salts are detergents that lower surface tension → emulsification.

Q11. Regarding vitamins †

a) Chemical nature of vitamin A is retinol **[T]**
Reason: Retinol (vitamin A) is the alcohol form.

b) Nicotinic acid is required for collagen synthesis **[F]**
Reason: Collagen synthesis needs vitamin C.

c) Active form of vitamin D is calcidiol **[F]**
Reason: Active form is calcitriol (1,25-DHCC); calcidiol is the storage form.

d) RDA of vitamin D in adults is 5 µg **[F]**
Reason: RDA of vitamin D in adults ≈ 600 IU (15 µg)/day.

e) Antioxidant vitamins are A, C and E **[T]**
Reason: Vitamins A, C, E are antioxidants.

Q12. Example of transcellular fluid

a) Blood **[F]**
Reason: Plasma is intravascular fluid, not transcellular.

b) Synovial fluid **[T]**
Reason: Synovial fluid is a transcellular fluid.

c) Fluid in aqueous humor **[T]**
Reason: Transcellular fluids (CSF, aqueous humor, synovial, pleural, peritoneal) are formed by epithelial transport.

d) Interstitial fluid **[F]**
Reason: Interstitial fluid is extracellular but not transcellular.

e) Pericardial fluid **[T]**
Reason: Pericardial fluid is a transcellular fluid.

Q13. Choose the correct answers †

a) 0.9% NaCl solution is a 1 N solution **[F]**
Reason: 0.9% NaCl is 'normal' saline, not 1 N (1 N NaCl = 5.85%).

b) 5% glucose solution is isotonic **[T]**
Reason: 5% glucose (278 mmol/L) is isotonic.

c) 1 M glucose solution produces 1 osmole **[T]**
Reason: 1 M glucose = 1 osmole/L (non-dissociating).

d) Anion gap is 18–20 mEq/L **[F]**
Reason: Normal anion gap is 8–16 mEq/L.

e) Plasma osmolarity is 280–296 mosm/L **[T]**
Reason: Normal plasma osmolarity is ~280–296 mosmol/L.

Q14. Deficiency of vitamins is associated with

a) Niacin – pellagra **[T]**
Reason: Niacin deficiency → pellagra (dermatitis, diarrhoea, dementia).

b) Biotin – beriberi **[F]**
Reason: Beriberi is thiamine deficiency.

c) Folic acid – neural tube defect **[T]**
Reason: Folate deficiency in early pregnancy → neural tube defects.

d) Vitamin C – megaloblastic anemia **[F]**
Reason: Vitamin C deficiency → scurvy; megaloblastic anaemia is B_{12}/folate.

e) Vitamin D – osteomalacia **[T]**
Reason: Vitamin D deficiency → rickets/osteomalacia.

Q15. Regarding protein †

a) Is a high-molecular-weight polypeptide **[T]**
Reason: Proteins are high-molecular-weight polypeptides.

b) Denatured by application of heat **[T]**
Reason: Heat breaks H-bonds and denatures protein.

c) Beta-pleated sheet is a tertiary structure **[F]**
Reason: β-pleated sheet is a secondary structure.

d) Exerts colloidal osmotic pressure **[T]**
Reason: Plasma proteins (albumin) exert oncotic pressure.

e) Concentration in plasma is 6–8 g/dL **[T]**
Reason: Normal plasma protein is 6–8 g/dL.

Q16. Folic acid has the following facts

a) The active form is dihydrofolate **[F]**
Reason: THF is the active form.

b) Is required for synthesis of nucleotides **[T]**

c) Acts as a carrier of single carbon unit **[T]**
Reason: THF carries one-carbon groups for purine/thymidylate synthesis.

d) RDA is 600–800 mg/day **[F]**
Reason: RDA of folate is ~400 µg/day (600 µg in pregnancy), not mg.

e) Deficiency causes macrocytic anemia **[T]**
Reason: Folate deficiency → macrocytic/megaloblastic anaemia.

Q17. Normal saline

a) Is a standard solution **[F]**
Reason: Normal saline is simply 0.9% NaCl isotonic solution.

b) Is 0.9% NaCl solution **[T]**
Reason: Normal saline = 0.9% NaCl (isotonic).

c) Means normal solution **[F]**
Reason: 'Normal' refers to physiological, not 1 N.

d) Is hypotonic in relation to plasma **[F]**
Reason: 0.9% NaCl is isotonic with plasma.

e) Used in the treatment of diarrhea **[T]**
Reason: Saline replaces fluid and Na^+ in diarrhoea.

Q18. Regarding trace elements

a) RDA is < 100 gm/day **[F]**
Reason: Trace elements are needed in mg (< 100 mg/day) not gm.

b) Fe is an example **[T]**
Reason: Iron is a trace element.

c) Iodine deficiency causes mental retardation **[T]**
Reason: Iodine is needed for thyroid hormone for brain development.

d) Daily requirement of iron is 10–20 mg for male **[T]**
Reason: Daily iron requirement is ~10 mg (men).

e) Zn acts as a cofactor of enzymes **[T]**
Reason: Zn is a cofactor of many enzymes.

Q19. The characteristics of kwashiorkor

a) Disease resulting from low protein intake **[T]**

b) There is severe emaciation **[F]**
Reason: Emaciation is marasmus.

c) There is hypoalbuminemia **[T]**

d) There is hepatomegaly **[T]**

e) Skin problem, hair change and loss of appetite **[T]**

Q20. In metabolic alkalosis

a) $[H^+]$ is decreased **[T]**
Reason: Alkalosis lowers $[H^+]$ and raises pH.

b) pH is increased **[T]**
Reason: Acidosis lowers pH; alkalosis raises it.

c) PCO_2 is increased (compensatory) **[T]**
Reason: Hypoventilation compensates metabolic alkalosis, raising PCO_2.

d) $[HCO_3^-]$ is increased **[T]**
Reason: HCO_3^- is raised in metabolic alkalosis (primary) and in compensated respiratory acidosis.

e) pH is < 7.35 **[F]**
Reason: Acidaemia is pH < 7.35; alkalosis has pH > 7.45.

November 2021 — Paper II

Q1. Enzymes required for cholesterol synthesis

a) HMG-CoA lyase **[F]**
Reason: HMG-CoA lyase is a ketogenesis enzyme.

b) Acetyl-CoA carboxylase **[F]**
Reason: Acetyl-CoA carboxylase is the rate-limiting enzyme of fatty acid synthesis.

c) HMG-CoA reductase **[T]**
Reason: HMG-CoA reductase is the rate-limiting enzyme.

d) Succinyl-CoA synthase **[F]**
Reason: Succinyl-CoA synthetase is a TCA enzyme.

e) HMG-CoA synthase **[T]**
Reason: HMG-CoA synthase (cytosolic) is in cholesterol synthesis.

Q2. HDL cholesterol

a) Synthesized from chylomicron **[F]**
Reason: Nascent HDL is made in liver/intestine.

b) Contains 50% apoprotein **[T]**
Reason: HDL is ~50% protein.

c) HDL size is largest among all lipoproteins **[F]**
Reason: HDL is the smallest and densest lipoprotein.

d) Atherogenic potential is (+++) **[F]**
Reason: HDL is anti-atherogenic (reverse cholesterol transport).

e) Metabolized in cardiac muscle **[F]**
Reason: HDL/LDL are metabolised in liver.

Q3. About blotting techniques

a) Northern blot is used for a specific RNA **[T]**
Reason: Northern detects RNA.

b) Western blot is used to detect DNA **[F]**
Reason: Western detects protein.

c) Southern blot detects a specific DNA segment **[T]**
Reason: Southern detects a DNA segment.

d) Gene expression can be detected by Northern blotting **[T]**
Reason: Northern blot shows mRNA expression.

e) DNA profile is visualized by Western blotting **[F]**
Reason: DNA fingerprint = Southern/PCR.

Q4. Regarding ketoacidosis

a) Low blood pH **[T]**
Reason: DKA: ketoacids lower pH.

b) Marked volume depletion occurs **[T]**
Reason: Osmotic diuresis → volume depletion.

c) ECF osmolarity is very high **[T]**

d) Plasma anion gap is normal **[F]**
Reason: Ketoanions raise the anion gap in DKA.

e) Blood acetoacetate is increased **[T]**
Reason: Ketogenesis raises acetoacetate.

Q5. Regarding glycogenesis

a) 3 ATP are produced **[F]**
Reason: Glycolysis yields ATP by substrate-level phosphorylation.

b) Glucose possesses higher energy than glycogen **[F]**
Reason: Glycogen is just a polymer of glucose; stored energy is not 'higher' in glucose per unit.

c) Rate-limiting enzyme is glycogen phosphorylase **[F]**
Reason: Glycogen synthase is rate-limiting for glycogenesis; phosphorylase for glycogenolysis.

d) Insulin inhibits glycogenesis **[F]**
Reason: Insulin stimulates glycogen synthase (glycogenesis).

e) Glycogenesis is stimulated by glucagon **[F]**
Reason: Glucagon stimulates glycogenolysis.

Q6. Choose the correct answer

a) All amino acids undergo transamination **[F]**
Reason: Lysine, threonine, proline, hydroxyproline do not undergo transamination.

b) Ammonia is removed by intestinal bacterial degradation **[F]**
Reason: Intestinal bacteria PRODUCE ammonia (urease); it is removed as urea/glutamine.

c) Oxidative deamination is a catabolic process **[T]**
Reason: Oxidative deamination (glutamate dehydrogenase) releases NH_3 from amino acids.

d) Excretion of NH_3 with urine as NH_4^+ **[T]**
Reason: Urinary NH_3 is excreted as NH_4^+.

e) Proline can be synthesized from the urea cycle **[F]**
Reason: Proline is made from glutamate; it is not a urea-cycle product.

Q7. About DNA replication

a) Is a semi-conservative process **[T]**
Reason: Each daughter DNA has one old and one new strand.

b) Requires elongation factors **[F]**
Reason: Elongation factors belong to translation.

c) Requires reverse transcriptase enzyme **[F]**
Reason: Replication uses DNA polymerase, not reverse transcriptase.

d) Polymerization of the new strand moves in the 3'→5' direction **[F]**
Reason: New strand grows 5'→3'.

e) DNA ligase is an enzyme of replication **[T]**
Reason: Ligase seals nicks in replication.

Q8. Features of Addison's disease are

a) Hypoglycemia **[T]**
Reason: Cortisol deficiency (Addison's) → hypoglycaemia.

b) Hyponatremia **[T]**
Reason: Aldosterone deficiency → Na^+ loss.

c) Hypocalcemia **[F]**
Reason: Addison's: calcium is normal/high, not low.

d) Hypopigmentation **[F]**
Reason: ↑ACTH/MSH → hyperpigmentation in Addison's.

e) Hypotension **[T]**
Reason: Aldosterone/cortisol lack → salt loss, hypotension; Cushing's causes hypertension.

Q9. Choose the correct answers

a) Blood osmolarity is 250 mOsm/L **[F]**
Reason: Normal plasma osmolarity is 280–296.

b) Microalbuminuria is 15 mg/day **[F]**
Reason: Microalbuminuria = 30–300 mg/day.

c) Normal blood urea is 40 mg/dL **[T]**

d) Rothera's nitroprusside test is positive in diabetic ketoacidosis **[T]**
Reason: Rothera's nitroprusside test detects acetoacetate/acetone.

e) Benedict's test is positive for sucrose **[F]**
Reason: Benedict's test detects reducing sugars (glucose, fructose, lactose, maltose), not sucrose.

Q10. Insulin

a) Is an exocrine pancreatic secretion **[F]**
Reason: Insulin is an endocrine secretion of β-cells.

b) Decreases ketosis **[T]**

c) Raises plasma potassium level in blood **[F]**
Reason: Insulin lowers plasma K^+.

d) Causes entry of glucose into skeletal muscle **[T]**
Reason: Insulin → GLUT-4 translocation.

e) Stimulates the release of free fatty acids from adipose tissue **[F]**
Reason: Insulin inhibits hormone-sensitive lipase.

Q11. TSH value †

a) 0.1 mU/L indicates thyrotoxicosis **[T]**
Reason: TSH < 0.1 mU/L suggests thyrotoxicosis.

b) > 10 mU/L indicates clinical hypothyroidism **[T]**
Reason: TSH > 10 mU/L with low T_4 = clinical hypothyroidism.

c) Can be used in diagnosis of thyroid disease **[T]**
Reason: TSH is the first-line thyroid screening test.

d) 5–20 mU/L indicates subclinical hypothyroidism **[T]**
Reason: Subclinical hypothyroidism: TSH 5–20 µIU/mL with normal T_3/T_4.

e) Should always be done in diagnosis of thyroid disease **[T]**
Reason: TSH is the first-line test for thyroid disease.

Q12. Regarding primary hypothyroidism †

a) Serum cholesterol increases **[T]**
Reason: Hypothyroidism reduces LDL receptors → ↑ cholesterol.

b) TSH always increases **[T]**
Reason: Primary hypothyroidism → raised TSH.

c) TSH is 5–20 µIU/mL in subclinical hypothyroidism **[T]**
Reason: Subclinical hypothyroidism: TSH 5–20 µIU/mL with normal T_3/T_4.

d) Strongly associated with infertility **[T]**
Reason: Hypothyroidism causes anovulation/menstrual disturbance → infertility.

e) T_3 and T_4 may be normal **[T]**
Reason: In subclinical hypothyroidism T_3/T_4 are normal.

Q13. In myocardial infarction, the enzymes/markers that rise within 4–6 hours

a) ALT **[F]**
Reason: ALT is the most specific for hepatocellular injury.

b) Troponin-I **[T]**
Reason: Cardiac troponins are the most specific MI markers.

c) AST **[F]**
Reason: AST is non-specific (heart, muscle, liver).

d) CPK-MB **[T]**

e) LDH **[F]**
Reason: LDH is a tetramer, so it has quaternary structure.

Q14. Choose the correct answers

a) Terminal codons are AUG, UAG and UGA **[F]**
Reason: AUG is the START codon; stop codons are UAA, UAG, UGA.

b) Mitochondrial DNA is maternal in origin **[T]**

c) Familial hypercholesterolemia is autosomal recessive **[F]**
Reason: FH is autosomal dominant (LDL receptor).

d) Haemophilia is an X-linked dominant disorder **[F]**
Reason: Haemophilia is X-linked RECESSIVE.

e) Mutation of gene may result in sudden termination of translation **[T]**

Q15. Blood glucose

a) Fasting hyperglycemia indicates insulin resistance **[T]**
Reason: Fasting hyperglycaemia = hepatic insulin resistance/deficiency.

b) Post-prandial hyperglycemia indicates insulin resistance **[T]**
Reason: Post-prandial rise indicates impaired insulin response.

c) Fasting < 70 mg/dL indicates hypoglycemia **[T]**
Reason: Fasting glucose < 70 mg/dL is hypoglycaemia.

d) Post-prandial > 140 mg/dL indicates hyperglycemia **[T]**
Reason: 2-h post-prandial > 140 mg/dL is abnormal.

e) Is increased in insulin therapy **[F]**
Reason: Insulin lowers blood glucose.

Q16. Regarding liver function tests †

a) Normal serum albumin level is 3–5.5 g/dL **[T]**
Reason: Normal albumin is 3.5–5.5 g/dL.

b) Serum ALT rises in hepatocellular jaundice **[T]**
Reason: ALT rises in hepatocellular jaundice.

c) Serum ALP rises in obstructive jaundice **[T]**
Reason: ALP rises in obstructive jaundice.

d) A:G ratio is reversed in chronic liver diseases **[T]**
Reason: A:G ratio is reversed (< 1) in chronic liver disease.

e) Conjugated bilirubin rises in prehepatic jaundice **[F]**
Reason: Prehepatic jaundice: unconjugated bilirubin rises.

Q17. The following statements are true

a) Triglycerides should be less than 40 mg/dL **[F]**
Reason: Desirable triglyceride < 150 mg/dL.

b) Down syndrome is an example of polyploidy **[F]**
Reason: Down syndrome is aneuploidy (trisomy), not polyploidy.

c) Non-disjunction may result in trisomy **[T]**
Reason: Non-disjunction gives trisomy/monosomy.

d) Turner syndrome karyotype is 47,XXY **[F]**
Reason: Turner = 45,X; 47,XXY is Klinefelter.

e) HbA_{1c} is more than 6.5% in DM **[T]**
Reason: HbA_{1c} > 6.5% is diagnostic of DM.

Q18. Examples of autosomal dominant disorders

a) Achondroplasia **[T]**
Reason: Achondroplasia is autosomal dominant.

b) Sickle cell anemia **[F]**
Reason: Sickle cell anaemia is autosomal recessive.

c) Phenylketonuria **[F]**
Reason: PKU is autosomal recessive.

d) Diabetes insipidus (nephrogenic) **[F]**
Reason: Nephrogenic diabetes insipidus is X-linked recessive.

e) Familial hypercholesterolemia **[T]**
Reason: FH is autosomal dominant (LDL receptor).

Q19. PCR

a) Requires RNA polymerase **[F]**
Reason: PCR uses DNA polymerase.

b) Is a hybridization followed by amplification **[T]**
Reason: PCR cycles: denaturation, primer annealing (hybridization), extension.

c) Requires a primer **[T]**
Reason: DNA polymerase needs a primer.

d) Polymerization occurs by formation of phosphodiester bond **[T]**
Reason: DNA polymerase forms 3'–5' phosphodiester bonds.

e) DNA ligase is necessary **[F]**
Reason: Ligase is not used in PCR.

Q20. Codons

a) Are consecutive bases of mRNA **[T]**

b) Are overlapped along the polynucleotide of mRNA **[F]**
Reason: The genetic code is non-overlapping.

c) Are responsible for synthesis of a specific amino acid **[T]**
Reason: Each codon specifies an amino acid.

d) UAG and UGA do not code for any amino acid **[T]**
Reason: UAA, UAG, UGA are stop codons.

e) AGU codes for methionine **[F]**
Reason: AUG = methionine (start); AGU = serine.

May 2021 — Paper I

Q1. Regarding buffer

a) Consists of a weak acid and its salt of strong base **[T]**
Reason: A buffer = weak acid + its salt with a strong base.

b) Is more effective when pH = pK **[T]**
Reason: Buffering is maximal at pH = pK (±1).

c) Regulates pH within seconds to minutes **[F]**
Reason: Chemical buffers act instantly; lungs take minutes and kidneys hours–days.

d) Only present extracellularly **[F]**
Reason: Buffers are present in both ECF and ICF.

e) The pK of the HCO_3^- buffer is 6.8 **[F]**
Reason: pK of the bicarbonate buffer is 6.1.

Q2. Colloid

a) Diameter is 1–200 mμ **[T]**
Reason: Colloid particle size is 1–200 millimicron (mμ = nm); crystalloids are < 1 nm.

b) Does not move in an electrical field **[F]**
Reason: Colloids carry charge and migrate in an electric field (electrophoresis).

c) Has an isoelectric pH **[T]**
Reason: Colloids such as proteins have a definite isoelectric pH.

d) Shows molecular movement in solution **[F]**
Reason: Colloids show Brownian movement (random particle movement); molecular/ionic movement is typical of crystalloids.

e) DNA is an example **[T]**
Reason: DNA is a macromolecule that forms colloidal solution.

Q3. The statements can be applied for the H-H equation

a) pH = pK + log salt/acid **[T]**
Reason: H-H equation: pH = pK + log [salt]/[acid].

b) pH = 6.1 + log HCO_3^-/H_2CO_3 **[T]**
Reason: Henderson–Hasselbalch for bicarbonate buffer: pH = 6.1 + log $[HCO_3^-]/[H_2CO_3]$.

c) pH = 6.1 + log $[HCO_3^-]/[H_2CO_3]$ **[T]**
Reason: Henderson–Hasselbalch for bicarbonate buffer: pH = 6.1 + log $[HCO_3^-]/[H_2CO_3]$.

d) pH = pK at half neutralization **[T]**
Reason: When acid is half-neutralized [salt] = [acid], log 1 = 0, so pH = pK.

e) Is a derivative of the law of mass action **[T]**
Reason: H-H equation is derived from the law of mass action applied to Ka of a weak acid.

Q4. Regarding carbohydrates

a) Are derivatives of polyhydric aldehydes and ketones **[T]**
Reason: Carbohydrates = polyhydroxy aldehydes/ketones and derivatives.

b) Show optical isomerism **[T]**
Reason: Asymmetric carbons give optical isomerism.

c) Fructose is an invert sugar **[F]**
Reason: Invert sugar is a glucose + fructose mixture from sucrose hydrolysis.

d) Sucrose exhibits mutarotation **[F]**
Reason: Sucrose has no free anomeric carbon, so no mutarotation.

e) Prevent ketosis **[T]**
Reason: Carbohydrate is anti-ketogenic (supplies oxaloacetate).

Q5. Quaternary structure is present in

a) LDH **[T]**
Reason: LDH is a tetramer, so it has quaternary structure.

b) Coagulated protein **[F]**
Reason: Coagulated protein is denatured; quaternary structure is lost.

c) Hemoglobin **[T]**
Reason: Hb is a tetramer (α2β2): quaternary structure.

d) Denatured protein **[F]**
Reason: Denatured protein loses its higher-order structure.

e) Myoglobin **[F]**
Reason: Myoglobin is a single polypeptide with tertiary structure only.

Q6. Choose the correct answer

a) All amino acids undergo transamination **[F]**
Reason: Lysine, threonine, proline, hydroxyproline do not undergo transamination.

b) Ammonia is removed by intestinal bacterial degradation **[F]**
Reason: Intestinal bacteria PRODUCE ammonia (urease); it is removed as urea/glutamine.

c) Oxidative deamination is a catabolic process **[T]**
Reason: Oxidative deamination (glutamate dehydrogenase) releases NH_3 from amino acids.

d) Excretion of NH_3 with urine as NH_4^+ **[T]**
Reason: Urinary NH_3 is excreted as NH_4^+.

e) Proline can be synthesized from the urea cycle **[F]**
Reason: Proline is made from glutamate; it is not a urea-cycle product.

Q7. Enzymes with their clinical significance

a) ALT is highly specific for liver **[T]**
Reason: ALT is more liver-specific than AST.

b) CPK-MM is highly specific for muscle **[T]**
Reason: CPK-MM is the skeletal muscle isoenzyme; CK-MB is cardiac.

c) SGPT is a heart disease marker **[F]**
Reason: ALT (SGPT) is the liver marker; AST rises in MI.

d) ALP is increased in prostate cancer **[F]**
Reason: ACP (not ALP) rises in prostate cancer.

e) Aldolase is increased in brain tumor **[F]**
Reason: Aldolase rises in muscle disease (muscular dystrophy).

Q8. Regarding fatty acids

a) Are aliphatic monocarboxylic acids **[T]**
Reason: Fatty acids are aliphatic monocarboxylic acids.

b) Are not bound to albumin in plasma **[F]**
Reason: Free fatty acids circulate bound to albumin.

c) Are simple lipids **[F]**
Reason: Fatty acids are components/derived lipids.

d) Require micelle for absorption **[T]**
Reason: Long-chain fatty acids need bile-salt micelles.

e) Can be synthesized from glucose **[T]**
Reason: Glucose → acetyl-CoA → fatty acids.

Q9. Osmosis

a) Depends on the concentration of the solutes **[T]**
Reason: Osmotic pressure depends on the number of solute particles.

b) Depends on the movement of the solvent **[T]**
Reason: Osmosis is by definition solvent movement.

c) Needs semipermeable membrane **[T]**
Reason: Osmosis = solvent movement across a semipermeable membrane.

d) Only the solutes can cross the membrane **[F]**
Reason: In osmosis the solvent moves; solutes cannot cross.

e) Both the solutes and the solvent cross the membrane **[F]**
Reason: In osmosis only the solvent crosses the semipermeable membrane.

Q10. Fat-soluble vitamins

a) Require bile salts for absorption from the gut **[T]**
Reason: Fat-soluble vitamins are absorbed in micelles.

b) May produce toxicity if taken in excess **[T]**
Reason: Fat-soluble vitamins (A, D) accumulate in the body.

c) Have no antioxidant properties **[F]**
Reason: Vitamins A, E (and carotenes) are antioxidants.

d) Vitamin K is required for blood coagulation **[T]**
Reason: Vitamin K γ-carboxylates factors II, VII, IX, X.

e) Are adequately synthesized in the body **[F]**
Reason: None is adequately synthesised except vitamin D (skin) partly.

Q11. Regarding coenzymes

a) Only protein in nature **[F]**
Reason: Coenzymes are NON-protein organic molecules.

b) Catalyze chemical reactions **[F]**
Reason: Enzymes catalyse; coenzymes only assist as carriers.

c) Most of them are derivatives of fat-soluble vitamins **[F]**
Reason: Most coenzymes derive from water-soluble (B-complex) vitamins.

d) Thiamine pyrophosphate is an example **[T]**
Reason: TPP (from vitamin B1) is a coenzyme for oxidative decarboxylation and transketolase.

e) NADPH is a coenzyme for reductive synthesis **[T]**
Reason: NADPH donates reducing equivalents for biosynthesis (fatty acid, cholesterol).

Q12. Following are the facts for gut hormones

a) CCK-PZ augments HCO_3^- secretion **[F]**
Reason: Secretin stimulates HCO_3^- secretion; CCK stimulates enzyme secretion.

b) Secretin increases enzyme-rich pancreatic secretion **[F]**
Reason: Secretin → HCO_3^--rich juice; CCK → enzyme-rich juice.

c) Gastrin causes contraction of gall bladder **[F]**
Reason: CCK causes gall bladder contraction.

d) Somatostatin reduces gastric and intestinal motility **[T]**

e) Gall bladder contraction is inhibited by CCK-PZ **[F]**
Reason: CCK stimulates gall bladder contraction.

Q13. Dietary fibres

a) Are oligosaccharides **[F]**
Reason: Dietary fibres are non-starch polysaccharides.

b) Are not absorbed from the gut **[T]**

c) Increase the incidence of colon cancer **[F]**
Reason: Fibre REDUCES colon cancer risk.

d) Lower the blood cholesterol level **[T]**

e) Prevent constipation **[T]**

Q14. In metabolic acidosis

a) $[H^+]$ decreased **[F]**
Reason: In acidosis $[H^+]$ rises and pH falls.

b) PCO_2 increased **[F]**
Reason: In metabolic acidosis PCO_2 falls (respiratory compensation); raised PCO_2 is respiratory acidosis.

c) HCO_3^- decreased **[T]**
Reason: Primary defect in metabolic acidosis is low HCO_3^-.

d) Normal anion gap only **[F]**
Reason: Metabolic acidosis may have high or normal anion gap.

e) pH is < 7.35 **[T]**
Reason: Acidaemia is pH < 7.35; alkalosis has pH > 7.45.

Q15. Regarding ECF

a) Is 40% of total body water **[F]**
Reason: ECF is ~1/3 (33%) of TBW; ICF is 2/3.

b) Concentration of Na^+ is 145 mEq/L **[T]**
Reason: ECF Na^+ is ~135–145 mEq/L.

c) Protein content is 35–55 g/L **[F]**
Reason: Protein content of ECF/plasma is 6–8 g/dL; interstitial fluid is much lower.

d) Maintains homeostasis **[T]**
Reason: ECF is the internal environment that maintains homeostasis.

e) Is less than ICF volume **[T]**
Reason: ECF volume (1/3 TBW) is less than ICF (2/3).

Q16. The characteristics of marasmus include

a) Inadequate protein and calorie intake **[T]**
b) Most commonly seen in first year of life **[T]**
c) Oedema is present **[F]**
Reason: Edema is a feature of kwashiorkor, not marasmus.
d) Severe growth retardation **[T]**
e) Body weight/age is 60–80% **[F]**
Reason: Marasmus: weight/age < 60% of standard (kwashiorkor 60–80% with oedema).

Q17. Radioactive isotope

a) Is a stable compound **[F]**
Reason: Radioisotopes are unstable.
b) Remains as a mixture in all natural sources **[F]**
Reason: Radioisotopes do not 'remain as a mixture'; they decay.
c) Breaks down constantly into more stable isotopes **[T]**
Reason: Radioisotopes decay to stable isotopes.
d) Is used to measure body fluid compartment volume **[T]**
Reason: Radioisotopes (tritium, I-131, Cr-51) measure fluid compartment volumes.
e) Is used for the diagnosis and treatment of hyperthyroidism **[T]**
Reason: I-131 is used to diagnose and treat hyperthyroidism.

Q18. Regarding digestion and absorption

a) Protein digestion begins in mouth **[F]**
Reason: Protein digestion begins in stomach (pepsin).
b) Fat emulsification occurs by bile salts **[T]**
Reason: Emulsification is one role; micelle formation for absorption is key.
c) Glucose is absorbed by Na^+-glucose cotransporter **[T]**
d) Fructose is absorbed by facilitated diffusion **[T]**
e) Cholagogue decreases bile flow **[F]**
Reason: Cholagogues INCREASE bile flow.

Q19. Choose the correct answers

a) Cell membrane is a phospholipid bilayer only **[F]**
Reason: Membrane also has proteins and cholesterol.
b) There are protein channels in the cell membrane **[T]**
c) Na^+-K^+ active transport releases 3 K^+ outside the cell **[F]**
Reason: Na^+-K^+ pump: 3 Na^+ OUT, 2 K^+ IN.
d) Ca^{2+} acts as a guard in the protein channel **[F]**
Reason: Ca^{2+} is not a channel guard; channels are gated by voltage/ligand.
e) Ion channels are required for generation of electrogenic potential **[T]**

Q20. Regarding iron

a) Absorption occurs in large gut **[F]**
Reason: Iron is absorbed in duodenum and proximal jejunum.
b) Is absorbed in ferric form **[F]**
Reason: Fe^{2+} (ferrous) is the absorbed form.
c) Vitamin D favors its absorption **[F]**
Reason: Vitamin C favours iron absorption.
d) Is stored as ferritin **[T]**
Reason: Intracellular iron is stored as ferritin.
e) Binds with transferrin **[T]**
Reason: Fe^{3+} circulates bound to transferrin.

May 2021 — Paper II

Q1. Fates of acetyl-CoA

a) Condenses with oxaloacetate **[T]**
Reason: Acetyl-CoA + OAA → citrate.

b) Converted to fatty acid **[T]**
Reason: Acetyl-CoA is the building block for FA synthesis.

c) Converted to amino acid **[F]**
Reason: Acetyl-CoA cannot be net-converted to amino acids.

d) Converted to ketone bodies **[T]**
Reason: Acetyl-CoA → ketone bodies in starvation.

e) Converted to glucose **[F]**
Reason: Acetyl-CoA cannot be converted to glucose in humans (no net).

Q2. Features of Addison's disease are

a) Hypoglycemia **[T]**
Reason: Cortisol deficiency (Addison's) → hypoglycaemia.

b) Hyponatremia **[T]**
Reason: Aldosterone deficiency → Na^+ loss.

c) Hypokalemia **[F]**
Reason: Addison's → HYPERkalaemia (aldosterone lack); Cushing's → hypokalaemia (mineralocorticoid effect of cortisol).

d) Hypopigmentation **[F]**
Reason: ↑ACTH/MSH → hyperpigmentation in Addison's.

e) Hypotension **[T]**
Reason: Aldosterone/cortisol lack → salt loss, hypotension; Cushing's causes hypertension.

Q3. Turner's syndrome is characterized by

a) Absence of ovary (streak gonads) **[T]**
Reason: Streak gonads in Turner.

b) Webbed neck **[T]**
Reason: Turner features.

c) Estrogen deficiency **[T]**
Reason: Streak ovaries → low oestrogen.

d) Association with primary hypothyroidism **[T]**
Reason: Down syndrome has an increased risk of hypothyroidism.

e) Hypergonadotropic hypogonadism **[T]**
Reason: Low oestrogen → high FSH/LH.

Q4. Genes

a) Contain alleles that cause the trait **[T]**
Reason: Genes have alleles.

b) Expression of 45,XO causes Turner syndrome **[F]**
Reason: Turner syndrome is a chromosomal (not gene) defect.

c) Female karyotype is 46,XY **[F]**
Reason: Female karyotype is 46,XX.

d) Contain code for enzyme synthesis **[T]**
Reason: Genes code for polypeptides/enzymes.

e) Can pair with non-complementary bases **[F]**
Reason: Base pairing is A–T, G–C only.

Q5. Causes of dwarfism are

a) Growth hormone deficiency **[T]**
Reason: GH deficiency → pituitary dwarfism.

b) End-organ unresponsiveness to GH **[T]**
Reason: Laron dwarfism: GH-receptor defect.

c) Destruction of anterior pituitary **[T]**
Reason: Loss of GH → dwarfism.

d) GHIH deficiency **[F]**
Reason: GHIH (somatostatin) deficiency would raise GH, not cause dwarfism.

e) Overproduction of IGF-1 **[F]**
Reason: Excess IGF-1 → gigantism/acromegaly, not dwarfism.

Q6. About DNA

a) Is present in nucleus **[T]**
Reason: DNA is in nucleus and mitochondria.

b) Introns are coding regions **[F]**
Reason: Introns are non-coding; exons code.

c) Length is measured by number of base pairs **[T]**
Reason: DNA length is in base pairs.

d) Present in mitochondria **[T]**
Reason: mtDNA is circular, maternal.

e) Contains ribose **[F]**
Reason: DNA contains deoxyribose.

Q7. Carbohydrate is metabolized by

a) Glycolysis **[T]**
Reason: Glycolysis: PFK-1 is rate limiting.

b) TCA cycle **[T]**

c) Gluconeogenesis **[T]**
Reason: Gluconeogenesis is an anabolic synthesis of glucose; occurs in fasting, liver/kidney, inhibited by insulin.

d) Pentose phosphate pathway **[T]**

e) Uronic acid pathway **[T]**

Q8. Features of Cushing's syndrome include

a) Hypokalemia **[T]**
Reason: Addison's → HYPERkalaemia (aldosterone lack); Cushing's → hypokalaemia (mineralocorticoid effect of cortisol).

b) Acidosis **[F]**
Reason: Acidosis shifts K^+ out of cells → hyperkalaemia, not hypokalaemia.

c) Hypotension **[F]**
Reason: Aldosterone/cortisol lack → salt loss, hypotension; Cushing's causes hypertension.

d) Positive dexamethasone suppression test **[F]**
Reason: Cushing's syndrome: cortisol is NOT suppressed by dexamethasone.

e) Muscular weakness **[T]**
Reason: Protein catabolism and hypokalaemia → weakness.

Q9. Impaired renal function causes

a) Increased blood ammonia **[F]**
Reason: Renal failure raises urea/creatinine; blood ammonia is raised in liver failure.

b) Increased prothrombin time **[F]**
Reason: Prolonged PT is a liver, not renal, feature.

c) Is associated with anemia **[T]**

d) Increased serum K^+ **[T]**

e) Decreased creatinine clearance **[T]**

Q10. Down syndrome

a) Karyotype is trisomy 22 **[F]**
Reason: Down syndrome is trisomy 21.

b) Is associated with primary hypothyroidism **[T]**
Reason: Down syndrome has an increased risk of hypothyroidism.

c) Is due to translocation (in all cases) **[F]**
Reason: ~95% of Down syndrome is trisomy; ~4% translocation.

d) In 75% of cases non-disjunction occurs in spermatozoa formation **[F]**
Reason: ~95% of non-disjunction is maternal.

e) Occurs due to meiotic non-disjunction **[T]**
Reason: Meiotic non-disjunction (mostly maternal) causes trisomy 21.

Q11. Ammonia

a) Is produced during oxidation of uric acid **[F]**
Reason: Ammonia is not produced during uric acid oxidation.

b) Is added to the amino acid pool **[F]**
Reason: Ammonia leaves the amino acid pool.

c) Is produced in transamination **[F]**
Reason: Transamination moves NH2 without releasing NH_3.

d) Is a toxic agent during uremia **[T]**
Reason: Hyperammonaemia is toxic.

e) Can combine with carbon dioxide to form carbamoyl phosphate **[T]**

Q12. Prostaglandin

a) Is a 20-carbon fatty acid derivative **[T]**
Reason: Cardiac muscle prefers fatty acids as fuel.

b) Is a cyclic endoperoxide **[T]**

c) Is a hormone **[F]**
Reason: Prostaglandins are local autacoids, not circulating hormones.

d) Prostacyclin constricts blood vessels and aggregates platelets **[F]**
Reason: PGI_2 vasodilates and inhibits platelet aggregation.

e) $PGF_{2\alpha}$ is a potent bronchodilator **[F]**
Reason: $PGF_{2\alpha}$ is a bronchoconstrictor.

Q13. Thyroid function test

a) Normal TSH level is 0.4–4.0 µIU/mL **[T]**
Reason: Normal TSH is ~0.4–4.5 µIU/mL.

b) When TSH is normal, T_3 and T_4 estimations are usually not carried out **[T]**
Reason: When TSH is normal, T_3/T_4 are normal in primary thyroid disease.

c) TSH > 20 µIU/mL indicates hyperthyroidism **[F]**
Reason: I-131 is used to diagnose and treat hyperthyroidism.

d) Normal TSH excludes hypothyroidism or hyperthyroidism **[T]**
Reason: A normal TSH excludes primary hypo/hyperthyroidism.

e) TSH, T_3 and T_4 levels decrease in secondary hypothyroidism **[T]**
Reason: Secondary hypothyroidism: low TSH, T_3, T_4.

Q14. The following statements are applicable for cortisol

a) Counter-shock (anti-stress) hormone **[T]**
Reason: Cortisol is the anti-stress hormone.

b) A life-saving hormone **[T]**
Reason: Cortisol/aldosterone deficiency is fatal.

c) Decreases protein breakdown **[F]**
Reason: Cortisol is catabolic for protein.

d) Excess secretion can cause central obesity **[T]**
Reason: Excess cortisol → truncal obesity.

e) Increases blood sugar mainly by glycogenolysis **[F]**
Reason: Cortisol raises glucose mainly via gluconeogenesis.

Q15. Regarding the respiratory chain

a) Can be inhibited by cyanide resulting in histotoxic hypoxia **[T]**
Reason: Cyanide blocks cytochrome oxidase → histotoxic hypoxia.

b) Reducing equivalents pass through coenzymes and carriers **[T]**
Reason: NADH/$FADH_2$ pass electrons through carriers.

c) Does not create a redox potential **[F]**
Reason: ETC carriers have increasing redox potential.

d) Drives a proton pump for oxidative phosphorylation **[T]**
Reason: Complexes I, III, IV pump H^+ to build the gradient for ATP synthase.

e) Produces ATP at substrate level **[F]**
Reason: Succinyl-CoA synthetase makes GTP/ATP at substrate level.

Q16. Regarding protein metabolism

a) Site of deamination is liver and kidney **[T]**
Reason: Deamination occurs mainly in liver and kidney.

b) Site of transamination is cytoplasm only **[F]**
Reason: Transamination occurs in cytosol and mitochondria.

c) Site of transamination is liver only **[F]**
Reason: Transamination occurs in many tissues.

d) CO_2 liberated by TCA cycle is used in urea cycle **[F]**
Reason: Urea-cycle CO_2 is from HCO_3^-/bicarbonate, not directly from TCA.

e) Dehydrogenation occurs in non-oxidative deamination **[F]**
Reason: Non-oxidative deamination uses dehydratase/lyase.

Q17. Ketone bodies

a) Are organic acids **[T]**
Reason: Acetoacetate and β-OHB are organic acids.

b) In excess can produce respiratory acidosis **[F]**
Reason: Ketone bodies cause METABOLIC acidosis.

c) Are produced in uncontrolled diabetes mellitus **[T]**
Reason: Lack of insulin → lipolysis → ketogenesis.

d) In diabetic ketosis they increase HCO_3^- **[F]**
Reason: Ketone bodies consume HCO_3^-.

e) Are identified by Rothera's nitroprusside test in urine **[T]**
Reason: Rothera's nitroprusside test detects acetoacetate/acetone.

Q18. Examples of autosomal dominant disorders

a) Achondroplasia **[T]**
Reason: Achondroplasia is autosomal dominant.

b) Sickle cell anemia **[F]**
Reason: Sickle cell anaemia is autosomal recessive.

c) Phenylketonuria **[F]**
Reason: PKU is autosomal recessive.

d) Diabetes insipidus (nephrogenic) **[F]**
Reason: Nephrogenic diabetes insipidus is X-linked recessive.

e) Familial hypercholesterolemia **[T]**
Reason: FH is autosomal dominant (LDL receptor).

Q19. Codons

a) Are three consecutive bases of mRNA **[T]**
Reason: Codon = 3 consecutive bases of mRNA.

b) Are overlapped along the polynucleotide of mRNA **[F]**
Reason: The genetic code is non-overlapping.

c) Are responsible for synthesis of a specific amino acid **[T]**
Reason: Each codon specifies an amino acid.

d) UAG and UGA do not code for any amino acid **[T]**
Reason: UAA, UAG, UGA are stop codons.

e) AGU codes for methionine **[F]**
Reason: AUG = methionine (start); AGU = serine.

Q20. Regarding lipoprotein the following statements are correct

a) Chylomicrons are absorbed via the portal circulation **[F]**
Reason: Chylomicrons enter lymph (lacteals), not the portal vein.

b) Chylomicron is catabolized via apo-E receptor **[T]**
Reason: Chylomicron remnants are taken up by liver via apo-E receptor.

c) VLDL is produced from liver **[T]**
Reason: VLDL is synthesised in liver.

d) Cholesteryl ester transfer protein transfers cholesteryl ester from HDL to LDL **[T]**
Reason: CETP moves cholesteryl ester from HDL to VLDL/LDL.

e) LDL cholesterol is a risk factor for coronary heart disease **[T]**
Reason: High LDL is a major coronary risk factor.